

**SingleRAN**

# **Base Station Equipment Reliability Feature Parameter Description**

|              |            |
|--------------|------------|
| <b>Issue</b> | Draft B    |
| <b>Date</b>  | 2026-01-30 |



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<https://securitybulletin.huawei.com/enterprise/en/security-advisory>

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# 1 Change History

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This chapter describes changes not included in the "Parameters", "Counters", "Glossary", and "Reference Documents" chapters. These changes include:

- Technical changes  
Changes in functions and their corresponding parameters
- Editorial changes  
Improvements or revisions to the documentation

## 1.1 SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

### Technical Changes

None

### Editorial Changes

Deleted descriptions of the impact relationships of RF module super dormancy with Intelligent Shutdown of TRX Due to PSU Failure and Shutdown of TRX Due to PSU Failure.

Added the requirements of the "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" function for adding physical cell IDs and neighboring cell lists. For details, see [10.3.4 Networking](#).

Revised descriptions of fault scenarios in the principles of the "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" function. For details, see [10.1 Principles](#) and [10.2.1 Benefits](#).

Revised descriptions of setting the **Fault Detection Duration** parameter. For details, see [10.4.1.1 Data Preparation](#) and [10.4.1.2 Using MML Commands](#).

Added descriptions of impact relationships of "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" with RF module deep dormancy and RF module super dormancy. For details, see [10.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

## 1.2 SRAN22.1 Draft A (2025-12-31)

This issue introduces the following changes to SRAN21.1 06 (2025-09-26).

### Technical Changes

| Change Description                                                                                                                                                                                                                                                                                                                                                                        | Parameter Change                                                                                                             | Base Station Model                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added support for "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios". For details, see: <ul style="list-style-type: none"><li>• <a href="#">2.4 Differences</a></li><li>• <a href="#">3 Overview</a></li><li>• <a href="#">10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)</a></li></ul> | Modified parameter:<br>Added the <b>DUAL_PLANE_TOB_SW</b> option to the <b>gNBRedundancy.gNBReliabilitySwitch</b> parameter. | 5900 series base stations (macro base stations)                                                                                                                                   |
| Added support for transmission link backup of main control boards by the UMPThb. For details, see <a href="#">5.3.3 Hardware</a> .                                                                                                                                                                                                                                                        | None                                                                                                                         | <ul style="list-style-type: none"><li>• 3900 and 5900 series base stations (macro base stations)</li><li>• DBS3900 LampSite and DBS5900 LampSite</li></ul>                        |
| Added support for Cold Backup of Main Control Boards by the UMPThb. For details, see <a href="#">4.3.3 Hardware</a> .                                                                                                                                                                                                                                                                     | None                                                                                                                         | <ul style="list-style-type: none"><li>• 5900 series base stations (macro base stations)</li><li>• DBS5900 LampSite</li></ul>                                                      |
| Excluded the BTS5929R from the base station models that are compatible with Intra-Board Baseband Resource Pool. For details, see <a href="#">8.3.3 Hardware</a> .                                                                                                                                                                                                                         | None                                                                                                                         | <ul style="list-style-type: none"><li>• 3900 and 5900 series base stations (macro base stations) excluding the BTS5929R</li><li>• DBS3900 LampSite and DBS5900 LampSite</li></ul> |

| Change Description                                                                                                                                                                                                                                                         | Parameter Change | Base Station Model                                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Added impact relationships of RF module super dormancy with Intelligent Shutdown of TRX Due to PSU Failure and Shutdown of TRX Due to PSU Failure. For details, see <a href="#">13.3.2.3 Function Impacts</a> .                                                            | None             | <ul style="list-style-type: none"> <li>3900 and 5900 series base stations (macro base stations)</li> <li>DBS3900 LampSite and DBS5900 LampSite</li> </ul> |
| Added impact relationships of adaptive baseband shutdown and power saving first with Inter-Board Baseband Resource Redundancy. For details, see <a href="#">7.3.2.3 Function Impacts</a> .                                                                                 | None             | <ul style="list-style-type: none"> <li>3900 and 5900 series base stations (macro base stations)</li> <li>DBS5900 LampSite</li> </ul>                      |
| Added support for X2 self-management in SA networking by transmission link backup of main control boards. For details, see: <ul style="list-style-type: none"> <li><a href="#">5.4.3.1 Data Preparation</a></li> <li><a href="#">5.4.3.2 Using MML Commands</a></li> </ul> | None             | <ul style="list-style-type: none"> <li>3900 and 5900 series base stations (macro base stations)</li> <li>DBS3900 LampSite and DBS5900 LampSite</li> </ul> |

## Editorial Changes

Changed the name of the function controlled by the **BB\_SHUTDOWN\_POWERSAVING\_SW** option of the **gNodeBParam.BaseStationDeepDormancySw** parameter to "baseband shutdown for energy saving" and revised related descriptions of function impacts. For details, see [7.3.2.3 Function Impacts](#).

Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

# 2 About This Document

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## 2.1 General Statements

### Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

#### NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

### Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

## 2.2 Applicable RAT

This document applies to GSM, UMTS, LTE FDD, LTE TDD, NB-IoT, and NR.

Unless otherwise specified, in this document, LTE and eNodeB always include FDD, TDD, and NB-IoT. In scenarios where they need to be distinguished, LTE FDD, LTE TDD, and LTE NB-IoT are used. The same rules apply to eNodeB.

## 2.3 Features in This Document

This document describes the following features.

| RAT     | Feature ID      | Feature Name                                              | Chapter/Section                                                     |
|---------|-----------------|-----------------------------------------------------------|---------------------------------------------------------------------|
| UMTS    | MRFD-210101     | System Redundancy                                         | <a href="#">4 Cold Backup of Main Control Boards</a>                |
| LTE FDD | LBFD-00202102   | Cell Re-build Between Baseband Processing Units           | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a> |
| NB-IoT  | MLBFD-12000418  | Cell Re-build Between Baseband Processing Units           | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a> |
| LTE TDD | TDLBFD-00202102 | Cell Re-build Between Baseband Processing Units           | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a> |
| LTE FDD | LBFD-00202104   | Intra-baseband Card Resource Pool (user level/cell level) | <a href="#">8 Intra-Board Baseband Resource Pool</a>                |
| LTE TDD | TDLBFD-00202104 | Intra-baseband Card Resource Pool (user level/cell level) | <a href="#">8 Intra-Board Baseband Resource Pool</a>                |
| GSM     | GBFD-117804     | Intelligent Shutdown of TRX Due to PSU Failure            | <a href="#">13 Power Supply Reliability</a>                         |
| NR      | FBFD-020101     | Reliability                                               | <a href="#">4 Cold Backup of Main Control Boards</a>                |
| NR      | FBFD-020101     | Reliability                                               | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a> |
| NR      | FBFD-020101     | Reliability                                               | <a href="#">9 Inter-BBP Active/Standby Backup (NR)</a>              |
| NR      | FBFD-020101     | Reliability                                               | <a href="#">11 Cold Backup of RF Modules (NR TDD)</a>               |

## 2.4 Differences

**Table 2-1** Differences between LTE FDD and LTE TDD

| Function Name                                   | Difference                | Chapter/Section                                                            |
|-------------------------------------------------|---------------------------|----------------------------------------------------------------------------|
| Cold Backup of Main Control Boards              | None                      | <a href="#">4 Cold Backup of Main Control Boards</a>                       |
| Transmission Link Backup of Main Control Boards | None                      | <a href="#">5 Transmission Link Backup of Main Control Boards (LTE/NR)</a> |
| Inter-Board Baseband Resource Redundancy        | None                      | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a>        |
| Intra-Board Baseband Resource Pool              | None                      | <a href="#">8 Intra-Board Baseband Resource Pool</a>                       |
| RRU Channel Cross Connection Under MIMO         | Supported only in LTE FDD | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>        |
| Power Supply Reliability                        | None                      | <a href="#">13 Power Supply Reliability</a>                                |
| Heat Dissipation Reliability for Fans           | None                      | <a href="#">14.2 Heat Dissipation Reliability for Fans</a>                 |
| Power Supply Redundancy                         | None                      | <a href="#">14.3 Power Supply Redundancy</a>                               |
| Anti-Misinsertion Design of Boards              | None                      | <a href="#">14.4 Anti-Misinsertion Design of Boards</a>                    |
| Overtemperature Protection for BBU Boards       | None                      | <a href="#">14.5 Overtemperature Protection for BBU Boards</a>             |
| Surge Protection Design                         | None                      | <a href="#">14.6 Surge Protection Design</a>                               |

**Table 2-2** Differences between LTE NB-IoT and LTE FDD

| Function Name                      | Difference | Chapter/Section                                      |
|------------------------------------|------------|------------------------------------------------------|
| Cold Backup of Main Control Boards | None       | <a href="#">4 Cold Backup of Main Control Boards</a> |



| Function Name                                   | Difference                | Chapter/Section                                                            |
|-------------------------------------------------|---------------------------|----------------------------------------------------------------------------|
| Transmission Link Backup of Main Control Boards | None                      | <a href="#">5 Transmission Link Backup of Main Control Boards (LTE/NR)</a> |
| Inter-Board Baseband Resource Redundancy        | None                      | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a>        |
| Intra-Board Baseband Resource Pool              | Supported only in LTE FDD | <a href="#">8 Intra-Board Baseband Resource Pool</a>                       |
| RRU Channel Cross Connection Under MIMO         | Supported only in LTE FDD | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>        |
| Power Supply Reliability                        | None                      | <a href="#">13 Power Supply Reliability</a>                                |
| Heat Dissipation Reliability for Fans           | None                      | <a href="#">14.2 Heat Dissipation Reliability for Fans</a>                 |
| Power Supply Redundancy                         | None                      | <a href="#">14.3 Power Supply Redundancy</a>                               |
| Anti-Misinsertion Design of Boards              | None                      | <a href="#">14.4 Anti-Misinsertion Design of Boards</a>                    |
| Overtemperature Protection for BBU Boards       | None                      | <a href="#">14.5 Overtemperature Protection for BBU Boards</a>             |
| Surge Protection Design                         | None                      | <a href="#">14.6 Surge Protection Design</a>                               |

**Table 2-3** Differences between NR FDD and NR TDD

| Function Name                                   | Difference | Chapter/Section                                                            |
|-------------------------------------------------|------------|----------------------------------------------------------------------------|
| Cold Backup of Main Control Boards              | None       | <a href="#">4 Cold Backup of Main Control Boards</a>                       |
| Transmission Link Backup of Main Control Boards | None       | <a href="#">5 Transmission Link Backup of Main Control Boards (LTE/NR)</a> |
| Inter-Board Baseband Resource Redundancy        | None       | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a>        |
| Inter-BBP Active/Standby Backup (Macro)         | None       | <a href="#">9 Inter-BBP Active/Standby Backup (NR)</a>                     |

| Function Name                                                                              | Difference               | Chapter/Section                                                                                                        |
|--------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------------------------------|
| Inter-BBP Active/Standby Backup (LampSite)                                                 | Supported only in NR TDD | <a href="#">9 Inter-BBP Active/Standby Backup (NR)</a>                                                                 |
| Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios | Supported only in NR TDD | <a href="#">10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)</a> |
| Cold Backup of RF Modules                                                                  | Supported only in NR TDD | <a href="#">11 Cold Backup of RF Modules (NR TDD)</a>                                                                  |
| RRU Channel Cross Connection Under MIMO                                                    | None                     | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>                                                    |
| Heat Dissipation Reliability for Fans                                                      | None                     | <a href="#">14.2 Heat Dissipation Reliability for Fans</a>                                                             |
| Power Supply Redundancy                                                                    | None                     | <a href="#">14.3 Power Supply Redundancy</a>                                                                           |
| Anti-Misinsertion Design of Boards                                                         | None                     | <a href="#">14.4 Anti-Misinsertion Design of Boards</a>                                                                |
| Overtemperature Protection for BBU Boards                                                  | None                     | <a href="#">14.5 Overtemperature Protection for BBU Boards</a>                                                         |
| Surge Protection Design                                                                    | None                     | <a href="#">14.6 Surge Protection Design</a>                                                                           |

**Table 2-4** Differences between NSA and SA

| Function Name                                   | Difference | Chapter/Section                                                            |
|-------------------------------------------------|------------|----------------------------------------------------------------------------|
| Cold Backup of Main Control Boards              | None       | <a href="#">4 Cold Backup of Main Control Boards</a>                       |
| Transmission Link Backup of Main Control Boards | None       | <a href="#">5 Transmission Link Backup of Main Control Boards (LTE/NR)</a> |
| Inter-Board Baseband Resource Redundancy        | None       | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a>        |
| Inter-BBP Active/Standby Backup                 | None       | <a href="#">9 Inter-BBP Active/Standby Backup (NR)</a>                     |

| Function Name                                                                              | Difference                      | Chapter/Section                                                                                                        |
|--------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios | Supported only in SA networking | <a href="#">10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)</a> |
| Cold Backup of RF Modules                                                                  | None                            | <a href="#">11 Cold Backup of RF Modules (NR TDD)</a>                                                                  |
| RRU Channel Cross Connection Under MIMO                                                    | None                            | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>                                                    |
| Heat Dissipation Reliability for Fans                                                      | None                            | <a href="#">14.2 Heat Dissipation Reliability for Fans</a>                                                             |
| Power Supply Redundancy                                                                    | None                            | <a href="#">14.3 Power Supply Redundancy</a>                                                                           |
| Anti-Misinsertion Design of Boards                                                         | None                            | <a href="#">14.4 Anti-Misinsertion Design of Boards</a>                                                                |
| Overtemperature Protection for BBU Boards                                                  | None                            | <a href="#">14.5 Overtemperature Protection for BBU Boards</a>                                                         |
| Surge Protection Design                                                                    | None                            | <a href="#">14.6 Surge Protection Design</a>                                                                           |

**Table 2-5** Differences between high frequency bands and low frequency bands

| Function Name                                   | Difference                            | Chapter/Section                                                            |
|-------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------|
| Cold Backup of Main Control Boards              | None                                  | <a href="#">4 Cold Backup of Main Control Boards</a>                       |
| Transmission Link Backup of Main Control Boards | None                                  | <a href="#">5 Transmission Link Backup of Main Control Boards (LTE/NR)</a> |
| Inter-Board Baseband Resource Redundancy        | None                                  | <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a>        |
| Inter-BBP Active/Standby Backup                 | Supported only in low frequency bands | <a href="#">9 Inter-BBP Active/Standby Backup (NR)</a>                     |

| Function Name                                                                              | Difference                            | Chapter/Section                                                                                                        |
|--------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios | Supported only in low frequency bands | <a href="#">10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)</a> |
| Cold Backup of RF Modules                                                                  | Supported only in low frequency bands | <a href="#">11 Cold Backup of RF Modules (NR TDD)</a>                                                                  |
| RRU Channel Cross Connection Under MIMO                                                    | None                                  | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>                                                    |
| Heat Dissipation Reliability for Fans                                                      | None                                  | <a href="#">14.2 Heat Dissipation Reliability for Fans</a>                                                             |
| Power Supply Redundancy                                                                    | None                                  | <a href="#">14.3 Power Supply Redundancy</a>                                                                           |
| Anti-Misinsertion Design of Boards                                                         | None                                  | <a href="#">14.4 Anti-Misinsertion Design of Boards</a>                                                                |
| Overtemperature Protection for BBU Boards                                                  | None                                  | <a href="#">14.5 Overtemperature Protection for BBU Boards</a>                                                         |
| Surge Protection Design                                                                    | None                                  | <a href="#">14.6 Surge Protection Design</a>                                                                           |

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

# 3 Overview

With reliability design, base station equipment can continue to provide services even when some parts are faulty. This avoids or reduces the impact on services caused by equipment faults and improves system reliability.

**Table 3-1** describes the base station equipment reliability features/functions supported by GSM, UMTS, LTE, and NR. In this table, "Y" means "supported" and "N" means "not supported."

**Table 3-1** Base station equipment reliability features/functions supported by each mode

| Feature/<br>Function<br>Name                    | Support<br>by GSM          | Support<br>by UMTS | Support<br>by LTE | Support<br>by NR | Reference                                                  |
|-------------------------------------------------|----------------------------|--------------------|-------------------|------------------|------------------------------------------------------------|
| Cold Backup of Main Control Boards              | GBTS: N<br>eGBTS (UMPT): Y | Y                  | Y <sup>a</sup>    | Y                | 4 Cold Backup of Main Control Boards                       |
| Transmission Link Backup of Main Control Boards | N                          | N                  | Y                 | Y                | 5 Transmission Link Backup of Main Control Boards (LTE/NR) |

| Feature/<br>Function Name                                                                  | Support<br>by GSM | Support<br>by UMTS | Support<br>by LTE                                  | Support<br>by NR                          | Reference                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------|-------------------|--------------------|----------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Inter-Board Baseband Resource Redundancy                                                   | Y <sup>a</sup>    | Y <sup>a</sup>     | Y                                                  | Y                                         | <ul style="list-style-type: none"> <li>GBTS/eGBTS/NodeB: <a href="#">6 Inter-Board Baseband Resource Redundancy (GSM/UMTS)</a></li> <li>eNodeB/gNodeB: <a href="#">7 Inter-Board Baseband Resource Redundancy (LTE/NR)</a></li> </ul>                                                |
| Intra-Board Baseband Resource Pool                                                         | Y <sup>b</sup>    | Y <sup>b</sup>     | Y (supported only in LTE FDD and TDD) <sup>b</sup> | Y                                         | <ul style="list-style-type: none"> <li><a href="#">8 Intra-Board Baseband Resource Pool</a></li> <li>eNodeB: This feature is enabled by default if the LBBPc is configured. For details about activation verification, see <a href="#">8.4.4 Activation Verification</a>.</li> </ul> |
| Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios | N                 | N                  | N                                                  | Y <sup>a</sup> (supported only in NR TDD) | <a href="#">10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)</a>                                                                                                                                                               |
| Cold Backup of RF Modules                                                                  | N                 | N                  | N                                                  | Y (supported only in NR TDD)              | <a href="#">11 Cold Backup of RF Modules (NR TDD)</a>                                                                                                                                                                                                                                |

| Feature/<br>Function<br>Name                       | Support<br>by GSM | Support<br>by UMTS | Support<br>by LTE                             | Support<br>by NR | Reference                                                                                                                                                                                                                                                                                                                                                                                       |
|----------------------------------------------------|-------------------|--------------------|-----------------------------------------------|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RRU Channel Cross Connection Under MIMO            | N                 | N                  | Y<br>(supported only in LTE FDD) <sup>a</sup> | Y <sup>b</sup>   | <a href="#">12 RRU Channel Cross Connection Under MIMO (LTE/NR)</a>                                                                                                                                                                                                                                                                                                                             |
| Power Supply Reliability                           | Y                 | Y                  | Y                                             | N                | <ul style="list-style-type: none"> <li>Power Supply Reliability for a BBU<sup>b</sup> and intelligent shutdown of carriers due to PSU failure: <a href="#">13 Power Supply Reliability</a></li> <li>Other functions involved in Power Supply Reliability for a base station: <i>Power Supply Management</i></li> </ul>                                                                          |
| RF Channel Cooperation                             | Y                 | Y                  | N                                             | N                | <ul style="list-style-type: none"> <li><a href="#">14.1 RF Channel Cooperation (GSM/UMTS)</a></li> <li>GBTS/eGBTS: <i>TRX Cooperation</i></li> <li>NodeB: <i>RRU Redundancy</i></li> </ul>                                                                                                                                                                                                      |
| Heat Dissipation Reliability for Fans <sup>b</sup> | Y                 | Y                  | Y                                             | Y                | <ul style="list-style-type: none"> <li><a href="#">14.2 Heat Dissipation Reliability for Fans</a></li> <li>The TCU, FMU, or BBU FAN needs to be configured. For details about the initial configuration of the TCU, FMU, or BBU FAN, see <i>3900 &amp; 5900 Series Base Station Initial Configuration Guide</i> in <i>3900 &amp; 5900 Series Base Station Product Documentation</i>.</li> </ul> |

| Feature/<br>Function<br>Name              | Support<br>by GSM | Support<br>by UMTS | Support<br>by LTE | Support<br>by NR | Reference                                                                                                                                                                                                              |
|-------------------------------------------|-------------------|--------------------|-------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power Supply Redundancy                   | Y                 | Y                  | Y                 | Y                | <b>14.3 Power Supply Redundancy</b> <ul style="list-style-type: none"> <li>Power Supply Redundancy of a base station: <i>Power Supply Management</i></li> <li>Power Supply Redundancy for a BBU<sup>b</sup></li> </ul> |
| Anti-Misinsertion Design of Boards        | Y                 | Y                  | Y                 | Y                | <b>14.4 Anti-Misinsertion Design of Boards</b>                                                                                                                                                                         |
| Overtemperature Protection for BBU Boards | Y                 | Y                  | Y                 | Y                | <b>14.5 Overtemperature Protection for BBU Boards</b>                                                                                                                                                                  |
| Surge Protection Design                   | Y                 | Y                  | Y                 | Y                | <b>14.6 Surge Protection Design</b>                                                                                                                                                                                    |

 **NOTE**

a: This function is a basic function of the base station and has no feature ID. It requires software configurations.

b: This function is a basic function of the base station and has no feature ID. It does not require any software configurations.



# 4 Cold Backup of Main Control Boards

---

## 4.1 Principles

When a base station is configured with only one main control board, services will be interrupted for a long period of time if this main control board is faulty. To support Cold Backup of Main Control Boards, two main control boards of the same type must be configured for the base station. When a fault is detected on the active main control board, the active and standby boards switch roles. Services carried on the original active board are interrupted but automatically recover within the expected period, which ensures reliability. For macro base stations, the expected period for automatic recovery is 4 to 7 minutes. For LampSite base stations, the expected period is 4 to 10 minutes, as a large number of pRRUs are used.

In the following scenarios, the service recovery required time is 2 minutes longer than the expected period for automatic recovery:

- The switchover between the boards is triggered manually by running the **SWP BRD** command. The service recovery required time equals the interval between the time when the command is executed and the time when services are restored.
- The switchover between the boards is triggered after the running active main control board is removed.
- In a secure networking scenario, the new active main control board does not have a digital certificate or the digital certificate is invalid or has expired. For details about secure networking scenarios, see *Transmission Security Overview*.

Cold Backup of Main Control Boards involves three processes: active/standby competition, data backup, and active/standby switchover.

### 4.1.1 Active/Standby Competition

After a BBU with two main control boards is powered on, the system automatically determines the active main control board using active/standby competition if both main control boards function properly. You can run the **DSP BRD** command to query the active/standby status of the main control boards. If

one main control board cannot be detected or is not functioning properly, the other main control board becomes the active one.

 **NOTE**

The main control boards in active/standby mode must be in their original system state. If a main control board has been used at a site, certain configuration data may be left in this board, which affects the active/standby competition process. As a result, the active/standby relationship may not be established as expected. For example, the main control board in slot 6 is expected to be the active board, and the main control board inserted later in slot 7 is expected to be the standby board for a site. If any configuration data of another site is left in the board in slot 7, it may serve as the active board after the active/standby competition. Therefore, if the main control board inserted later is not in its original system state and has residual configuration data, run the **ACT CFGFILE** command to activate the minimum configuration before inserting the board.

## 4.1.2 Data Backup

The main control and transmission board adopts the cold backup mode. Therefore, the active and standby main control boards do not need to synchronize data in real time. Therefore, you only need to ensure that the static data (such as configuration data, software, and logs) between the active and standby main control boards is synchronized. The standby board does not back up the signaling and service data on the active board.

Data backup consists of initial backup and routine backup, which are described as follows:

- Initial backup: After the base station and the active and standby main control boards are started, the base station compares the files on the two boards. Then, the base station copies the files that are unique on the active board to the standby board and removes unnecessary files from the standby board. During initial backup, configuration data, software data, and logs are all backed up.
- Routine backup: After the base station completes initial backup, the base station periodically compares the files on the active and standby main control boards (every 5 minutes by default). Then, the base station copies the files that are unique on the active board to the standby board.

 **NOTE**

- During a fault-triggered active/standby switchover, the base station copies only configuration data on the active board to the standby board before the reset to minimize service interruption duration. Other data is not backed up. As a result, data updated between the previous periodic backup and the fault occurrence may be lost. However, this impact is negligible because the data backup period is brief and the purpose of the active/standby switchover is to ensure service continuity.
- If an active/standby switchover is triggered during a routine backup, the system backs up data before performing the active/standby switchover. In this case, services are interrupted for 1 to 2 minutes more than that for a regular active/standby switchover.
- During the data backup of the active and standby boards, ensure that the standby board is running properly. If you remove, insert, replace, or reset the standby board during the backup, the backup may fail. In this case, you need to reset the active board to ensure the backup can be performed properly. The **DSP BKPSTATUS** command can be executed to query the data backup status of the active and standby boards.

### 4.1.3 Active/Standby Switchover

An active/standby switchover between the two main control boards is triggered in one of the following scenarios:

- Fault-triggered: When the active main control board experiences a key component fault, running exception, or file system damage, the system automatically triggers an active/standby switchover.
- Command-triggered: A user delivers an MML command to trigger an active/standby switchover.
- Others: When the status of the Ethernet ports carrying the IP addresses on the active main control board are inactive and the status of the corresponding Ethernet ports on the standby main control board are active, the system automatically triggers a switchover.

The prerequisites and methods for active/standby switchover vary with triggering conditions, as described in [Table 4-1](#).

**Table 4-1** Prerequisites and methods for active/standby switchover

| Switchover Type            | Prerequisites                                                                                                                                                               | Method                                            | Remarks                                                                                                                                                                                                                                                                |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fault-triggered switchover | The standby main control board is functioning properly, the links of the standby board are normal, the standby board has no major hardware faults, and the clock is locked. | The system automatically triggers the switchover. | When the active main control board experiences major faults, services carried on this board must be switched over to the standby main control board to ensure the normal operation of the base station. Therefore, the switchover prerequisites are relatively simple. |

| Switchover Type              | Prerequisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Method                                                                                                                | Remarks                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Command-triggered switchover | <ul style="list-style-type: none"> <li>The standby main control board is functioning properly, the links of the standby board are normal, the standby board has no major hardware faults, and the clock is locked.</li> <li>The clock locking status of the board can be queried by running the <b>DSP CLKSTAT</b> command.</li> <li>The backup status of the active and standby main control boards is <b>Idle</b><sup>a</sup>. The backup status can be queried by running the <b>DSP BKPSTATUS</b> command.</li> <li>The base station is not performing a software upgrade (including downloading and activating software packages or patches).</li> <li>More than 3 minutes have elapsed since the last active/standby switchover. This is to prevent frequent switchovers.</li> </ul> | A user delivers a command to trigger the switchover. For details, see <a href="#">4.4 Operation and Maintenance</a> . | Before a user delivers a command to trigger an active/standby switchover, the base station works properly. The impact of the switchover on the base station must be minimized. Therefore, the switchover prerequisites are relatively complex. For example, when KPIs of the base station drastically deteriorate but the active/standby switchover is not automatically triggered, a manual active/standby switchover is recommended. |

| Switchover Type | Prerequisites                                                                                                                                                               | Method                                            | Remarks                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Others          | The standby main control board is functioning properly, the links of the standby board are normal, the standby board has no major hardware faults, and the clock is locked. | The system automatically triggers the switchover. | When the status of the Ethernet ports carrying the IP addresses on the active main control board are inactive and the status of the corresponding Ethernet ports on the standby main control board are active, services carried on the active board must be switched over to the standby board to ensure normal operation of the base station. Therefore, the switchover prerequisites are relatively simple. |

#### NOTE

a: The command-triggered switchover cannot be performed before the initial or routine backup between the active and standby main control boards is completed. Perform the command-triggered switchover after the hardware installation is complete and the base station has been running for more than two hours. In active/standby scenarios, you are not allowed to perform the following operations to change the settings of **NODE.WM (LTE eNodeB, 5G gNodeB)**, **NODE.PRODUCTTYPE**, **NODE.WMEXTENSION (LTE eNodeB, 5G gNodeB)**, and **NODE.LTEMODE (LTE eNodeB, 5G gNodeB)**. Otherwise, a main control RAT change may occur, the active/standby relationship between the two main control boards may be canceled, and unexpected execution results may occur.

- Run the **SET NODE** command to change the settings of the preceding parameters.
- Run the **ACT CFGFILE** command with **MOD** set to **LEAST** to activate the minimum configuration file of the target base station.
- Run the **ACT CFGFILE** command with **MOD** set to **XML** to activate the configuration file of the target base station.

During an active/standby switchover, ensure that the standby main control board is working properly. If any exception occurs during the switchover, such as power-off of the standby board, the switchover may fail and the active main control board cannot be configured. In this case, reset the active board to ensure its proper operation.

## 4.2 Network Analysis

### 4.2.1 Benefits

None

## 4.2.2 Impacts

The impacts between this function and other functions are described in [4.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 4.3 Requirements

### 4.3.1 Licenses

None

### 4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 4.3.2.1 Prerequisite Functions

None

#### 4.3.2.2 Mutually Exclusive Functions

None

#### 4.3.2.3 Function Impacts

None

### 4.3.3 Hardware

- Only UMPT boards support Cold Backup of Main Control Boards.  
Cold backup of UMPT boards applies to macro and LampSite base stations that use the BBU3910, BBU3900, BBU5910, or BBU5900, including NodeBs, eNodeBs, gNodeBs, and co-MPT base stations. It is implemented in the same way on macro and LampSite base stations.
- To support Cold Backup of Main Control Boards, two identical main control boards are required in the same BBU. For example, the two boards are both UMPTb1. For LTE TDD, the main control boards must be UMPTe, UMPTg, UMPTga, UMPTHa, or UMPTHb series boards.

#### NOTE

In scenarios where two UMPT boards work in cold backup mode, if the spare UMPT board and the UMPT board to be replaced are of different RATs, the spare UMPT board automatically operates in the RAT of the replaced UMPT board and works in active/standby mode with the other original UMPT board after the replacement.

- Two transmission cables are required, with one connected to the main control board in slot 6 and the other connected to the main control board in slot 7.
- In a co-MPT multimode base station where two UMPT boards work in cold backup mode, the standby UMPT board can work as a signaling extension board for LTE or NR automatically but not for GSM or UMTS.

 **NOTE**

When the standby UMPT board works as a signaling extension board for LTE or NR, the LTE or NR services carried on this board will be interrupted if it experiences a hardware fault or is removed. When the active UMPT board becomes faulty and the active and standby UMPT boards switch roles, only the new active UMPT board provides signaling processing capability.

- In UCIU+UMPT BBU interconnection scenarios, only the primary BBU supports Cold Backup of Main Control Boards.
- In UMPT+UMPT BBU interconnection scenarios, the primary BBU does not support Cold Backup of Main Control Boards

## 4.3.4 Networking

- UMPT boards with the active/standby transmission capability are supported only in Ethernet networks.
- UMPT boards with the active/standby transmission capability do not support HUB transmission cascading.
- Cold backup of main control boards has the following restrictions on the secure networking planning:
  - If UMPT boards work in cold backup mode and IPsec and digital certificate authentication are used, the Certificate Authority (CA) configuration on the base station must allow the standby main control board to access the CA server and obtain the digital certificate after an active/standby board switchover.
  - If UMPT boards work in cold backup mode, and IPsec and digital certificate authentication are used, the **IDTYPE** parameter can be set to **IPV4**, **FQDN**, or **DN** for the base station. When the **IDTYPE** parameter is set to **FQDN** and the security gateway (SeGW) uses the digital certificate whitelist to verify the base station identity, the SeGW adds both the active and standby main control boards to the whitelist.
  - Cold backup does not support IPv6-related features, direct IPsec, or the new transmission model.
- The clock source must be the GNSS clock.

## 4.3.5 Others

None

# 4.4 Operation and Maintenance

## 4.4.1 When to Use

In GSM and UMTS networks, if customers have high requirements on the reliability of the entire base station, it is recommended that this feature be

enabled by using the HAGRP MO to improve the reliability of the base station. This feature is not recommended for LTE and NR. Examples are as follows:

- Services recover within several minutes after the main control board experiences a hardware fault.
- The UMPT supports multimode co-MPT and multi-RAT concurrency. When this main control board experiences a severe hardware fault, services of the multiple RATs are interrupted. Enabling this feature can improve the reliability of the base station.

## 4.4.2 Precautions

- After an **HAGRP** MO is configured, users must enter the value of the **Logic Slot No.** parameter specified in the **HAGRP** MO as the slot number when running the **ADD XXX**, **RMV XXX**, **MOD XXX**, **LST XXX**, or **DSP XXX** commands related to transmission or clock. The output of a **DSP XXX** command contains the number of the slot housing the active main control board.
- Adding or removing an **HAGRP** MO changes the connection between the base station and the transport network and resets the base station.

## 4.4.3 Data Configuration

### 4.4.3.1 Data Preparation

- Configuring main control boards  
Main control boards in active/standby mode must be configured on the same logical NE. [Table 4-2](#) describes the key parameters that must be set for configuring main control boards.

**Table 4-2** Key parameters for configuring main control boards

| Parameter Name | Parameter ID                                           | Setting Notes                                                                         |
|----------------|--------------------------------------------------------|---------------------------------------------------------------------------------------|
| Cabinet No.    | <b>CN</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>0</b> .                                                      |
| Subrack No.    | <b>SRN</b> parameter in the <b>ADD BRD</b> MML command | Set this parameter to <b>0</b> .                                                      |
| Slot No.       | <b>SN</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>6</b> and <b>7</b> for two main control boards.              |
| Board Type     | <b>BT</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>UMPT</b> . Boards in slots 6 and 7 must be of the same type. |

- Configuring an **HAGRP** MO  
[Table 4-3](#) describes the key parameters that must be set in the **HAGRP** MO.



**Table 4-3** Key parameters that must be set in the **HAGRP** MO

| Parameter Name | Parameter ID     | Setting Notes                                |
|----------------|------------------|----------------------------------------------|
| Cabinet No.    | <b>HAGRP.CN</b>  | None                                         |
| Subrack No.    | <b>HAGRP.SRN</b> | None                                         |
| Logic Slot No. | <b>HAGRP.SN</b>  | Set this parameter to <b>6</b> or <b>7</b> . |

### 4.4.3.2 Using MML Commands

Before using MML commands, refer to [4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

**Step 1** Run the **DSP BRD** command to query whether the slot number of the main control board that is configured with transport data is 6 or 7.

```
DSP BRD;
```

**Step 2** Run the **ADD BRD** command to add a main control board with a different slot number as the standby main control board. The default slot number of an existing main control board is 7.

```
ADD BRD: CN=0, SRN=0, SN=6, BT=UMPT;
```

**Step 3** Run the **ADD HAGRP** command to add a transport high availability (HA) group.

```
ADD HAGRP: CN=0, SRN=0, SN=7;
```

----End

#### NOTE

- When **GTRANSPARA.TRANSFCGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD** after the standby main control board is added, delete the data that is automatically generated on the standby main control board, including the Ethernet port data, which corresponds to the **ETHPORT** MO, and the default transmission resource group data, which corresponds to the **RSCGRP** MO.
- The preceding steps do not apply to separate-MPT base stations that are running properly. If you need to activate the Cold Backup of Main Control Boards feature for such a base station, contact Huawei engineers.

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Step 1** Run the **LST HAGRP** command to query the logical slot number of the transport HA group, for example, 6.

```
LST HAGRP;;
```

- Step 2** Run the **DSP BRD** command to query the slot number of the active main control board. Check whether the slot number of the active main control board (for example, 7) is the same as the logical slot number of the transport HA group.

```
DSP BRD;;
```

- Step 3** (Optional) If the slot number of the active main control board is different from the logical slot number of the transport HA group, run the **SWP BRD** command to switch over the active and standby main control boards. After this command is executed, the base station will be reset.

```
SWP BRD;;
```

- Step 4** Run the **RMV HAGRP** command to remove the transport HA group. After this command is executed, the base station will be reset.

```
RMV HAGRP: CN=0, SRN=0, SN=6;
```

- Step 5** Run the **RMV BRD** command to remove the main control board (standby main control board) whose slot number is different from the slot number of the transport HA group.

```
RMV BRD: CN=0, SRN=0, SN=7;
```

----End

### 4.4.3.3 Using the MAE-Deployment

This section only describes the operations on the MAE-Deployment for multimode base stations. The operations for single-mode base stations such as eGBTS, NodeB, eNodeB, and gNodeB are similar to those for multimode base stations.

It is recommended that Cold Backup of Main Control Boards be activated on the MAE-Deployment.

This section describes the steps for activating Cold Backup of Main Control Boards for a base station that meets the following requirements:

- A main control board has been configured in the base station and the transport data has been configured for this board.
- The base station is operating properly.

## Using the MAE-Deployment to Perform Single Configuration

- Step 1** Modify the base station configurations.

Configure the MOs on the MAE-Deployment as instructed in [Table 4-4](#) when **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**.

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

**Table 4-4** MOs to be configured on the MAE-Deployment

| SN | MO      | Configuration Description                                                                                                                                                        |
|----|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | MPT     | Add a standby main control board.                                                                                                                                                |
| 2  | ETHPORT | Delete the <b>ETHPORT</b> MOs that are automatically configured on the standby main control board.                                                                               |
| 3  | RSCGRP  | Delete the <b>RSCGRP</b> MOs that are automatically configured on the standby main control board.                                                                                |
| 4  | HAGRP   | Add a transport HA group. The value of the <b>Logic Slot No.</b> parameter must be the same as the slot number of the main control board that is configured with transport data. |

**Step 2** Export the base station deployment list. For detailed operations, see section "Preparing a Data Configuration File and a Deployment List" in *3900 & 5900 Series Base Station Commissioning Guide*.

**Step 3** Upload the base station deployment list to the MAE-Access so that the DHCP server-related configurations are updated. For detailed operations, see section "Creating an NE Commissioning Task" in *3900 & 5900 Series Base Station Commissioning Guide*.

**Step 4** Deliver incremental configurations.

----End

## Using the MAE-Deployment to Perform Batch Modification

 **NOTE**

When configuring this function on the MAE-Deployment, perform a single configuration first, and then perform a batch modification if required. You must perform single configuration for a parameter before batch modification of the parameter using the MAE-Deployment. You are advised to perform batch modification before logging out of the parameter setting interface.

**Step 1** After creating a planned data area, choose **Advanced > Data Management > Customize Summary Data File** from the main menu, to customize a summary data file for batch configuration.

 **NOTE**

For context-sensitive help on a current task in the client, click .

**Step 2** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk**

**Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.

- Step 3** In the summary data file, set the parameters related to the **MPT** MO for the standby main control board, and close the file.
- Step 4** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.
- Step 5** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.
- When **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **OLD**, delete the data that is automatically generated on the standby main control board, including the Ethernet port data, which corresponds to the **ETHPORT** MO, and the default transmission resource group data, which corresponds to the **RSCGRP** MO.
- Step 6** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.
- Step 7** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.
- Step 8** In the summary data file, set the parameters related to the **HAGRP** MO, and close the file.
- Step 9** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.
- Step 10** After the data verification is completed, choose **Area Management > Planned Area Management > Export Incremental Scripts** to export and activate the incremental scripts in the planned data area.

----End

## Using the MAE-Deployment to Perform Deactivation

- Step 1** Configure the MOs described in [Table 4-5](#) on the MAE-Deployment configuration interface. For detailed operations, see Feature Configuration Using the MAE-Deployment.

**Table 4-5** MOs to be configured on the MAE-Deployment

| SN | MO    | Configuration Description                                                                                                                 |
|----|-------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | HAGRP | Remove the <b>HAGRP</b> MO.                                                                                                               |
| 2  | MPT   | Remove the main control board (standby main control board) whose slot number is different from the slot number of the transport HA group. |

- Step 2** Export the base station deployment list. For detailed operations, see section "Preparing a Data Configuration File and a Deployment List" in *3900 & 5900 Series Base Station Commissioning Guide*.
- Step 3** Upload the base station deployment list to the MAE-Access so that the DHCP server-related configurations on the MAE-Access are updated. For detailed operations, see section "Creating an NE Commissioning Task" in *3900 & 5900 Series Base Station Commissioning Guide*.
- Step 4** Deliver incremental configurations.

----End

## 4.4.4 Activation Verification

- Step 1** Check the board status.

Run the **DSP BRD** command to check whether the boards in slots 6 and 7 are normal. If the value of **Standby Status** is **Active** for one board and is **Standby** for the other board and the value of **Availability Status** is **Normal** for both boards, the two boards are normal.

- Step 2** Check whether a high availability (HA) group (that is, a group of active and standby transmission boards) has been configured.

Run the **LST HAGRP** to check the configuration. If configuration of the transport HA group is available, the two boards with the active/standby transmission capability have been configured to work in active/standby mode.

- Step 3** Check the initial backup status of the board.

Run the **DSP BKPSTATUS** command to query the values of **Backup Switch** and **Backup Status**. If **Backup Switch** is **On**, cold backup of main control boards is activated. If **Backup Status** is **Idle**, initial backup is completed.

### NOTE

After the standby main control board is configured, initial backup is automatically performed. This initial backup process takes approximately 150 minutes. You can run the **DSP BKPSTATUS** command to query the backup progress.

- Step 4** Switch the active and standby roles of the main control boards.

Run the **SWP BRD** command to perform the switch. Upon execution of this command, the base station will automatically perform routine backup, reset, and switchover between the active and standby main control boards. This process takes approximately 30 minutes. Base station resets will interrupt ongoing services.

**Step 5** Verify the active/standby switchover.

When the base station becomes normal after the reset, and the connections to the LMT and MAE-Access are restored, perform step 1 to step 3 to check whether the active and standby roles of the main control boards have been switched.

----End

## 4.4.5 Network Monitoring

None

# 5 Transmission Link Backup of Main Control Boards (LTE/NR)

---

## 5.1 Principles

In transmission link backup of main control boards, two main control boards of the same model are configured in the same BBU subrack of a base station, and transmission links are configured on each main control board, providing transmission link protection. The transmission links are considered as the base station's backhaul interfaces and can work simultaneously. If one of the transmission links becomes faulty, the base station detects the fault and triggers a transmission link switchover to transfer services to a working link, ensuring transmission reliability. The S1, NG, X2, Xn, eX2, eXn, and CMP service interfaces support transmission link backup of main control boards.

The following transmission links must be configured on the main control boards: an OM link, a control-plane link, a user-plane link, and a certificate management link, which are used to transmit management-plane data, control-plane data, user-plane data, and certificate management data, respectively. The configurations are as follows:

- OM link: Active and standby OM channels are configured. An active OM channel and a standby OM channel are separately configured on the two main control boards. There is no coupling between the active/standby states of the OM channels and the active/standby states of the main control boards, allowing for flexible selection of main control boards. Normally, OM data flows are transmitted over the active OM channel. If the active OM channel is faulty, OM data flows are switched over to the standby one.
- Control-plane link: Control-plane SCTP multihoming is configured. Two IP addresses for control-plane SCTP multihoming are separately configured on the two main control boards and work in active/standby mode. There is no coupling between the active/standby states of the addresses and the active/standby states of the main control boards, allowing for flexible selection of main control boards. Normally, signaling packets are transmitted over the active SCTP link. If the active SCTP link is faulty, signaling packets are switched over to the standby one.

- User-plane link: User plane backup is configured. Two user-plane IP addresses are configured on the two main control boards, respectively. Each IP address can be used for setup of a user-plane transmission link. The base station selects a user-plane transmission link in polling or by active/standby mode to set up a user-plane bearer.
- A certificate management link can be configured to work as an active/standby certificate management channel using methods listed below. An active certificate management channel and a standby certificate management channel are separately configured on the two main control boards. There is no coupling between the active/standby states of the certificate management channels and those of the main control boards, allowing for flexible selection of main control boards.
  - Using the IP address of an OM channel. In normal cases, certificate management services are carried on the link between the source IP address of the active OM channel and the Certificate Authority (CA) address. If the active OM channel becomes faulty, certificate management services are carried on the link between the source IP address of the standby OM channel and the CA address.
  - Using a specified IP address as the source IP address for certificate management. In normal cases, certificate management services are carried on the link between the active source IP address and the CA address. If the active certificate management link becomes faulty, certificate management services are carried on the link between the standby source IP address and the CA address.
  - Using the source IP address for certificate application and that for certificate update, which are separately specified. In normal cases of certificate application, certificate management services are carried on the link between the active source IP address for application and the CA address. If the active certificate management link becomes faulty, certificate management services are carried on the link between the standby source IP address for application and the CA address. In normal cases of certificate update, certificate management services are carried on the link between the active source IP address for update and the CA address. If the active certificate management link becomes faulty, certificate management services are carried on the link between the standby source IP address for update and the CA address.

 **NOTE**

- For details about the principles of active and standby OM channels, control-plane SCTP multihoming, and user plane backup, see *IPv4 Transmission* or *IPv6 Transmission*.
- The user-plane links over the X2/Xn/eX2/eXn interfaces between base stations can work only in load-sharing mode, meaning that the links cannot work in active/standby mode.

If a transmission link is faulty, for example, a transmission port/transmission device/optical fiber is faulty, the protection against the service interface fault is as follows:

- OM link: Application-layer heartbeat detection is used between the base station and the MAE. A fault in the active OM channel triggers a switchover to the standby one, causing an interruption for about 5 minutes.
- Control-plane link: The base station uses the SCTP heartbeat detection mechanism. Upon detecting a fault in the active SCTP link, the base station



triggers a switchover to the standby SCTP link for signaling packet forwarding, causing an interruption for about 2 minutes.

- User-plane link: By default, the base station uses the GTP-U detection mechanism to check user-plane link connectivity. If a fault in the transmission link of a user-plane host is detected, the faulty user-plane host will not be selected for bearer setup and bearers will be set up on normal user-plane hosts in subsequent setup procedures. In this case, the faulty user-plane link is interrupted, causing a switchover from the faulty transmission link to the standby one. This causes a service interruption for about four minutes. User-plane link status fast awareness based on the physical link status can be enabled to facilitate fast switchover of user-plane links by setting the **GUPTRANSPARA.UPSTATFASW (5G gNodeB, LTE eNodeB)** parameter to **ENABLE**. The switchover causes an interruption of 10 seconds or less.
- Certificate management link: The certificate management link is not a persistent link. If the active certificate management link fails to be set up, an active/standby switchover is triggered, causing an attempt to set up a standby certificate management link.

#### NOTE

- A transmission fault triggers only a switchover of transmission links on the main control boards, not an active/standby switchover of the main control boards.
- When all EP paths set up on the user plane of the local end are faulty, for example, a transmission port or the next-hop router is faulty, the base station starts a user-plane transmission link switchover. If only some EP paths on the user plane are faulty, a user-plane transmission link switchover is not triggered.
- If the transmission interruption duration longer than 1 minute is acceptable, it is recommended that the **GUPTRANSPARA.UPSTATFASW (5G gNodeB, LTE eNodeB)** parameter be set to **DISABLE**. In this case, the GTP-U detection mechanism is used to implement user-plane link detection.
- The change in the status of an Ethernet port from UP to DOWN triggers a fast user-plane link switchover provided that the **GUPTRANSPARA.UPSTATFASW (5G gNodeB, LTE eNodeB)** parameter is set to **ENABLE**.
- If IPv4-over-IPv6 IPsec or IPv6-over-IPv4 IPsec is configured for user-plane links, user-plane link status fast awareness based on the physical link status does not take effect for user-plane links.

## 5.2 Network Analysis

### 5.2.1 Benefits

With transmission link backup of main control boards, the base station speeds up switchovers of transmission links if a transmission link is faulty. In this way, services can be restored within a short period of time.

### 5.2.2 Impacts

The impacts between this function and other functions are described in [5.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 5.3 Requirements

### 5.3.1 Licenses

None

### 5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 5.3.2.1 Prerequisite Functions

None

#### 5.3.2.2 Mutually Exclusive Functions

None

#### 5.3.2.3 Function Impacts

None

### 5.3.3 Hardware

- Transmission link backup of main control boards applies only to macro base stations and LampSite base stations that use the BBU3910, BBU5910, or BBU5900.
- Only the UMPTb, UMPTe, UMPTg, UMPTHa, and UMPTHb series boards support this function. When an IEEE 1588V2 clock source is used in transmission link backup of main control boards, only the UMPTe, UMPTg, UMPTHa, and UMPTHb series boards support this function.

### 5.3.4 Networking

- Transmission link backup requires that the main control boards be deployed in the same BBU subrack, but not in different subracks.
- Transmission link backup supports only Ethernet transmission.
- Transmission link backup does not apply to transmission cascading or co-transmission scenarios. For example, after reconstruction from a single main control board to two main control boards: If panel cascading or backplane co-transmission is configured on the main control board before the reconstruction, passerby data forwarding is not supported after a switchover of the main control boards.

- It is recommended that the cable outlet ports of the two main control boards be both optical or electrical ports and configured with the same data rate. This prevents services from being greatly affected after a fault-triggered switchover.
- Control-plane SCTP multihoming requires that all core network or base station devices support SCTP multihoming.
- User-plane GTP-U detection is used to detect faults for reliability. The peer transmission device must support user-plane link status fast awareness based on the physical link status when this function is enabled for user-plane links to facilitate fast switchovers.
- Source IP-based routing needs to be configured on the base station as uplink service flows with different source IP addresses need to be sent over different ports.
- The peer base station to be connected with the local one over the X2/Xn/eX2/eXn interface must support the multi-IP address function. If the peer base station does not support the multi-IP address function, the transmission link backup function does not take effect in multi-IP address configuration scenarios.
- When IPv6 and IPv4 are configured as non-default VRF routing domains, inter-board routing and forwarding are not supported. In this case, the service IP addresses of the OM, control plane, and user plane must be on the same main control board as the outbound interface corresponding to the next-hop gateway IP address for service flow routing and forwarding. In addition, these service IP addresses must be configured on the primary BBU.
- Transmission link backup applies only to deployment by PnP in non-secure networking, but not secure networking.
- When multiple IP addresses are configured for the user plane, IPsec tunnels to the SeGW can be manually configured and direct IPsec networking is not supported.
- Transmission link backup does not support tunnel reliability functions in IPsec scenarios. Such functions include IPsec tunnel backup, SeGW redundancy, and IPsec redirection.
- Transmission link backup supports only the endpoint configuration mode, not the link configuration mode.
- The IP address of the OMCH must have been configured on the base station. The IP address can be the IP address of a port or a loopback interface.
- Inter-board routing and forwarding configuration is not allowed. The board that sends or receives packets must be the one that corresponds to the IP address configured for service packets.
- In IPv4 and IPv6 dual-stack configuration for X2 scenarios, at least one pair of an IPv4 address and an IPv6 address must be configured on the same board.
- The clock source must be a GNSS or IEEE 1588V2 clock.
  - It is recommended that the GNSS antenna system be connected to both the active and standby main control boards to improve the GNSS antenna access reliability and prevent manual cable connection adjustment during the active/standby main control board switchover if a GNSS clock is used as the clock source.
  - If an IEEE 1588V2 clock is used as the clock source, both the IEEE 1588V2 clock synchronization links must be available.

## 5.3.5 Others

None

## 5.4 Operation and Maintenance

### 5.4.1 When to Use

In LTE and NR networks, if customers have high requirements on the reliability of the entire base station, it is recommended that this feature be enabled. Cold Backup of Main Control Boards is not recommended. Examples are as follows:

- It is recommended that this function be enabled if services need to be restored within several minutes after a main control board hardware fault or transmission link fault occurs.
- The UMPT supports multimode co-MPT and multi-RAT concurrency. When this main control board experiences a severe hardware fault, services of the multiple RATs are interrupted. It is recommended that this function be enabled if the reliability of the base station in this scenario needs to be improved.

### 5.4.2 Precautions

- It is recommended that the cable outlet ports of the two main control boards be both optical or electrical ports and configured with the same data rate. This prevents services from being greatly affected after a fault-triggered switchover.
- If multiple IP addresses are configured for a service interface, it is recommended that all IP addresses used for communication be reachable between the base station and the core network and between base stations. Otherwise, ALM-25954 User Plane Fault is reported due to failures in the self-setup of user-plane links.

### 5.4.3 Data Configuration

#### 5.4.3.1 Data Preparation

- Configuring main control boards  
During main control board configuration, the two boards (active and standby main control boards) must be configured on the same logical NE. [Table 5-1](#) describes the key parameters that must be set for configuring main control boards.

**Table 5-1** Key parameters for configuring main control boards

| Parameter Name | Parameter ID                                          | Setting Notes                    |
|----------------|-------------------------------------------------------|----------------------------------|
| Cabinet No.    | <b>CN</b> parameter in the <b>ADD BRD</b> MML command | Set this parameter to <b>0</b> . |

| Parameter Name | Parameter ID                                                  | Setting Notes                                                                         |
|----------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Subrack No.    | <b><i>SRN</i></b> parameter in the <b>ADD BRD</b> MML command | Set this parameter to <b>0</b> .                                                      |
| Slot No.       | <b><i>SN</i></b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>6</b> and <b>7</b> for two main control boards.              |
| Board Type     | <b><i>BT</i></b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>UMPT</b> . Boards in slots 6 and 7 must be of the same type. |

- Configure transmission links.

To implement transmission link backup of main control boards, ports, VLANs, IP addresses, and IP routes must be planned and configured on the two main control boards. For details about how to configure active and standby OM channels, see *IP eRAN Engineering Guide* or *IP NR Engineering Guide*. For details about data configuration for control-plane SCTP multihoming and user plane backup in LTE and NR NSA, see *S1 and X2 Self-Management* or *X2 and S1 Self-Management in NSA Networking*. For details about data configuration for control-plane SCTP multihoming in LTE and NR SA, see *X2 Self-Management in SA Networking*. For details about data configuration for control-plane SCTP multihoming and user plane backup in NR SA, see *NG and Xn Self-Management*. If IPsec is enabled, IPsec-related security parameters must be planned and configured. For details, see *IPsec*.

**Table 5-2** describes the parameters used for activating transmission link backup.

**Table 5-2** Key parameters related to transmission links

| Parameter Name                               | Parameter ID                                           | Setting Notes                                                                                                                                                                                            |
|----------------------------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| User-plane Path Status Fast Awareness Switch | <b>GUPTRANSPARA.UPSTATFASW (5G gNodeB, LTE eNodeB)</b> | To enable user-plane link status fast awareness based on the physical link status, set this parameter to <b>ENABLE</b> .                                                                                 |
| Transmission Configuration Mode              | <b>GTRANSPARA.TRANSCFGMODE (LTE eNodeB, 5G gNodeB)</b> | In transmission link backup, control-plane and user-plane links support only the new transmission model, meaning that <b>GTRANSPARA.TRANSCFGMODE (LTE eNodeB, 5G gNodeB)</b> must be set to <b>NEW</b> . |

#### NOTE

- For details about how to configure active and standby OM channels, see *IP eRAN Engineering Guide* or *IP NR Engineering Guide*.
- For details about data configuration for control-plane SCTP multihoming and user plane backup, see *S1 and X2 Self-Management*, *X2 and S1 Self-Management in NSA Networking*, or *X2 Self-Management in SA Networking*.
  - In transmission link backup, control-plane and user-plane links support only the new transmission model, meaning that **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**.
  - In transmission link backup, transmission establishment for control-plane and user-plane links supports only the endpoint mode, meaning that control-plane links are configured using the **SCTPHOST** MO and user-plane links are configured using the **USERPLANEHOST** MO.

### 5.4.3.2 Using MML Commands

Before using MML commands, refer to [5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

- Step 1** Run the **DSP BRD** command to query whether the slot number of the main control board that is configured with transport data is 6 or 7.

```
DSP BRD;
```

- Step 2** Run the **ADD BRD** command to add a main control board with a different slot number as the standby main control board.

```
ADD BRD: CN=0, SRN=0, SN=6, BT=UMPT;
```

- Step 3** The following are examples of some major transmission link configurations. For details, see *IP eRAN Engineering Guide* in *eRAN Feature Documentation* or *IP NR Engineering Guide* in *5G RAN Feature Documentation*.

```
//Configuring Ethernet ports on the two main control boards
SET ETHPORT: CN=0, SRN=0, SN=6, SBT=BASE_BOARD, PN=1, PORTID=1, PA=FIBER, SPEED=1000M,
DUPLEX=FULL, FC=OPEN, FERAT=10, FERDT=10, RXBCPKTALMOCRTHD=322, RXBCPKTALMCLRTHD=290;
SET ETHPORT: CN=0, SRN=0, SN=7, SBT=BASE_BOARD, PN=3, PORTID=2, PA=FIBER, SPEED=1000M,
DUPLEX=FULL, FC=OPEN, FERAT=10, FERDT=10, RXBCPKTALMOCRTHD=322, RXBCPKTALMCLRTHD=290;
//Configuring VLAN interfaces on the two main control boards and the associated VLANs
ADD INTERFACE: ITFID=0, ITFTYPE=VLAN, PT=ETH, PORTID=1, VLANID=100, VRFIDX=0, MTU4=1500,
ARPPROXY=ENABLE;
ADD INTERFACE: ITFID=1, ITFTYPE=VLAN, PT=ETH, PORTID=2, VLANID=200, VRFIDX=0, MTU4=1500,
ARPPROXY=ENABLE;
//Configuring the interface IP addresses of the Ethernet ports on the two main control boards (IPv4 as an
example)
ADD IPADDR4: ITFID=0, IP="10.1.1.1", MASK="255.255.255.0", VRFIDX=0;
ADD IPADDR4: ITFID=1, IP="20.1.1.1", MASK="255.255.255.0", VRFIDX=0;
//Configuring source IP address-based routes for the two main control boards
```

```

ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="10.1.1.1", RTTYPE=NEXTHOP, NEXTHOP="10.1.1.2", PREF=60;
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="20.1.1.1", RTTYPE=NEXTHOP, NEXTHOP="20.1.1.2", PREF=60;
//Configuring active and standby OM channels on the two main control boards
SET GTRANSPARA: OMCHSWITCHBACK=ENABLE, OMCHSBWAITTIME=60;
ADD OMCH: FLAG=MASTER, BEAR=IPv4, IP="10.1.1.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BRT=NO, CHECKTYPE=AUTO_UDPSESSION;
ADD OMCH: FLAG=SLAVE, BEAR=IPv4, IP="20.1.1.1", MASK="255.255.255.255", PEERIP="100.100.100.1",
PEERMASK="255.255.255.0", BRT=NO, CHECKTYPE=NONE;
//Configuring user-plane IP addresses on the two main control boards to specify user-plane transmission
links
ADD USERPLANEHOST: UPHOSTID=0, IPVERSION=IPv4, LOCIPV4="10.1.1.1", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="20.1.1.1", IPSECSWITCH=DISABLE;
//Adding two user-plane hosts to the specified user-plane endpoint group
ADD UPHOST2EPGRP: EPGROUPID=0, UPHOSTID=0;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Configuring active and standby certificate management links on the two main control boards with the IP
address of the OM channel as the source IP address
ADD CA: CANAME="C=cn,S=sc,L=cd,O=Huawei,OU=Wireless,CN=enbca", URL="http://192.168.1.167:80",
SIGNALG=SHA256, MODE=DEFAULT_MODE, POLICYID=0, CERTREQSW=DEFAULT, OCSPURL="http://
192.168.1.167:80";
//Enabling user-plane GTP-U detection
MOD GTPU: TIMEOUTTH=5000, TIMEOUTCNT=3, DSCP=46, STATICCHK=ENABLE,
ULGTPUSNPADSW=DISABLE, DLGTPUSNCHKSW=DISABLE;
//Optional) Configuring user-plane fast switchover
SET GUPTRANSPARA: UPSTATFASW=ENABLE;
//Configuring an SCTP link on the base station to specify the link to transmit signaling. The IP address
specified by SIGIP1V4 is used as the IP address of the active SCTP link, and the IP address specified by
SIGIP2V4 is used as the IP address of the standby SCTP link.
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="10.1.1.1", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="20.1.1.1", SIGIP2SECSWITCH=DISABLE, PN=1024, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTEMPLATEID=0, DTLSPOLICYID=NULL;

```

- Step 4** Run MML commands to configure transmission interfaces. For details about MML-based configurations for control-plane SCTP multihoming and user plane backup in LTE and NR NSA, see *S1 and X2 Self-Management* or *X2 and S1 Self-Management in NSA Networking*. For details about MML-based configurations for control-plane SCTP multihoming in LTE and NR SA, see *X2 Self-Management in SA Networking*. For details about MML-based configurations for control-plane SCTP multihoming and user plane backup in NR SA, see *NG and Xn Self-Management*. If IPsec is enabled, IPsec-related security parameters must be planned and configured. For details, see *IPsec*.

----End

#### NOTE

The preceding steps do not apply to separate-MPT multimode base stations that are running properly. To enable transmission link backup of main control boards for such a base station, contact Huawei engineers.

In scenarios where an existing link needs to be reconstructed into a transmission backup link, the **BLK SCTPPEER** or **ADD BLKSCTPPEERIPV4** command needs to be first executed to block an SCTP peer or SCTP peer IPv4 address, respectively, after the SCTP link backup is complete. And two minutes later, the **UBL SCTPPEER** or **RMV BLKSCTPPEERIPV4** command is then executed to unblock the SCTP peer or SCTP peer IPv4 address.

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Step 1** Run the MML command **DSP BRD** to query the slot number of the active main control board. For example, the slot number is 7.

DSP BRD;;

**Step 2** Run the MML command **RMV BRD** to remove the main control board (standby main control board) whose slot number (for example, 6) is different from that of the active main control board.

RMV BRD: CN=0, SRN=0, SN=6;

----End

### 5.4.3.3 Using the MAE-Deployment

This section only describes the operations on the MAE-Deployment for multimode base stations. The operations for single-mode base stations such as eGBTS, NodeB, eNodeB, and gNodeB are similar to those for multimode base stations.

It is recommended that Cold Backup of Main Control Boards be activated on the MAE-Deployment.

This section describes the steps for activating Cold Backup of Main Control Boards for a base station that meets the following requirements:

- A main control board has been configured in the base station and the transport data has been configured for this board.
- The base station is operating properly.

### Using the MAE-Deployment to Perform Single Configuration

**Step 1** Modify the base station configurations.

Configure related MOs on the MAE-Deployment as instructed in [Table 5-3](#) when **GTRANSPARA.TRANS CFGMODE (LTE eNodeB, 5G gNodeB)** is set to **NEW**.

For detailed operations, see Feature Configuration Using the MAE-Deployment.

#### NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

**Table 5-3** MOs to be configured on the MAE-Deployment

| SN | MO                          | Configuration Description                                                                              |
|----|-----------------------------|--------------------------------------------------------------------------------------------------------|
| 1  | MPT                         | Used to add a standby main control board.                                                              |
| 2  | ETHPORT                     | Used to remove the Ethernet ports that are automatically configured on the standby main control board. |
| 3  | INTERFACE                   | Used to add an IP interface corresponding to a port.                                                   |
| 4  | IPADDR4/IPADDR6             | Used to add an IPv4 or IPv6 address for a port.                                                        |
| 5  | SRCIPROUTE4/<br>SRCIPROUTE6 | Used to add an IPv4 or IPv6 source-based route.                                                        |



| SN | MO            | Configuration Description             |
|----|---------------|---------------------------------------|
| 6  | OMCH          | Used to add active and standby OMCHs. |
| 7  | SCTPHOST      | Used to add an SCTP link.             |
| 8  | USERPLANEHOST | Used to add a user-plane link.        |

**Step 2** Export the base station deployment list. For detailed operations, see section "Preparing a Data Configuration File and a Deployment List" in *3900 & 5900 Series Base Station Commissioning Guide*.

**Step 3** Upload the base station deployment list to the MAE-Access so that the DHCP server-related configurations are updated. For detailed operations, see section "Creating an NE Commissioning Task" in *3900 & 5900 Series Base Station Commissioning Guide*.

**Step 4** Deliver incremental configurations.

----End

## Using the MAE-Deployment to Perform Batch Modification

### NOTE

When configuring this function on the MAE-Deployment, perform a single configuration first, and then perform a batch modification if required. You must perform single configuration for a parameter before batch modification of the parameter using the MAE-Deployment. You are advised to perform batch modification before logging out of the parameter setting interface.

**Step 1** After creating a planned data area, choose **Advanced > Data Management > Customize Summary Data File** from the main menu, to customize a summary data file for batch configuration.

### NOTE

For context-sensitive help on a current task in the client, click .

**Step 2** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.

**Step 3** In the summary data file, set the parameters related to the **MPT** MO for the standby main control board, and close the file.

**Step 4** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.

**Step 5** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.

- Step 6** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.
- Step 7** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Export Base Station Bulk Configuration Data**, to export the existing NE data stored on the MAE-Deployment to the customized summary data file.
- Step 8** Specify the transport parameters to be reconfigured in the summary data file and close the file.
- Step 9** On the main menu, choose **GSM Application/UMTS Application/LTE Application/NR Application > Network Adjustment > Import Base Station Bulk Configuration Data**, to import the data in the summary data file into the MAE-Deployment, and then start the data verification.
- Step 10** After the data verification is completed, choose **Area Management > Planned Area Management > Export Incremental Scripts** to export and activate the incremental scripts in the planned data area.
- End

## Using the MAE-Deployment to Perform Deactivation

- Step 1** Configure the MOs described in [Table 5-4](#) on the MAE-Deployment configuration interface. For detailed operations, see Feature Configuration Using the MAE-Deployment.

**Table 5-4** MOs to be configured on the MAE-Deployment

| SN | MO            | Configuration Description                                                                                                                 |
|----|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | OMCH          | Remove active and standby OMCHs.                                                                                                          |
| 2  | SCTPHOST      | Remove an SCTP link.                                                                                                                      |
| 3  | USERPLANEHOST | Remove a user-plane link.                                                                                                                 |
| 4  | MPT           | Remove the main control board (standby main control board) whose slot number is different from the slot number of the transport HA group. |

- Step 2** Export the base station deployment list. For detailed operations, see section "Preparing a Data Configuration File and a Deployment List" in *3900 & 5900 Series Base Station Commissioning Guide*.
- Step 3** Upload the base station deployment list to the MAE-Access so that the DHCP server-related configurations on the MAE-Access are updated. For detailed operations, see section "Creating an NE Commissioning Task" in *3900 & 5900 Series Base Station Commissioning Guide*.
- Step 4** Deliver incremental configurations.
- End

## 5.4.4 Activation Verification

### Step 1 Check the board status.

Run the **DSP BRD** command to check whether the boards in slots 6 and 7 are normal. If the value of **Standby Status** is **Active** for one board and is **Standby** for the other board and the value of **Availability Status** is **Normal** for both boards, the two boards are normal.

### Step 2 Check the status of the Ethernet port.

Run the **DSP ETHPORT** command to check the status of the transmission ports on the two main control boards.

Expected result:

- The values of **Port Status** and **Physical Layer Status** are **UP**, indicating that the physical port has been successfully activated.
- The values of **Local Speed** and **Peer Speed** are the same.
- The values of **Local Duplex** and **Peer Duplex** are the same.

### Step 3 Query the route status.

Run the **DSP SRCIPROUTE4** or **DSP SRCIPROUTE6** command to check the route status.

Expected result: The value of **Valid State** is **Valid**, indicating that the route has taken effect.

### Step 4 Query the IP address status.

Run the **PING** MML command to ping the peer IP address, with the local IP address set to the one indicated by **IPADDR4/IPADDR6**.

Expected result: The address is pinged successfully, indicating that the route and IP address have taken effect.

### Step 5 Check the status of SCTP dual-homing links.

Run the **DSP SCTPLNK** command to query the SCTP link status.

Check the values of **SCTP Link Status** and **Work Address Flag**. If the value of **SCTP Link Status** is **Up**, the link is normal. **Work Address Flag** indicates whether the working SCTP link is active or standby.

### Step 6 Query the status of the OM channel.

Run the **DSP OMCH** MML command to check the status of the active and standby OM channels. An OM channel is activated and in use if the value of **OM Channel Status** is **Normal** and the value of **Used State** is **In Use**.

### Step 7 Check the initial backup status of the board.

Run the **DSP BKPSTATUS** MML command to query the values of **Backup Switch** and **Backup Status**. If **Backup Switch** is **On**, cold backup of main control boards is activated. If **Backup Status** is **Idle**, initial backup is completed.

 **NOTE**

After the standby main control board is configured, the **DSP BKPSTATUS** MML command can be repeatedly executed to query the progress of initial backup. The initial backup process takes approximately 150 minutes.

**Step 8** Switch the active and standby roles of the main control boards.

Run the **SWP BRD** command to perform the switch with **Cabinet No.**, **Subrack No.**, and **Slot No.** set to default values for main control boards. Upon execution of this command, the base station will automatically perform routine backup, reset, and switchover between the active and standby main control boards. This process takes approximately 30 minutes. Base station resets will interrupt ongoing services.

**Step 9** Verify the active/standby switchover.

When the base station becomes normal after the reset, and the connections to the LMT and MAE-Access are restored, perform step 1 to step 3 to check whether the active and standby roles of the main control boards have been switched.

----End

## 5.4.5 Network Monitoring

None

# 6 Inter-Board Baseband Resource Redundancy (GSM/UMTS)

---

## 6.1 Principles

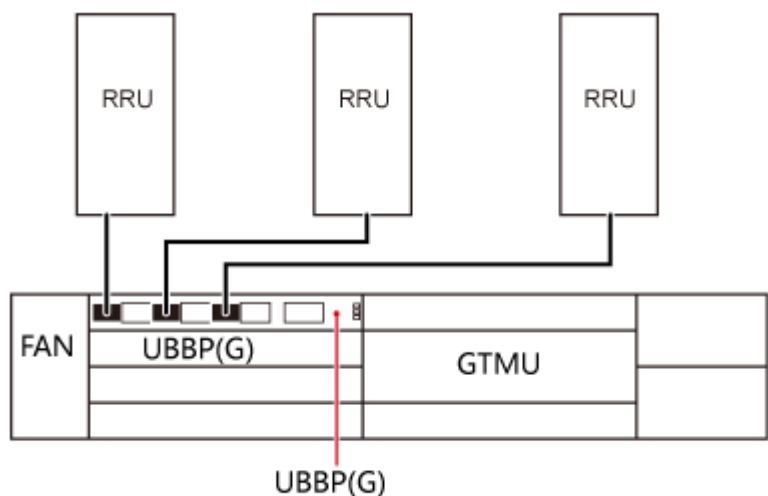
When a BBP fails, the cells served by this faulty BBP will be affected. Inter-Board Baseband Resource Redundancy allows multiple BBPs to be configured in a base station. This enables the cells or carriers served by a faulty BBP to be reestablished on another operational BBP with available resources, which improves base station reliability.

To implement this feature, a base station must be equipped with at least two BBPs and these two BBPs must be installed in the same BBU.

### Inter-Board Baseband Resource Redundancy for GSM

**Figure 6-1** illustrates a GBTS S2/2/2 configuration scenario where the GBTS is configured with two UBBP\_G boards. If one UBBP\_G board fails due to a hardware or communication port fault, the GBTS can detect and identify the fault and then attempt to reestablish the carriers served by the faulty UBBP\_G board on another UBBP\_G board with available baseband resources. Services carried on the BCCH carrier preferentially recover.

**Figure 6-1** GBTS S2/2/2



C1801C3021

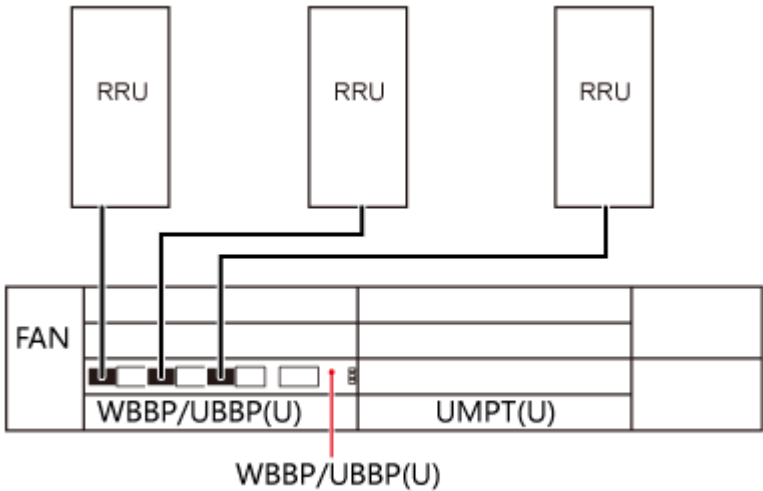
#### NOTICE

- For GSM, only the UBBP board supports inter-board baseband resource redundancy. It is recommended that the two UBBP boards be configured in slots 0 and 1.
- Inter-board baseband resource redundancy for GSM does not require CPRI-based topologies and is supported as long as two UBBP boards are configured. However, the inter-board cold backup ring topology and hot backup ring topology are not supported in GSM. For details, see *RF Unit and Topology Management*.

## Inter-Board Baseband Resource Redundancy for UMTS

**Figure 6-2** illustrates a NodeB S1/1/1 configuration scenario. A NodeB is configured with at least two WBBP or UBBP\_U boards. If one board is faulty due to a hardware or communication port fault, the NodeB can detect and locate the fault and then attempt to reestablish the cells served by the faulty board on another WBBP or UBBP\_U board with available baseband resources. Services recover within 20s.

Figure 6-2 NodeB S1/1/1



NOTE

Inter-board baseband resource redundancy for UMTS does not require CPRI-based topologies and is supported as long as two BBPs are configured. However, the hot backup ring topology is not supported in UMTS. For details, see *RF Unit and Topology Management*.

6.2 Network Analysis

6.2.1 Benefits

None

6.2.2 Impacts

The impacts between this function and other functions are described in [6.3.2.3 Function Impacts](#).  
This function has no impact on the network.

6.3 Requirements

6.3.1 Licenses

None

6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 6.3.2.1 Prerequisite Functions

None

### 6.3.2.2 Mutually Exclusive Functions

None

### 6.3.2.3 Function Impacts

None

## 6.3.3 Hardware

- The NodeB is configured with two or more WBBP or UBBP boards, which are installed in slots 0 to 5.
- The GBTS or eGBTS is configured with two or more UBBP boards. If two UBBP boards are configured, it is recommended that they be installed in slots 0 and 1.
- This feature applies only to macro base stations that use the BBU3900, BBU3910, BBU5910, or BBU5900.

## 6.3.4 Others

None

## 6.4 Operation and Maintenance

### 6.4.1 When to Use

It is recommended that this feature be activated in scenarios with high requirements for reliability. For example, services are required to recover within 30s after a baseband processing unit becomes faulty. If multiple UBBPs are configured in a multimode base station, it is recommended that each RAT be deployed on all UBBPs. This improves system reliability by using the inter-board baseband resource redundancy.

### 6.4.2 Data Configuration

#### 6.4.2.1 Data Preparation

[Table 6-1](#) describes the key parameters that must be set in the **BBP** MO for GSM (eGBTS) and UMTS.



**Table 6-1** Key parameters that must be set in the **BBP** MO

| Parameter Name          | Parameter ID                                           | Setting Notes                                                                                                        |
|-------------------------|--------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| Cabinet No.             | <b>CN</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>0</b> .                                                                                     |
| Subrack No.             | <b>SRN</b> parameter in the <b>ADD BRD</b> MML command | Set this parameter to <b>0</b> .                                                                                     |
| Slot No.                | <b>SN</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>0</b> and <b>1</b> for two BBPs in an eGBTS. There is no special requirement for the NodeB. |
| Board Type              | <b>BT</b> parameter in the <b>ADD BRD</b> MML command  | Set this parameter to <b>UBBP</b> for the eGBTS, and to <b>WBBP</b> or <b>UBBP</b> for the NodeB as required.        |
| Base Band Work Standard | <b>BBP.BBWS</b>                                        | Set this parameter to the RATs for which the UBBP board provides baseband processing resources.                      |

**Table 6-2** describes the key parameters that must be set in the **BTSBRD** MO for GSM (GBTS).

**Table 6-2** Key parameters that must be set in the **BTSBRD** MO for GSM (GBTS)

| Parameter Name | Parameter ID      | Setting Notes                                             |
|----------------|-------------------|-----------------------------------------------------------|
| Cabinet No.    | <b>BTSBRD.CN</b>  | Set this parameter to <b>0</b> .                          |
| Subrack No.    | <b>BTSBRD.SRN</b> | Set this parameter to <b>0</b> .                          |
| Slot No.       | <b>BTSBRD.SN</b>  | Set this parameter to <b>0</b> and <b>1</b> for two BBPs. |
| Board Type     | <b>BTSBRD.BT</b>  | Set this parameter to <b>UBBP</b> .                       |

**Table 6-3** describes the key parameters that must be set in the **BASEBANDEQM** MO. For GSM, the **BASEBANDEQM** MO is required by the eGBTS and is not required by the GBTS.

**Table 6-3** Key parameters that must be set in the **BASEBANDEQM** MO

| Parameter Name            | Parameter ID                        | Setting Notes                                                                                                                                                                 |
|---------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Baseband Equipment ID     | <b>BASEBANDEQM.BASEBANDEQMID</b>    | For an eGBTS, set this parameter to <b>0</b> . For a NodeB, you need to configure two baseband equipment IDs ( <b>1</b> and <b>2</b> ).                                       |
| Baseband Equipment Type   | <b>BASEBANDEQM.BASEBANDEQMTYPE</b>  | For an eGBTS, set this parameter to <b>ULDL</b> . For a NodeB, set this parameter to <b>UL</b> and <b>DL</b> for baseband equipment 1 and baseband equipment 2, respectively. |
| UMTS UL Demodulation Mode | <b>BASEBANDEQM.UMTSDEMMODE</b>      | For an eGBTS, this parameter is invalid and should be set to <b>NULL</b> . For a NodeB, set this parameter based on the network plan.                                         |
| Baseband Equipment Board  | <b>BASEBANDEQM.BASEBANDEQMBOARD</b> | Set this parameter based on the cabinet number, subrack number, and slot number of the BBP.                                                                                   |

### 6.4.2.2 Using MML Commands

Before using MML commands, refer to [6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

#### NodeB

- Run the **ADD BRD** command to add BBPs.  
ADD BRD: CN=0, SRN=0, SN=2, BT=UBBP, BBWS=GSM-0&UMTS-1&LTE\_FDD-0&LTE\_TDD-0;  
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=GSM-0&UMTS-1&LTE\_FDD-0&LTE\_TDD-0;
- Run the **ADD BASEBANDEQM** command to add all BBPs requiring backup to the same baseband equipment. Set the UL and DL baseband equipment types, respectively.  
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMTYPE=DL, SN1=2, SN2=1;  
ADD BASEBANDEQM: BASEBANDEQMID=0, UMTSDEMMODE=DEM\_2\_CHAN, BASEBANDEQMTYPE=UL, CN1=0, SRN1=0, SN1=2, CN2=0, SRN2=0, SN2=1;

### GBTS

1. Run the **ADD BTSBRD** command to add two UBBPs.  
ADD BTSBRD: IDTYPE=BYID, BTSID=120, CN=0, SRN=0, SN=0, BT=UBBP;  
ADD BTSBRD: IDTYPE=BYID, BTSID=120, CN=0, SRN=0, SN=1, BT=UBBP;

### eGBTS

1. Run the **ADD BRD** command to add UBBPs.  
ADD BRD: CN=0, SRN=0, SN=0, BT=UBBP, BBWS=GSM-1&UMTS-0&LTE\_FDD-0&LTE\_TDD-0;  
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=GSM-1&UMTS-0&LTE\_FDD-0&LTE\_TDD-0;
2. Run the **ADD BASEBANDEQM** command to add the baseband equipment.  
ADD BASEBANDEQM: BASEBANDEQMID=0, UMTSDEMMode=NULL, BASEBANDEQMType=ULDL,  
CN1=0, SRN1=0, SN1=0, CN2=0, SRN2=0, SN2=1;

## Deactivation Command Examples

This feature does not need to be deactivated.

### 6.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

## 6.4.3 Activation Verification

1. Run the following commands to query the carriers deployed on each BBP.
  - **DSP BTSBRDBBINFO** command for the GBTS
  - **DSP GTRX** command for the eGBTS
  - **DSP ULOCCELLRES** for the NodeB
2. Run the **BLK BRD** command to block one BBP. Cells served by this board will be dynamically reestablished on another operational BBP and services will automatically recover.

## 6.4.4 Network Monitoring

None

# 7 Inter-Board Baseband Resource Redundancy (LTE/NR)

---

## 7.1 Principles

When a BBP is faulty, the cells served by the BBP will be affected. When multiple BBPs are configured in a base station, Inter-Board Baseband Resource Redundancy enables the cells or carriers served by a faulty BBP to be reestablished on another operational BBP with available resources, which improves base station reliability.

To implement this feature, a base station must be equipped with at least two BBPs and these two BBPs must be installed in the same BBU.

### Macro

In CPRI scenarios, Inter-Board Baseband Resource Redundancy for LTE/NR is only supported in the inter-board cold backup ring topology and hot backup ring topology. It is required that baseband equipment not be specified. For details about the ring topology types supported by LTE and NR, see *RF Unit and Topology Management*.

In eCPRI scenarios, if RF modules work in cold backup mode, Inter-Board Baseband Resource Redundancy for LTE/NR is supported in the intra-board load sharing topology and inter-board load sharing topology. For details about the load sharing topology types supported by LTE and NR, see *RF Unit and Topology Management*.

In eCPRI scenarios, if RF modules are not in cold backup mode, Inter-Board Baseband Resource Redundancy for NR is only supported in the inter-board load sharing topology. It is required that baseband equipment not be specified. For details about the load sharing topology types supported by NR, see *RF Unit and Topology Management*.

The topology type is specified by the **RRUCHAIN.TT** parameter, and the backup mode is specified by the **RRUCHAIN.BM** parameter.

**NOTE**

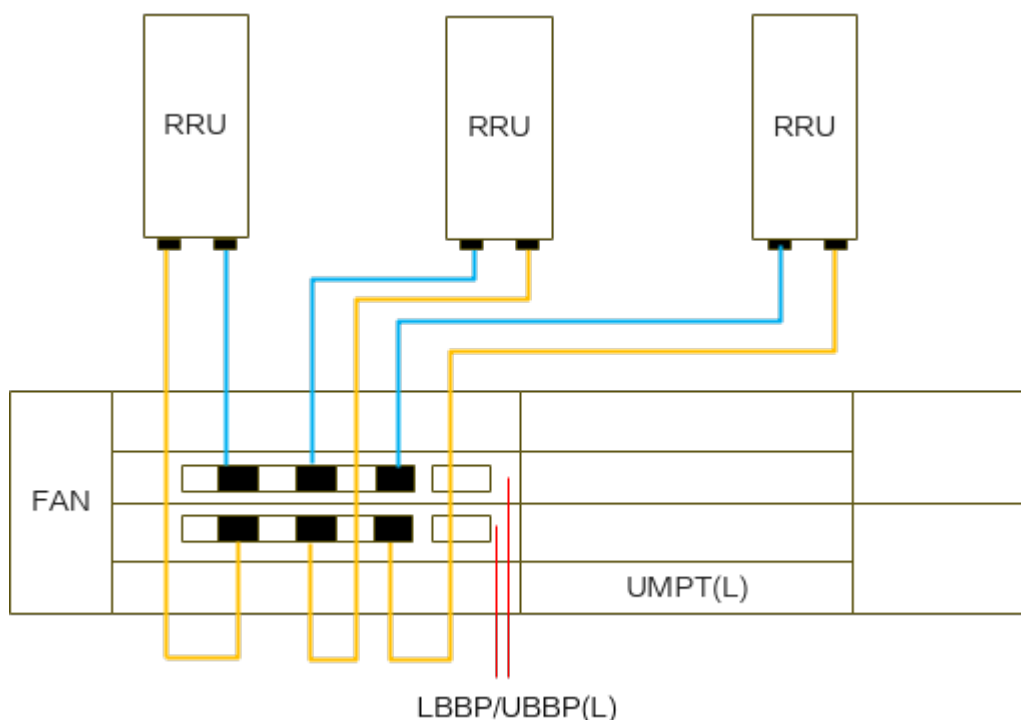
The implementation on the gNodeB is the same as that on the eNodeB. The following part uses the eNodeB as an example.

**Figure 7-1** illustrates a 3x10 MHz 2T2R configuration scenario for the eNodeB. In this scenario, the eNodeB is configured with two LBBP or UBBP(L) boards and the two BBPs connect to the same RRUs so that an inter-board one-level cold backup ring topology or hot backup ring topology is formed. If one LBBP or UBBP(L) board is faulty due to a hardware or communication port fault, the eNodeB can detect and identify the fault. The impacts of an LBBP or UBBP(L) board failure on cells served by the faulty board in the following two topologies are different:

- In the inter-board cold backup ring topology, one or more cells served by the faulty LBBP or UBBP(L) board will be reestablished on another LBBP or UBBP(L) board that connects to the same RRUs as that connected to the faulty board.
- In the inter-board hot backup ring topology, the cells served by the faulty LBBP or UBBP(L) board will be reestablished on the target LBBP or UBBP(L) board, but will not if the CPRI port is not operational.

The target LBBP or UBBP(L) board is chosen based on the available resources of all candidate target LBBP or UBBP(L) boards. In **Figure 7-1**, the blue lines indicate the communication channels between the source board and RRUs, and the orange lines indicate the communication channels between the target board and the RRUs.

**Figure 7-1** LTE 3x10 MHz 2T2R



#### NOTICE

An LBBPc board can only work as a backup for another LBBPc board. An LBBPd board and a UBBP(L) board can work as a backup for each other.

When an LBBP is used, the working mode must be configured by using the **BBP.WM** parameter.

When a UBBP(L) is used, the baseband working mode must be configured by using the **BBP.BBWS** parameter.

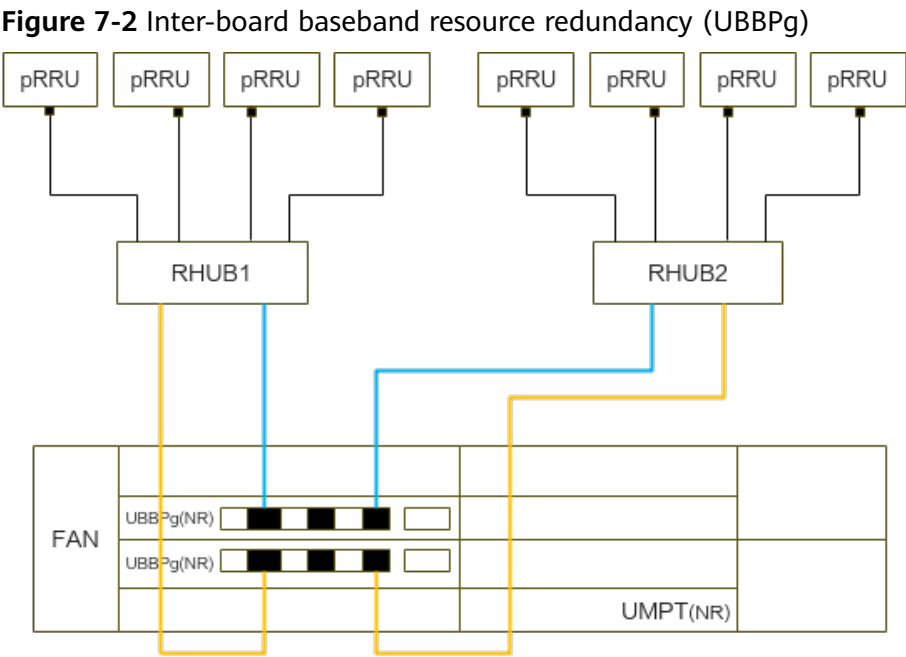
## LampSite

LampSite base stations support Inter-Board Baseband Resource Redundancy. For LampSite base stations, if the inter-board trunk chain topology is used and cells are distributed massive MIMO cells, the UBBPg(NR) boards support Inter-Board Baseband Resource Redundancy. The topology type is specified by the **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** parameter, and the backup mode is specified by the **RRUCHAIN.BM (LTE eNodeB, 5G gNodeB)** parameter. For details about support for the inter-board trunk chain topology in NR, see *RF Unit and Topology Management*. For details about a distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD)* in *5G RAN Feature Documentation*.

**Figure 7-2** shows a one-level inter-board trunk chain dual-fiber topology when the gNodeB is configured with two UBBPg boards connected to the same RHUB. If one UBBPg board is faulty or its communication port is faulty, the gNodeB can detect and identify the fault. A faulty UBBPg board has the following impacts on the cells served by it:

- If the communication port is faulty, the base station will attempt to reestablish and recover the cells through the communication port of the target UBBPg board that is connected to the same RHUB as the faulty UBBPg board.
- If the UBBPg board is faulty, the cells will be reestablished on the target UBBPg board.

The target UBBPg board is chosen based on the available resources of all candidate target UBBPg boards. Inter-Board Baseband Resource Redundancy is only supported in single-carrier scenarios. **Figure 7-2** shows a scenario where eight pRRUs, two RHUBs, and two UBBPg boards are used. Each RHUB is connected to BBPs through two fibers in an inter-board trunk chain topology. The blue and orange lines indicate the communication channels between RHUBs and the source and target UBBPg boards, respectively. A carrier of a distributed massive MIMO cell is configured on the pRRUs.



7.2 Network Analysis

7.2.1 Benefits

None

7.2.2 Impacts

The impacts between this function and other functions are described in [7.3.2.3 Function Impacts](#).

This function has no impact on the network.

7.3 Requirements

7.3.1 Licenses

None

7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

7.3.2.1 Prerequisite Functions

| RAT                  | Function Name                                | Function Switch                                                                                  | Reference                                                                                |
|----------------------|----------------------------------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Distributed massive MIMO (low-frequency TDD) | <b>DM_MIMO_SERVICE_SWITC</b><br>H option of the <b>NRDUCellAlgoSwitch.DmMimoSwitch</b> parameter | <i>Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation</i> |

7.3.2.2 Mutually Exclusive Functions

None

7.3.2.3 Function Impacts

| RAT                         | Function Name                       | Function Switch                                                                                        | Reference                                                                         | Description                                                                                                                                                                           |
|-----------------------------|-------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR FDD Low-frequency NR TDD | Adaptive baseband shutdown          | <b>BBP.CELLDEPLOY</b> set to <b>ADAPT_BBP_SHUTDOWN</b>                                                 | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | Adaptive baseband shutdown does not take effect in distributed massive MIMO cells served by LampSite base stations when Inter-Board Baseband Resource Redundancy is enabled.          |
| NR FDD Low-frequency NR TDD | Baseband shutdown for energy saving | <b>BB_SHUTDOWN_POWERSAVING_SW</b> option of the <b>gNodeBParam.BaseStationDeepDormancySw</b> parameter | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | Baseband shutdown for energy saving does not take effect in distributed massive MIMO cells served by LampSite base stations when Inter-Board Baseband Resource Redundancy is enabled. |



| RAT                         | Function Name      | Function Switch                                         | Reference                                                                    | Description                                                                                                                                                          |
|-----------------------------|--------------------|---------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR FDD Low-frequency NR TDD | Power saving first | <b>BBP.CELLDEPLOY</b> set to <b>ADAPT_BB_P_SHUTDOWN</b> | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Document</i> | Power saving first does not take effect in distributed massive MIMO cells served by LampSite base stations when Inter-Board Baseband Resource Redundancy is enabled. |

## 7.3.3 Hardware

### Macro

- This feature applies to macro base stations that use the BBU3910, BBU3900, BBU5910, or BBU5900.
- The eNodeB is configured with two or more LBBP/UBBP boards.
- The gNodeB is configured with two or more UBBP boards.
- An LBBPc board can only work as a backup for another LBBPc board. An LBBPd board and a UBBP board can work as a backup for each other.
- If the cold backup ring topology is used, there is no requirement on slots for configuring BBPs. If the BBU3900 and hot backup ring topology are used, the BBPs must be configured in the slots specified in the following table.

| If...                                                                                   | Then...                                                                                                                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Two LBBPc boards are to be configured                                                   | One LBBPc board must be configured in slot 2 or 3, and the other LBBPc board can be configured in any slot from slot 0 to slot 5. A maximum of three hot backup rings are supported.<br><br>An LBBPc board can be configured with only one hot backup LBBPc board at a time. |
| Two UBBP boards, two LBBPd boards, or one LBBPd and one UBBP board are to be configured | One BBP must be configured in slot 2 or 3, and the other BBP can be configured in any slot from slot 0 to slot 5. A maximum of six hot backup rings are supported.                                                                                                           |

- The UBBPfw board must be configured in slot 0, 2, or 4. A maximum of four cold or hot backup rings are supported.

## LampSite

- This feature applies to LampSite base stations that use the BBU5900 or BBU5910.
- The gNodeB is configured with two or more BBPs.
- Both the UBBPg boards and BBPs that support distributed massive MIMO cells are compatible with Inter-Board Baseband Resource Redundancy. For details about BBPs that support distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)* in *5G RAN Feature Documentation*.
- If the inter-board trunk chain dual-fiber topology is used, there is no requirement on slots for configuring BBPs.
- The inter-board trunk chain dual-fiber topology supports only one-level RHUBs.
- The length difference between the two links of the trunk chain must not exceed 90 m.
- Except for the RHUB592x, RHUB593x, and RHUB39xx, all RHUBs that support the inter-board trunk chain topology, such as the RHUB596x, support Inter-Board Baseband Resource Redundancy. For details about support for the inter-board trunk chain topology, see *RF Unit and Topology Management*.
- All NR-capable pRRUs, except the pRRU592x, pRRU593x, and pRRU595x, support Inter-Board Baseband Resource Redundancy.

### 7.3.4 Others

None

## 7.4 Operation and Maintenance

### 7.4.1 When to Use

When a BBP fails or some resources of the BBP become unavailable, the cells served by the failed BBP cannot be dynamically reestablished on the BBP due to resource insufficiency. With Inter-Board Baseband Resource Redundancy for LTE/NR, the cells served by the failed BBP can be reestablished on another operational BBP with available resources or on a backup BBP, if available.

This ensures that services can recover automatically, reduces service interruption duration, and improves eNodeB/gNodeB reliability.

### 7.4.2 Data Configuration

#### 7.4.2.1 Data Preparation

[Table 7-1](#) describes the key parameters that must be set for configuring BBPs.

**Table 7-1** Key parameters for configuring BBPs

| Parameter Name          | Parameter ID                                           | Setting Notes                                                                                                                                       |
|-------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Cabinet No.             | <b>CN</b> parameter in the <b>ADD BRD MML</b> command  | Set this parameter to <b>0</b> .                                                                                                                    |
| Subrack No.             | <b>SRN</b> parameter in the <b>ADD BRD MML</b> command | Set this parameter to <b>0</b> .                                                                                                                    |
| Slot No.                | <b>SN</b> parameter in the <b>ADD BRD MML</b> command  | Set this parameter to <b>1</b> and <b>3</b> for two BBPs, respectively.                                                                             |
| Board Type              | <b>BT</b> parameter in the <b>ADD BRD MML</b> command  | Set this parameter to <b>LBBP</b> for an LBBP board. Set this parameter to <b>UBBP</b> for a UBBP board.                                            |
| Work Mode               | <b>BBP.WM (LTE eNodeB, 5G gNodeB)</b>                  | This parameter must be specified for an LBBP board. For LTE FDD, set this parameter to <b>FDD</b> . For LTE TDD, set this parameter to <b>TDD</b> . |
| Base Band Work Standard | <b>BBP.BBWS (LTE eNodeB, 5G gNodeB)</b>                | This parameter must be specified for a UBBP board. Set this parameter to the RATs for which the UBBP board provides baseband processing resources.  |

**Table 7-2** describes the key parameters that must be set in the **RRUCHAIN** MO.

**Table 7-2** Key parameters for configuring the RRU chain

| Parameter Name | Parameter ID                               | Setting Notes                                                                                                                                                                                                                                      |
|----------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Topo Type      | <b>RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)</b> | <ul style="list-style-type: none"> <li>Macro: Set this parameter to <b>RING</b> when the CPRI protocol is used, and to <b>LOADBALANCE</b> when the eCPRI protocol is used.</li> <li>LampSite: Set this parameter to <b>TRUNK_CHAIN</b>.</li> </ul> |
| Backup Mode    | <b>RRUCHAIN.BM (LTE eNodeB, 5G gNodeB)</b> | <ul style="list-style-type: none"> <li>Macro: Set this parameter to <b>COLD</b> or <b>HOT</b> based on the configuration of the ring topology.</li> <li>LampSite: Set this parameter to <b>COLD</b> for a trunk chain.</li> </ul>                  |

| Parameter Name   | Parameter ID                                          | Setting Notes                                       |
|------------------|-------------------------------------------------------|-----------------------------------------------------|
| Head Cabinet No. | <b>RRUCHAIN.HCN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  | Set these parameters based on hardware connections. |
| Head Subrack No. | <b>RRUCHAIN.HSRN</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) |                                                     |
| Head Slot No.    | <b>RRUCHAIN.HSN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  |                                                     |
| Head Port No.    | <b>RRUCHAIN.HPN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  |                                                     |
| Tail Cabinet No. | <b>RRUCHAIN.TCN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  |                                                     |
| Tail Subrack No. | <b>RRUCHAIN.TSRN</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) |                                                     |
| Tail Slot No.    | <b>RRUCHAIN.TSN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  |                                                     |
| Tail Port No.    | <b>RRUCHAIN.TPN</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  |                                                     |

**Table 7-3** describes the key parameters that must be set in the **RHUB** MO for LampSite base stations.

**Table 7-3** Key parameters for configuring the RHUB

| Parameter Name | Parameter ID                                     | Setting Notes                                                                                                                                              |
|----------------|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RRU Chain No.  | <b>RHUB.RCN</b> ( <i>LTE eNodeB, 5G gNodeB</i> ) | Set this parameter to the actual ring number.                                                                                                              |
| RHUB Position  | <b>RHUB.PS</b> ( <i>LTE eNodeB, 5G gNodeB</i> )  | In the inter-board trunk chain topology, the RHUB should be installed in position 0. (The inter-board trunk chain topology supports only one-level RHUBs.) |

 **NOTE**

For LampSite scenarios, the configurations of pRRUs in the inter-board trunk chain topology are the same as those of pRRUs in common topologies. For details about the topologies of pRRUs in LampSite scenarios, see section "LampSite Branch Chain/Ring Topology" in *RF Unit and Topology Management*.

**Table 7-4** describes the key parameters that must be set in the **RRU MO** for macro base stations.

**Table 7-4** Key parameters for configuring the RF unit

| Parameter Name | Parameter ID                           | Setting Notes                                                                                                                                                                                                                                                                                          |
|----------------|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RRU Chain No.  | <b>RRU.RCN (LTE eNodeB, 5G gNodeB)</b> | Set this parameter to the actual ring number.                                                                                                                                                                                                                                                          |
| RRU Position   | <b>RRU.PS (LTE eNodeB, 5G gNodeB)</b>  | In the cold backup ring topology, RRUs should be installed into different positions. For example, the two RRUs in a second-level ring should be installed in positions 0 and 1, respectively.<br><br>In the hot backup ring topology, the RRU in a first-level ring should be installed in position 0. |

## 7.4.2.2 Using MML Commands

Before using MML commands, refer to [7.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

You must remove the configurations of all boards before modifying the configurations related to subracks and cabinets. Therefore, you must perform the operations described below locally. You are not advised to use MML commands to initially configure new base stations.

Before delivery of base stations, the main equipment has been preconfigured. When MML commands are used to configure base stations, you need to consider the impact of original main equipment configurations. Before reconfiguring the base station type and cabinets, query the configurations of the main equipment by running the **LST ENODEBFUNCTION** and **LST CABINET** commands and remove configurations of all equipment. Remove and add configurations in the following order:

- Macro
  - Configuration removal: **RMV CELL (LTE)/RMV NRDUCELL (NR), RMV SECTOR, MOD RRUCHAIN, RMV RRU, RMV RRUCHAIN, RMV BRD**
  - Configuration addition: **ADD BRD, ADD RRUCHAIN, MOD RRUCHAIN, ADD RRU, MOD RRUCHAIN**
- LampSite
  - Configuration removal: **RMV NRDUCELL, RMV SECTOR, RMV RRU, RMV RRUCHAIN, RMV RHUB, RMV RRUCHAIN, RMV BRD**
  - Configuration addition: **ADD BRD, ADD RRUCHAIN, ADD RHUB, ADD RRUCHAIN, ADD RRU**

 **NOTE**

The execution of **RMV BRD** or **ADD BRD** may lead to a base station reset.

## Activation Command Examples (Macro)

**Step 1** Run the **ADD BRD** command to add a board.

**Step 2** Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

- In the case of setting up a cell in the cold backup ring topology, set **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** to **RING** and **RRUCHAIN.BM (LTE eNodeB, 5G gNodeB)** to **COLD** when configuring the RRU chain/ring.
- In the case of setting up a cell in the hot backup ring topology, set **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** to **RING** and **RRUCHAIN.BM (LTE eNodeB, 5G gNodeB)** to **HOT** when configuring the RRU chain/ring.
- In the case of setting up a cell in the load sharing topology, set **RRUCHAIN.TT (LTE eNodeB, 5G gNodeB)** to **LOADBALANCE** when configuring the RRU chain/ring.

**Step 3** Run the **MOD RRUCHAIN** command to add breakpoints on the RRU ring before adding the RRU.

**Step 4** Run the **ADD RRU** command to add an RRU.

**Step 5** Run the **MOD RRUCHAIN** command to remove breakpoints on the RRU ring after adding all the RRUs.

----End

### Scenario 1: Setting up a Cell in the Cold Backup Ring Topology

```
//Configuring the baseband working mode when a UBBP is used
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
ADD BRD: CN=0, SRN=0, SN=3, BT=UBBP, BBWS=NR-1;
//Setting the working mode to FDD for LTE FDD, and to TDD for LTE TDD when an LBBP is used. The
following uses TDD as an example.
ADD BRD: CN=0, SRN=0, SN=1, BT=LBBP, WM=TDD;
ADD BRD: CN=0, SRN=0, SN=3, BT=LBBP, WM=TDD;
ADD RRUCHAIN: RCN=0, TT=RING, BM=COLD, HSN=1, HPN=0, TSN=3, TPN=0;
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=LRRU, RS=LO, RXNUM=2, TXNUM=2;
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

### Scenario 2: Setting up a Cell in the Hot Backup Ring Topology

```
//Configuring the baseband working mode when a UBBP is used
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
```

```
ADD BRD: CN=0, SRN=0, SN=3, BT=UBBP, BBWS=NR-1;  
//Setting the working mode to FDD for LTE FDD, and to TDD for LTE TDD when an LBBP is used. The  
following uses TDD as an example.  
ADD BRD: CN=0, SRN=0, SN=1, BT=LBBP, WM=TDD;  
ADD BRD: CN=0, SRN=0, SN=3, BT=LBBP, WM=TDD;  
ADD RRUCHAIN: RCN=0, TT=RING, BM=HOT, HSN=1, HPN=0, TSN=3, TPN=0;  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=LRRU, RS=LO, RXNUM=2, TXNUM=2;  
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

### Scenario 3: Setting up a Cell in the Load Sharing Topology

```
//Configuring the baseband working mode when a UBBP is used  
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;  
ADD BRD: CN=0, SRN=0, SN=3, BT=UBBP, BBWS=NR-1;  
//Setting the working mode to FDD for LTE FDD, and to TDD for LTE TDD when an LBBP is used. The  
following uses TDD as an example.  
ADD BRD: CN=0, SRN=0, SN=1, BT=LBBP, WM=TDD;  
ADD BRD: CN=0, SRN=0, SN=3, BT=LBBP, WM=TDD;  
ADD RRUCHAIN: RCN=0, TT=LOADBALANCE, HSN=1, HPN=0, TSN=3, TPN=0;  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=NO, RXNUM=16, TXNUM=16,  
MNTMODE=NORMAL, RFDPCWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF,  
TEMPERATUREALMDetENHSW=OFF, DCUVADSW=OFF;  
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

## Activation Command Examples (LampSite)

**Step 1** Run the **ADD BRD** command to add a board.

**Step 2** Run the **ADD RRUCHAIN** command to add an RRU chain/ring.

In the case of setting up a cell in the cold backup trunk chain topology in LampSite scenarios, set **RRUCHAIN.TT** (*LTE eNodeB, 5G gNodeB*) to **TRUNK\_CHAIN** and **RRUCHAIN.BM** (*LTE eNodeB, 5G gNodeB*) to **COLD** when configuring the RRU chain/ring.

**Step 3** Run the **ADD RHUB** command to add an RHUB.

**Step 4** Run the **ADD RRUCHAIN** command to add an RHUB branch chain/ring.

**Step 5** Run the **ADD RRU** command to add an RRU.

----End

### Setting up a Cell in the Inter-Board Trunk Chain Topology

```
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;  
ADD BRD: CN=0, SRN=0, SN=3, BT=UBBP, BBWS=NR-1;  
ADD RRUCHAIN: RCN=0, TT=TRUNK_CHAIN, AT=LOCALPORT, CONNPORTNUM=2, CONNSN1=1,  
CONNPN1=0, CONNSN2=3, CONNPN2=0;  
ADD RRUCHAIN: RCN=1, TT=TRUNK_CHAIN, AT=LOCALPORT, CONNPORTNUM=2, CONNSN1=1,  
CONNPN1=1, CONNSN2=3, CONNPN2=1;  
ADD RHUB: CN=0, SRN=60, SN=0, RCN=0, PS=0;  
ADD RHUB: CN=0, SRN=70, SN=0, RCN=1, PS=0;  
ADD RRUCHAIN: RCN=61, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=60, HSN=0, HPN=0;  
ADD RRUCHAIN: RCN=62, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=60, HSN=0, HPN=1;  
ADD RRUCHAIN: RCN=71, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=70, HSN=0, HPN=6;  
ADD RRUCHAIN: RCN=72, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=70, HSN=0, HPN=7;  
ADD RRU: CN=0, SRN=61, SN=0, TP=BRANCH, RCN=61, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;  
ADD RRU: CN=0, SRN=62, SN=0, TP=BRANCH, RCN=62, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;  
ADD RRU: CN=0, SRN=71, SN=0, TP=BRANCH, RCN=71, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;  
ADD RRU: CN=0, SRN=72, SN=0, TP=BRANCH, RCN=72, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;
```

## Deactivation Command Examples

To deactivate Inter-Board Baseband Resource Redundancy, remove the preceding configurations. Add breakpoints on the RRU or RFU ring according to the MML command prompts and MML command help before removing the RRU or RFU. To remove the RRU or RFU from a ring, you need to interrupt the logical connection between the RRU or RFU and the BBP and then remove the physical connections.

### NOTE

- To remove an RRU from a ring, you need to interrupt the logical connection between the RRU or RFU and the BBP. Run the **MOD RRUCHAIN** command to add two breakpoints to the ring. The position of the first breakpoint is 0, and the position of the second breakpoint is the number of RRU cascading levels.
- Deleting main equipment (by running the **RMV BRD** command) may lead to a base station reset.

**Step 1** Run the **RMV CELL (LTE)/RMV NRDUCELL (NR)** command to remove the cell.

**Step 2** Run the **RMV SECTOR** command to remove the sector.

**Step 3** (For macro) Run the **MOD RRUCHAIN** command to add breakpoints to the RRU ring before removing the RRU.

**Step 4** Run the **RMV RRU** command to remove the RRU.

**Step 5** Run the **RMV RRUCHAIN** command to remove the RRU ring.

**Step 6** (For LampSite) Run the **RMV RHUB** command to remove the RHUB.

**Step 7** Run the **RMV BRD** command to remove the board.

----End

### eNodeB (Macro)

```
RMV CELL: LocalCellId=0;  
RMV SECTOR: SECTORID=0;  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=1;  
RMV RRU: CN=0, SRN=60, SN=0;  
RMV RRUCHAIN: RCN=0;  
RMV BRD: CN=0, SRN=0, SN=2;
```

### gNodeB (Macro)

```
RMV NRDUCELL: NrDuCellId=0;  
RMV SECTOR: SECTORID=0;  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=1;  
RMV RRU: CN=0, SRN=60, SN=0;  
RMV RRUCHAIN: RCN=0;  
RMV BRD: CN=0, SRN=0, SN=2;
```

### gNodeB (LampSite)

```
RMV NRDUCELL: NrDuCellId=0;  
RMV SECTOR: SECTORID=0;  
RMV RRU: CN=0, SRN=61, SN=0;  
RMV RRUCHAIN: RCN=61;  
RMV RHUB: CN=0, SRN=60, SN=0;  
RMV RRUCHAIN: RCN=0;  
RMV BRD: CN=0, SRN=0, SN=2;
```



### 7.4.2.3 Using the MAE-Deployment

This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

## 7.4.3 Activation Verification

### NOTE

(For macro) The implementation method on the gNodeB is the same as that on the eNodeB. The following part uses the eNodeB as an example.

### Scenario 1: Setting up a Cell in the Cold Backup Ring Topology (Macro)

On the MAE-Access or LMT, perform the following steps to check whether Inter-Board Baseband Resource Redundancy has taken effect.

**Step 1** Check the status of the cell.

Run the **DSP CELL** command to check the information about a cell. If **Cell instance state** is **Normal**, the cell has been activated. Then, check the value of **Primary BBP information** to identify the BBP on which the cell is established.

**Step 2** Check whether the cell is functional.

If the UEs can successfully access the cell, the cell is functional.

**Step 3** Run the **BLK BRD** command to block the BBP that serves the cell and to trigger the cell to be reestablished on another BBP.

**Step 4** Perform steps 1 and 2.

If the parameter values and operation results are the same as those in the preceding steps, and the cell has been reestablished on another BBP, this feature has taken effect and cell reestablishment has succeeded.

----End

### Scenario 2: Setting up a Cell in the Hot Backup Ring Topology (Macro)

On the MAE-Access or LMT, perform the following steps to check whether Inter-Board Baseband Resource Redundancy has taken effect.

**Step 1** Check the status of the cell.

Run the **DSP CELL** command to check the information about a cell. If **Cell instance state** is **Normal**, the cell has been activated. Then, check the value of **Primary BBP information** to identify the BBP on which the cell is established.

**Step 2** Check whether the cell is functional.

If the UEs can successfully access the cell and the data rates of the UDP services are stable, the cell is functional.

**Step 3** Switch the CPRI port for the cell.

Run the **BLK CPRIPORT** command to block the CPRI port on which the cell is established. Alternatively, remove the CPRI optical fiber from the BBP to trigger

the cell to use a CPRI port on another BBP. It can be observed that the cell reestablishment is not triggered but the UEs can still retain their connections, and the UDP data rates do not fluctuate greatly.

**Step 4** Check the status of the cell.

Run the **DSP CELL** command to check the information about a cell. If **Cell instance state** is **Normal**, the cell has been activated. Check the value of **Primary BBP information** to determine the BBP on which the cell is currently established. If the BBP on which the cell is established remains unchanged, this feature has taken effect.

 **NOTE**

When hot backup fails due to insufficient resources, cold backup starts. In this case, UEs served by the cell are released and the cell is reestablished on another BBP.

----End

### Scenario 3: Setting up a Cell in the Inter-Board Load Sharing Topology (Macro)

On the MAE-Access or LMT, perform the following steps to check whether Inter-Board Baseband Resource Redundancy has taken effect.

**Step 1** Check the status of the cell.

Run the **DSP CELL** command to check the information about a cell. If **Cell instance state** is **Normal**, the cell has been activated. Then, check the value of **Primary BBP information** to identify the BBP on which the cell is established.

**Step 2** Check whether the cell is functional.

If the UEs can successfully access the cell and the data rates of the UDP services are stable, the cell is functional.

**Step 3** Switch the CPRI port for the cell.

Run the **BLK CPRIPORT** command to block the CPRI port on which the cell is established. Alternatively, remove the CPRI optical fiber from the BBP to trigger cell establishment on another BBP.

**Step 4** Check the status of the cell.

Run the **DSP CELL** command to check the information about a cell. If **Cell instance state** is **Normal**, the cell has been activated. Check the value of **Primary BBP information** to determine the BBP on which the cell is currently established. If the BBP on which the cell is established remains unchanged, this feature has taken effect.

 **NOTE**

If the new BBP does not have idle resources, cell reestablishment cannot be triggered.

----End

## Scenario 4: Setting up a Cell in the Hot Backup Ring Topology (LampSite)

On the MAE-Access or LMT, perform the following steps to check whether Inter-Board Baseband Resource Redundancy has taken effect.

**Step 1** In the topology view on the MAE, check whether each RHUB is connected to the active and standby BBPs through two optical fibers. LampSite inter-board baseband resource redundancy requires that each RHUB connects to the active and standby BBPs through two optical fibers.

**Step 2** Check the baseband equipment ID of the cell.

Run the **LST NRDUCELLTRP** command and check whether the value of **Baseband Equipment ID** in the command output is **255**. If the value is **255**, the baseband equipment used by the cell is unspecified. LampSite inter-board baseband resource redundancy does not require a specified baseband equipment.

Run the **LST GNODEBPARAM** command and check whether **Auto Cell Bind Baseband Equipment Switch** is set to **OFF**. If not, run the **MOD GNODEBPARAM** command and change the value to **OFF**.

**Step 3** Check the status of the cell.

Run the **DSP NRDUCELL** command to check the information about a cell. If the value of **NR DU Cell State** is **Normal**, the cell has been activated.

**Step 4** Check the serving BBP information of the cell.

Run the **DSP NRDUCELLANTENNA** command to check the information about the serving BBP indicated by the value of **Served BBP Information**.

**Step 5** Check whether the cell is functional.

If the UEs can successfully access the cell, the cell is functional.

**Step 6** Change the BBP that serves the cell.

Run the **BLK BRD** command to block the BBP on which the cell is established, and trigger the cell to be reestablished on another BBP.

**Step 7** Perform steps 3 to 5.

If the parameter values and operation results are the same as those in the preceding steps, and the cell has been reestablished on another BBP, this feature has taken effect and cell reestablishment has succeeded.

----End

## 7.4.4 Network Monitoring

None

# 8 Intra-Board Baseband Resource Pool

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## 8.1 Principles

### 8.1.1 Overview

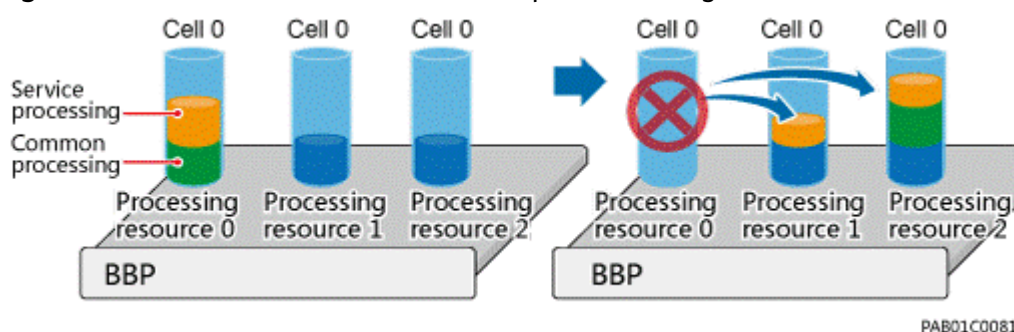
A single BBP forms a resource pool of processing capabilities. Services of cells or carriers are evenly deployed on a BBP according to the processing capabilities of each processing unit and traffic volume. That is, processing resources are aggregated into a resource pool and shared by user data of multiple cells or carriers. If a processing unit is faulty, services carried on the processing unit are interrupted and then reestablished on other processing units with available resources. If a processing unit is overloaded or the resources for the processing unit are exhausted, the base station can transfer users on the processing resource to other resources. This improves system reliability.

### 8.1.2 Intra-Board Cell-Level Resource Pool

#### Intra-Board Cell-Level Resource Pool for a Single Cell

For GSM and LTE, when a BBP allocates several resources to a single cell for load sharing (as shown in [Figure 8-1](#)), the common processing parts, for example, RACH detection, on a faulty processing resource can be transferred to other normal resources. This process ensures service continuity and quick automatic service recovery because it does not require manual intervention and generally takes less than 500 ms.

**Figure 8-1** Intra-board cell-level resource pool for a single cell



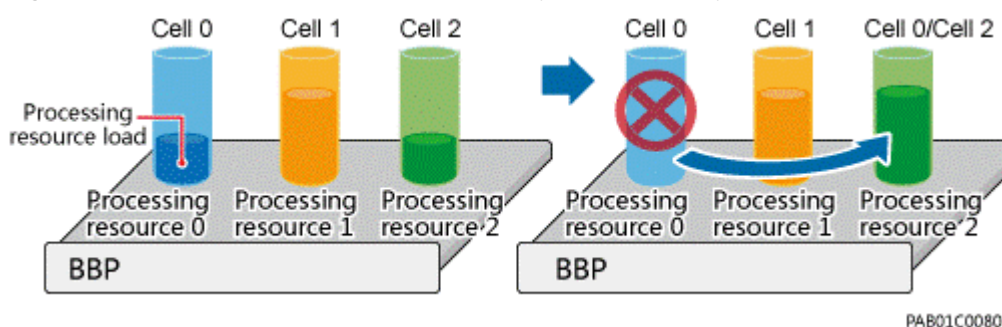
For UMTS, when a BBP allocates several resources to a single cell for load sharing, the common processing parts of the cell can use only one resource. If this resource fails, the cells served by this resource can be reestablished on other normal processing resources within 20s. This ensures service recovery.

## Intra-Board Cell-Level Resource Pool for Multiple Cells

When a BBP allocates several processing resources to multiple cells (as shown in [Figure 8-2](#)), the cells served by a faulty processing resource can be reestablished on other normal processing resources. This ensures that services are recovered after a short interruption.

The intra-board cell-level resource pool for multiple cells is supported by GSM, UMTS, LTE, and NR.

**Figure 8-2** Intra-board cell-level resource pool for multiple cells



### 8.1.3 Intra-Board User-Level Resource Pool

When multiple processing resources are available for one cell, the resources can be dynamically allocated to UEs that access the cell. When a UE accesses the cell, the BBP dynamically allocates resources to the UE based on the resource load.

The intra-board user-level resource pool is supported by UMTS, LTE, and NR, but not supported by GSM.

If a BBP in an eNodeB or gNodeB provides multiple processing resources for one cell, multiple UEs that attempt to access the cell can share these processing resources. When the cell has a small number of UEs, more processing resources can be allocated to a single UE, increasing the data rate per UE. After accessing the cell successfully, a UE cannot use other resources on the BBP.

If a BBP in a NodeB provides multiple processing resources for one cell, multiple UEs that attempt to access the cell can share these processing resources. However, a single UE can use only one processing resource. After accessing the cell successfully, the UE can use other resources on the BBP if the UE attribute must be modified.

## 8.2 Network Analysis

### 8.2.1 Benefits

None

### 8.2.2 Impacts

The impacts between this function and other functions are described in [8.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 8.3 Requirements

### 8.3.1 Licenses

None

### 8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 8.3.2.1 Prerequisite Functions

None

#### 8.3.2.2 Mutually Exclusive Functions

None

#### 8.3.2.3 Function Impacts

None

### 8.3.3 Hardware

This function is implemented in the same way on macro base stations and LampSite base stations. The BTS5929R does not support this function.

## 8.3.4 Others

None

## 8.4 Operation and Maintenance

### 8.4.1 When to Use

This feature is enabled by default.

### 8.4.2 Precautions

None

### 8.4.3 Data Configuration

#### 8.4.3.1 Data Preparation

None

#### 8.4.3.2 Using MML Commands

This feature is activated by default.

To observe this feature, see the engineering guidelines section in *Cell Management*. This feature has different requirements on cell configurations as described below.

### Scenario 1: Intra-Board Cell-Level Resource Pool

Configure three 2T2R, three 1T2R, or three 1T1R cells for Intra-Board Cell-Level Resource Pool for multiple cells.

Configure one 4T4R cell with a bandwidth of 20 MHz or one 4T4R cell with a bandwidth of 15 MHz for Intra-Board Cell-Level Resource Pool for a single cell.

### Scenario 2: Intra-Board User-Level Resource Pool

Configure one 4T4R cell with a bandwidth of 20 MHz or one 4T4R cell with a bandwidth of 15 MHz.

For details, see the engineering guidelines section in *Cell Management*.

#### 8.4.3.3 Using the MAE-Deployment

This feature is activated by default.

## 8.4.4 Activation Verification

### Scenario 1: Intra-Board Cell-Level Resource Pool

The prerequisite for activation verification is that a digital signal processor (DSP) chip on the BBP is faulty and ALM-26200 Board Hardware Fault is reported. Perform the following steps on the MAE-Access or LMT to check whether Intra-Board Baseband Resource Pool has taken effect:

 **NOTE**

If any hardware unit on the BBP is faulty, ALM-26200 Board Hardware Fault will be reported. Intra-Board Resource Pool recovers services only when the DSP chip is faulty but not in all hardware fault scenarios.

**Step 1** Run the **DSP CELL** command to check the information about a cell.

If **Cell instance state** is **Normal**, the cell has been activated. The command output shows that only one BBP serves the cell.

**Step 2** Check whether the cell is functional.

If the UEs can successfully access the cell, the cell is functional.

----End

### Scenario 2: Intra-Board User-Level Resource Pool

The prerequisites for activation verification are that the cell establishment is completed, at least two UEs have accessed the cell, and the UEs are performing uplink services. Perform the following operation on the MAE-Access or LMT to check whether Intra-Board Baseband Resource Pool has taken effect:

Run the **DSP CPUUSAGE** command with **Cabinet No.**, **Subrack No.**, and **Slot No.** set to those of the BBP to query the DSP chip loads.

The DSP chip loads depend on the number of UEs and the traffic volume of services. If the loads of three DSP chips on the BBP are similar, Intra-Board Resource Pool has taken effect.

## 8.4.5 Network Monitoring

None



# 9 Inter-BBP Active/Standby Backup (NR)

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## 9.1 Principles

When Inter-BBP Active/Standby Backup is enabled, the base station configures the active and standby relationship for two BBPs serving the same cell. In this case, the cell can recover quickly when a BBP is faulty, which improves base station reliability. Assuming that a cell is served by two BBPs configured with the active and standby relationship and the primary BBP becomes faulty, the gNodeB detects the fault and triggers a cell switchover. The cell is then reestablished on the secondary BBP. The Inter-BBP Active/Standby Backup function applies only to low-frequency scenarios.

To implement this function, a base station must be equipped with an even number of BBPs of the same model and these BBPs must work in NR mode and be installed in the same BBU.

For a base station where Inter-BBP Active/Standby Backup is configured, it is recommended that the BBPs be bound according to the following requirements. Otherwise, a switchover failure may be caused.

- All cells served by the base station must be bound to BBPs to prevent dynamic changes in the boards where the cells are deployed.
- It is recommended that cells working in the Inter-BBP Active/Standby Backup mode not share the BBP with other cells.
- It is recommended that the active and standby BBPs configured for the **NrDuCellTrp** MOs under the same **NrDuCell** MO be the same. This prevents Inter-BBP Active/Standby Backup from being affected by inter-BBP resource restrictions.

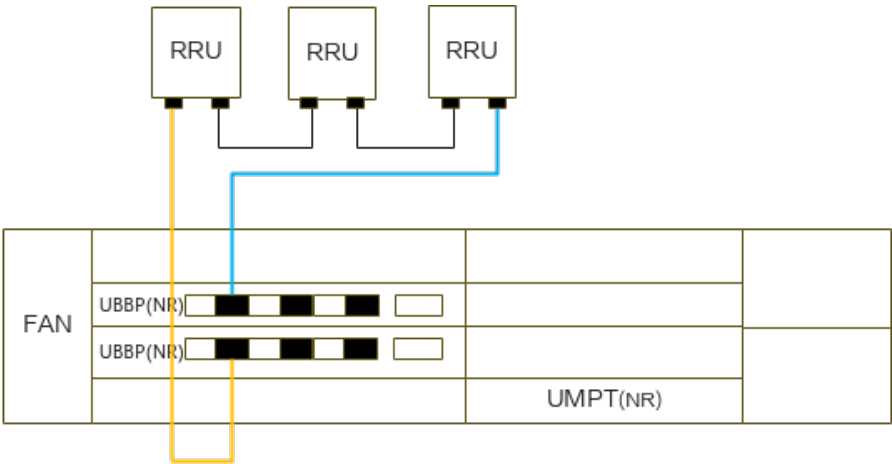
The active/standby switchover takes a longer time in the cell where Inter-BBP Active/Standby Backup is configured if all the following conditions are met: (1) The BBP for the cell where Inter-BBP Active/Standby Backup is configured or an optical fiber connected to the BBP is faulty. (2) The BBP for the cell where this function is not configured, an optical fiber connected to the BBP, or the RF module for the cell is faulty. (3) The two cells are served by the same gNodeB.

Macro (NR)

Inter-BBP Active/Standby Backup for macro base stations is supported by both common cells and distributed massive MIMO cells.

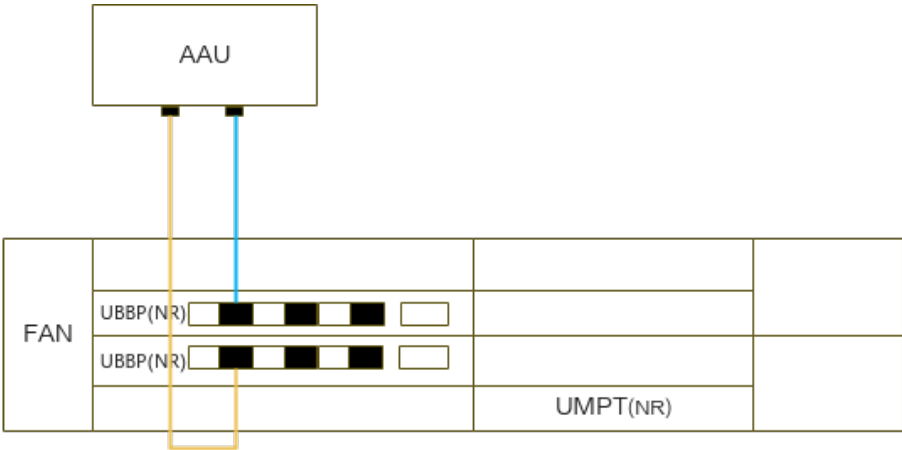
- For a common cell, Inter-BBP Active/Standby Backup supports the CPRI-based inter-board cold backup ring topology and the eCPRI-based inter-board load sharing topology (single-mode).
  - Figure 9-1** shows the CPRI-based inter-board cold backup ring topology. Two BBPs of the same type are configured for the NR network. The ring head and ring tail of an RRU chain/ring are connected to different BBPs, forming an inter-board cold backup ring topology. The blue and orange lines indicate the communication channels between the RRU and the primary and secondary BBPs, respectively.

**Figure 9-1** CPRI-based inter-board cold backup ring topology for a common cell



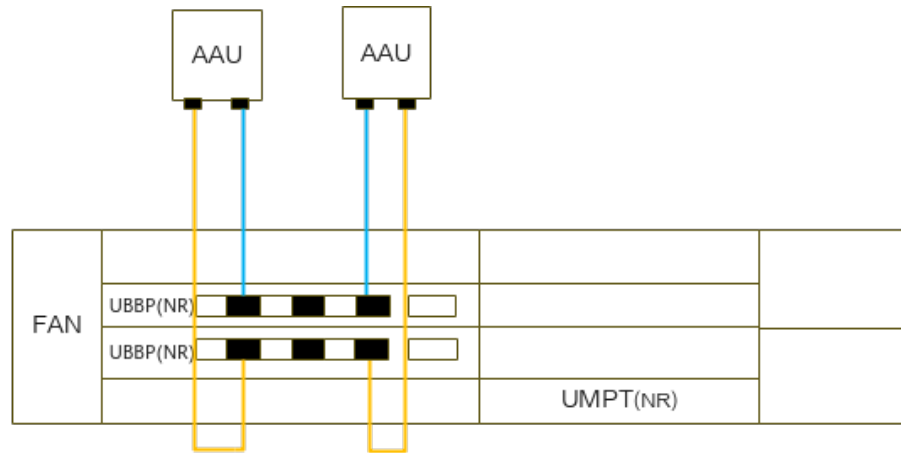
- Figure 9-2** shows the eCPRI-based inter-board load sharing topology (single-mode). An AAU (Active Antenna Unit) is connected to two BBPs to form an inter-board load sharing topology. The blue and orange lines indicate the communication channels between the AAU and the primary and secondary BBPs, respectively.

**Figure 9-2** eCPRI-based inter-board load sharing topology for a common cell



- For a distributed massive MIMO cell, Inter-BBP Active/Standby Backup supports the CPRI-based inter-board load sharing topology (single-mode). **Figure 9-3** shows the CPRI-based inter-board load sharing topology. Two BBPs of the same type are configured for the NR network. An RRU is connected to CPRI ports on two BBPs to form an inter-board load sharing topology. The blue and orange lines indicate the communication channels between the RRU and the primary and secondary BBPs, respectively. Distributed massive MIMO cells are available only in low-frequency NR TDD. For details, see *Distributed Antenna Solutions (Low-Frequency TDD)* in *5G RAN Feature Documentation*.

**Figure 9-3** CPRI-based inter-board load sharing topology for a distributed massive MIMO cell



For details about the CPRI-based inter-board cold backup ring topology, eCPRI-based inter-board load sharing topology (single-mode), and CPRI-based inter-board load sharing topology, see *RF Unit and Topology Management*. The primary and secondary BBPs are specified by the **NRDUCellTrp.BasebandEqmId** and **NRDUCellTrp.SecondaryBasebandEqmId** parameters, respectively.

Each cell is configured with a primary BBP and a secondary BBP, and is preferentially deployed on the primary BBP.

When the primary BBP or an optical fiber connected to the BBP is faulty, the gNodeB quickly detects the fault and triggers the corresponding cell switchover.

- When an optical fiber fault occurs, the cell switchover of common cells and that of distributed massive MIMO cells are as follows:
  - For a common cell, if the optical fiber is faulty, the cell using the optical fiber attempts to reestablish the cell preferentially on the original BBP. If the attempt fails, the cell is switched over to the secondary BBP. In eCPRI load sharing networking, during the cell reestablishment attempt on the original BBP, run the **DSP NRDUCELL** command. The command output displays the cause of cell unavailability, for example, insufficient eCPRI bandwidth.
  - For a distributed massive MIMO cell, if all optical fibers connected to the primary BBP serving the cell are disconnected, the cell becomes unavailable, causing the cell to be switched over to the secondary BBP.
- If the BBP is faulty, the affected cell is immediately switched over to the secondary BBP.

After the switchover, the cell is always deployed on the secondary BBP, even when the base station is reset and a cell reestablishment is required. In this case, run the **STR NRDUCELLREDEPLOY** command to manually trigger a cell reestablishment on the primary BBP.

 **NOTE**

- To implement Inter-BBP Active/Standby Backup, specifications of both the primary and secondary BBPs must meet the configuration requirements of all cells. Otherwise, the switchover may fail in some cells because the specifications of the BBPs do not meet the requirements.
- In eCPRI-based inter-board load sharing, ensure that the bandwidth of a single optical fiber meets the eCPRI bandwidth requirements of all cells served by the AAU. For details, see *eCPRI*.

## LampSite (NR TDD)

In LampSite scenarios, Inter-BBP Active/Standby Backup takes effect when all of the following conditions are met:

- The cell is a distributed massive MIMO cell.
- The inter-board trunk chain topology is used.

The inter-board trunk chain topology is specified by the **RRUCHAIN.TT** parameter, and the backup mode is specified by the **RRUCHAIN.BM** parameter. For details about support for the inter-board trunk chain topology in NR, see *RF Unit and Topology Management*. For details about requirements of a distributed massive MIMO cell, see *Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation*.

The primary and secondary BBPs are specified when the master transmission reception point (TRP) of a distributed massive MIMO cell served by a pRRU is configured using the **NRDUCellTrp** MO to implement Inter-BBP Active/Standby Backup. When the **NRDUCellTrp.TrpType** parameter is set to **MASTER\_DM\_MIMO** for a TRP, this TRP is the master TRP of a distributed massive MIMO cell. In this case, the primary and secondary BBPs are specified by the **NRDUCellTrp.BasebandEqmId** and **NRDUCellTrp.SecondaryBasebandEqmId** parameters, respectively. When the **NRDUCellTrp.TrpType** parameter is set to **SLAVE** for a TRP, this TRP is a slave TRP. Primary and secondary BBPs cannot be configured for the TRP. A distributed massive MIMO cell can be configured with only one master TRP and multiple slave TRPs.

Inter-BBP Active/Standby Backup can be applied to single-carrier and dual-carrier scenarios.

- **Figure 9-4** shows a single-carrier scenario where six pRRUs, two RHUBs, and two UBBPg(NR) boards are used. Each RHUB is connected to BBPs through two fibers in an inter-board trunk chain topology. The blue and orange lines indicate the communication channels between RHUBs and the primary and secondary BBPs, respectively. A carrier of a distributed massive MIMO cell is configured on the pRRUs. A distributed massive MIMO cell supports a maximum of 12 TRPs, and requires two UBBPg(NR) boards. The **NRDUCellTrp** MO is used to configure the master TRP for the distributed massive MIMO cell on a pRRU, as well as the active and standby relationship between the two UBBPg(NR) boards.

- **Figure 9-5** shows a dual-carrier scenario where six pRRUs, three RHUBs, and four UBBPg(NR) boards are used. Each RHUB is connected to BBPs through two fibers in an inter-board trunk chain topology. The blue and orange lines indicate the communication channels between RHUBs and the primary and secondary BBPs, respectively. Carriers of two distributed massive MIMO cells are configured on the pRRUs. A distributed massive MIMO cell supports a maximum of eight TRPs, and requires two UBBPg(NR) boards. The **NRDUCellTrp** MO is used to configure the master TRP for each distributed massive MIMO cell on a pRRU, as well as the active and standby relationship between the two UBBPg(NR) boards for each distributed massive MIMO cell.

Figure 9-4 Single-carrier scenario

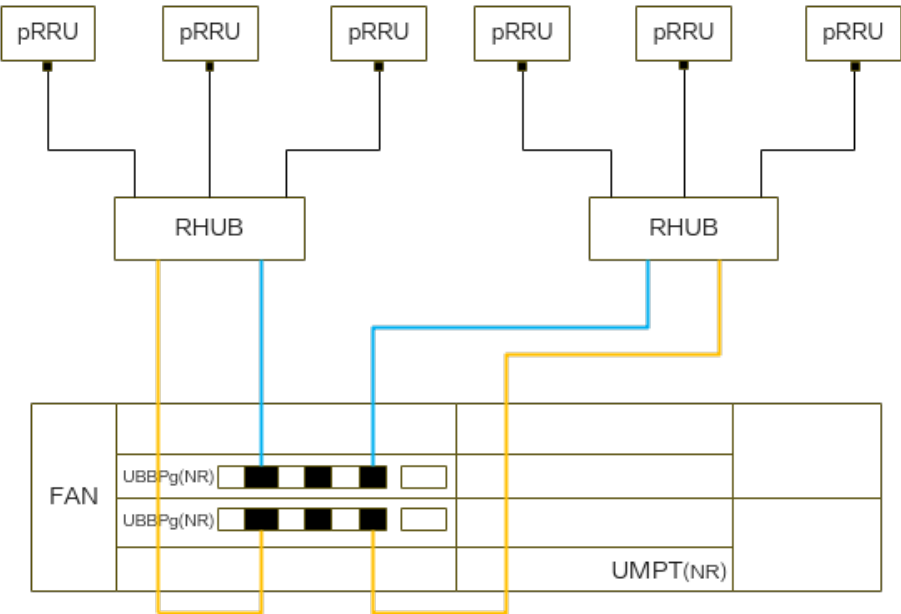
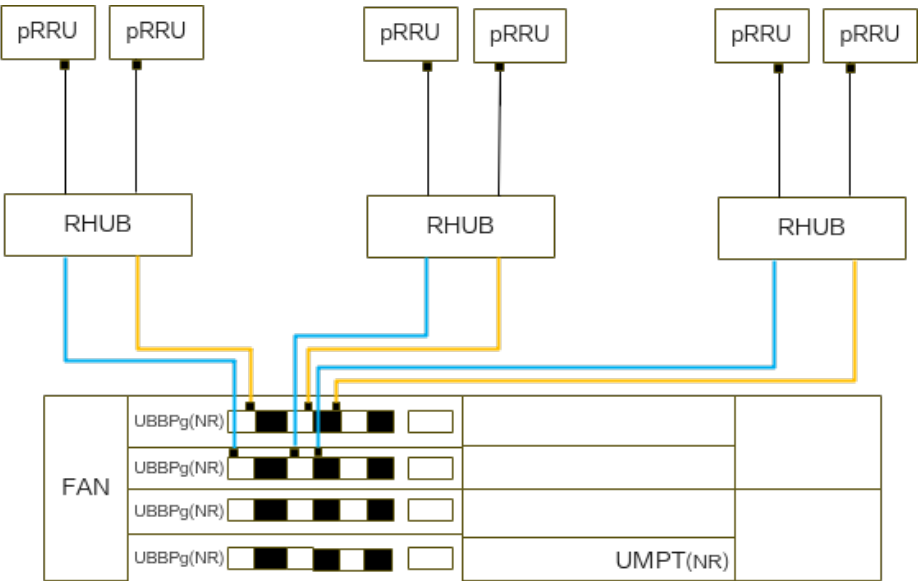


Figure 9-5 Dual-carrier scenario



Each cell is configured with a primary BBP and a secondary BBP, and the cell is preferentially deployed on the primary BBP.

When the primary BBP for a distributed massive MIMO cell or an optical fiber connected to the BBP is faulty, the gNodeB quickly detects the fault and triggers a cell switchover.

- If the optical fiber is faulty, the cell is preferentially reestablished on the original BBP.
- If the BBP is faulty, the cell is immediately switched over to the secondary BBP.

After the switchover, the cell is always deployed on the secondary BBP, even when the base station is reset and a cell reestablishment is required. To deploy the cell on the primary BBP, you need to manually trigger the cell reestablishment.

 **NOTE**

To implement Inter-BBP Active/Standby Backup, specifications of both the primary and secondary BBPs must meet the configuration requirements of the corresponding distributed massive MIMO cell. Otherwise, the switchover may fail.

## 9.2 Network Analysis

### 9.2.1 Benefits

When services of a cell are affected due to a BBP fault, the gNodeB can use the Inter-BBP Active/Standby Backup function to quickly reestablish the cell on another BBP.

### 9.2.2 Impacts

The impacts between this function and other functions are described in [9.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 9.3 Requirements

### 9.3.1 Licenses

None

### 9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 9.3.2.1 Prerequisite Functions

| RAT                                                                                                               | Function Name                         | Function Switch                                                                                       | Reference                                                                                |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Low-frequency NR TDD                                                                                              | Distributed massive MIMO <sup>a</sup> | <b>DM_MIMO_SERVICE_SWITCH</b> option of the <b>NRDUCellAlgoSwitch</b> . <b>DmMimoSwitch</b> parameter | <i>Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation</i> |
| a. Inter-BBP Active/Standby Backup requires this function in LampSite base stations, but not macro base stations. |                                       |                                                                                                       |                                                                                          |

### 9.3.2.2 Mutually Exclusive Functions

| RAT                            | Function Name                          | Function Switch                                                                                                                                                     | Reference |
|--------------------------------|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| NR FDD<br>Low-frequency NR TDD | Base Station Deep Dormancy             | <b>BB_SHUTDOWN_POWERSAVING_SW</b> option of the <b>gNodeBParam.BaseStationDeepDormancySw</b> parameter                                                              | None      |
| NR FDD<br>Low-frequency NR TDD | Baseband Processing Unit Deep Dormancy | <b>BBP_DEEP_DORMANCY_SW</b> option of the <b>gNBWPwrSavingAlgo.DormancySwitch</b> parameter is enabled and <b>BBP.PWRSVGSW</b> is set to <b>BBP_DEEP_DORMANCY</b> . | None      |

### 9.3.2.3 Function Impacts

None

## 9.3.3 Hardware

### Macro (NR)

- Inter-BBP Active/Standby Backup applies to macro base stations that use the BBU3910, BBU5910, or BBU5900.
- The gNodeB must be configured with an even number of BBPs. The models of primary and secondary BBPs must be the same.
- For details about BBPs and RF modules that support distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD) in 5G RAN Feature Documentation* (only applicable to ToB scenarios).

LampSite (NR TDD)

- Inter-BBP Active/Standby Backup applies to LampSite base stations that use the BBU5900 or BBU5910.
- The gNodeB must be configured with an even number of BBPs. The models of primary and secondary BBPs must be the same.
- Both the UBBPg boards and BBPs that support distributed massive MIMO cells support Inter-BBP Active/Standby Backup. For details about BBPs that support distributed massive MIMO cells, see *Distributed Antenna Solutions (Low-Frequency TDD)* in *5G RAN Feature Documentation*.
- If the inter-board trunk chain dual-fiber topology is used, there is no requirement on slots for configuring BBPs.
- The inter-board trunk chain dual-fiber topology supports only one-level RHUBs.
- The length difference between the two fibers connected to the same RHUB must not exceed 90 m.
- Except for the RHUB592x, RHUB593x, and RHUB39xx, all RHUBs that support the inter-board trunk chain topology support Inter-BBP Active/Standby Backup. For details about support for the inter-board trunk chain topology, see *RF Unit and Topology Management*.
- All NR-capable pRRUs, except the pRRU592x, pRRU593x, and pRRU595x, support Inter-BBP Active/Standby Backup.

9.3.4 Others

None

9.4 Operation and Maintenance

9.4.1 When to Use

Inter-BBP Active/Standby Backup ensures that services of a cell recover quickly, reduces service interruption duration, and improves reliability.

9.4.2 Data Configuration

9.4.2.1 Data Preparation

[Table 9-1](#) describes the key parameters that must be set in the **BaseBandEqm** MO to add secondary baseband equipment.

Table 9-1 Key parameters for configuring baseband equipment

| Parameter Name        | Parameter ID                      | Setting Notes                                  |
|-----------------------|-----------------------------------|------------------------------------------------|
| Baseband Equipment ID | <b>BaseBandEqm.BASEBANDEQ MID</b> | Two baseband equipment IDs must be configured. |



| Parameter Name                | Parameter ID                            | Setting Notes                                                   |
|-------------------------------|-----------------------------------------|-----------------------------------------------------------------|
| Baseband Equipment Type       | <b>BaseBandEqm.BASEBANDEQ<br/>MTYPE</b> | The value <b>ULDL</b> is recommended.                           |
| Cabinet No. of Process Unit 1 | <b>BaseBandEqm.SN1</b>                  | Set this parameter to the number of the slot that houses BBP 1. |

**Table 9-2** describes the key parameters that must be set in the **NRDUCellTrp** MO to bind secondary baseband equipment.

**Table 9-2** Key parameters for configuring the NR DU cell TRP

| Parameter Name                  | Parameter ID                              | Setting Notes                                                                                                                                                           |
|---------------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NR DU Cell TRP ID               | <b>NRDUCellTrp.NrDuCellTrpld</b>          | Set this parameter based on the network plan.                                                                                                                           |
| TRP Type                        | <b>NRDUCellTrp.TrpType</b>                | Macro: Use the default value <b>DEFAULT</b> .<br>LampSite: Secondary baseband equipment ID can be configured only when this parameter is set to <b>MASTER_DM_MIMO</b> . |
| NR DU Cell ID                   | <b>NRDUCellTrp.NrDuCellId</b>             | Set this parameter based on the network plan.                                                                                                                           |
| Baseband Equipment ID           | <b>NRDUCellTrp.BasebandEqmId</b>          | It is recommended that this parameter be set to the ID of the primary baseband equipment for the NR DU cell TRP.                                                        |
| Secondary Baseband Equipment ID | <b>NRDUCellTrp.SecondaryBasebandEqmId</b> | It is recommended that this parameter be set to the ID of the secondary baseband equipment for the NR DU cell TRP.                                                      |

**Table 9-3** describes the key parameters that must be set in an **RRUCHAIN** MO.

**Table 9-3** Key parameters for configuring the RRU chain

| Parameter Name   | Parameter ID             | Setting Notes                                                                                                                                                                                                                                                                       |
|------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Topo Type        | <b>RRUCHAIN.TT</b>       | Macro: Set this parameter to <b>RING</b> for common cells in CPRI networking and to <b>LOADBALANCE</b> for common cells in eCPRI networking. Set it to <b>LOADBALANCE</b> for distributed massive MIMO cells.<br><br>LampSite: Set this parameter to <b>TRUNK_CHAIN</b> .           |
| Backup Mode      | <b>RRUCHAIN.BM</b>       | Macro: Set this parameter to <b>COLD</b> when the CPRI-based inter-board cold backup ring topology is used. This parameter is not required for the inter-board load sharing topology.<br><br>LampSite: This parameter does not need to be set for inter-board trunk chain topology. |
| Protocol Type    | <b>RRUCHAIN.PROTOCOL</b> | Set this parameter to <b>CPRI</b> or <b>eCPRI</b> based on the protocol type of the chain/ring. If the protocol type of the chain/ring is the same as that of the <b>EQUIPMENT MO</b> , you can use the default value <b>DEFAULT</b> .                                              |
| Head Cabinet No. | <b>RRUCHAIN.HCN</b>      | Set these parameters based on hardware connections.                                                                                                                                                                                                                                 |
| Head Subrack No. | <b>RRUCHAIN.HSRN</b>     |                                                                                                                                                                                                                                                                                     |
| Head Slot No.    | <b>RRUCHAIN.HSN</b>      |                                                                                                                                                                                                                                                                                     |
| Head Port No.    | <b>RRUCHAIN.HPN</b>      |                                                                                                                                                                                                                                                                                     |
| Tail Cabinet No. | <b>RRUCHAIN.TCN</b>      |                                                                                                                                                                                                                                                                                     |
| Tail Subrack No. | <b>RRUCHAIN.TSRN</b>     |                                                                                                                                                                                                                                                                                     |
| Tail Slot No.    | <b>RRUCHAIN.TSN</b>      |                                                                                                                                                                                                                                                                                     |
| Tail Port No.    | <b>RRUCHAIN.TPN</b>      |                                                                                                                                                                                                                                                                                     |

### 9.4.2.2 Using MML Commands

Before using MML commands, refer to [9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples (Macro)

Scenario 1: Inter-BBP Active/Standby Backup is enabled for common cells using the CPRI-based inter-board cold backup ring topology.

**Step 1** Run the **ADD BRD** command to add a board.

```
ADD BRD: CN=0, SRN=0, SN=0, BT=UBBP, BBWS=NR-1;  
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
```

**Step 2** Run the **ADD RRUCHAIN**, **ADD RRU**, and **MOD RRUCHAIN** commands to add an RRU chain/ring, add an RRU/AAU, and modify the RRU chain/ring.

```
//Adding an RRU chain/ring, adding an RRU, and modifying the RRU chain/ring in CPRI networking  
//Adding an RRU chain/ring  
ADD RRUCHAIN: RCN=0, TT=RING, BM=COLD, HSRN=0, HSN=0, HPN=0, TSRN=0, TSN=1, TPN=0,  
PROTOCOL=CPRI, CR=AUTO, USERDEFERATENEGOSW=OFF;  
//Configuring breakpoints  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;  
//Adding an RRU  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=NO, RXNUM=4, TXNUM=4,  
MNTMODE=NORMAL, RFDPCWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF;  
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=0, PS=1, RT=MRRU, RS=NO, RXNUM=4, TXNUM=4,  
MNTMODE=NORMAL, RFDPCWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF;  
ADD RRU: CN=0, SRN=62, SN=0, TP=TRUNK, RCN=0, PS=2, RT=MRRU, RS=NO, RXNUM=4, TXNUM=4,  
MNTMODE=NORMAL, RFDPCWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF;  
//Removing the breakpoints  
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

**Step 3** Run the **ADD BASEBANDEQM** command to add the baseband equipment.

```
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMType=ULDL, UMTSDEMMODE=NULL, SN1=0;  
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMType=ULDL, UMTSDEMMODE=NULL, SN1=1;
```

**Step 4** Run the **ADD SECTOR** and **ADD NRDUCELL** commands to add sectors and NR DU cells.

```
//Adding a sector (CPRI)  
ADD SECTOR: SECTORID=60, ANTNUM=4, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0,  
ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60, ANT3SN=0, ANT3N=R0C, ANT4CN=0,  
ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE, SECTOREQMID=60;  
//Adding an NR DU cell  
ADD NRDUCELL: NrDuCellId=60, NrDuCellName="DUCELL60", DuplexMode=CELL_FDD, CellId=60,  
PhysicalCellId=22, FrequencyBand=N1, DINarfcn=424000, UIBandwidth=CELL_BW_20M,  
DLBandwidth=CELL_BW_20M, TrackingAreaId=0, TaOffset=0TC, SsbFreqPos=5285, SsbPeriod=MS80,  
Sib1Period=MS80, LogicalRootSequenceIndex=100, SmtcPeriod=MS80, SmtcDuration=MS3,  
CustomizedBwConfigInd=NOT_CONFIG, DeployPosition=DEPLOY_INTEGRATED;
```

**Step 5** Run the **ADD NRDUCELLTRP** command to add an NR DU cell TRP.

```
//Adding an NR DU cell TRP and specifying the primary and secondary baseband equipment (CPRI)  
ADD NRDUCELLTRP: NrDuCellTrpId=60, TrpType=DEFAULT, NrDuCellId=60, TxRxMode=4T4R,  
BasebandEqmId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=380,  
CpriCompression=2_COMPRESSION, FrAndDuplexMode=FDD, SecondaryBasebandEqmId=1;
```

- Step 6** Run the **ADD NRDUCELLCOVERAGE** and **ADD NRCELL** commands to add NR DU cell coverage areas and an NR cell.

```
//Adding NR DU cell coverage areas
ADD NRDUCELLCOVERAGE: NrDuCellTrpld=60, NrDuCellCoverageId=0, SectorEqmId=60;
//Adding an NR cell
ADD NRCELL: NrCellId=60, CellName="NRCELL", CellId=60, FrequencyBand=N1, DuplexMode=CELL_FDD;
```

- Step 7** Run the **ACT NRCELL** command to activate the NR cell.

```
//Activating the NR cell
ACT NRCELL: NrCellId=60;
```

----End

Scenario 2: Inter-BBP Active/Standby Backup is enabled for common cells using the eCPRI-based inter-board load sharing topology (single-mode).

- Step 1** Run the **ADD BRD** command to add boards.

```
ADD BRD: CN=0, SRN=0, SN=0, BT=UBBP, BBWS=NR-1;
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
```

- Step 2** Run the **ADD RRUCHAIN**, **ADD RRU**, and **MOD RRUCHAIN** commands to add an RRU chain/ring, add an RRU/AAU, and modify the RRU chain/ring.

```
//Adding an RRU chain/ring and an AAU, and modifying the RRU chain/ring in eCPRI networking
//Adding an RRU chain/ring
ADD RRUCHAIN: RCN=0, TT=LOADBALANCE, HSRN=0, HSN=3, HPN=3, TSRN=0, TSN=3, TPN=4,
PROTOCOL=eCPRI, CR=AUTO, USERDEFERATENEGOSW=OFF;
//Configuring breakpoints
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;
//Adding an AAU
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=NO, RXNUM=16, TXNUM=16,
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF,
TEMPERATUREALMDetenHSW=OFF, DCUVADSW=OFF;
//Removing the breakpoints
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;
```

- Step 3** Run the **ADD BASEBANDEQM** command to add the baseband equipment.

```
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMType=ULDL, UMTSDEMMode=NULL, SN1=0;
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMType=ULDL, UMTSDEMMode=NULL, SN1=1;
```

- Step 4** Run the **ADD SECTOR** and **ADD NRDUCELL** commands to add a sector and an NR DU cell.

```
//Adding a sector (eCPRI)
ADD SECTOR: SECTORID=60, ANTNUM=0, CREATESECTOREQM=FALSE;
ADD SECTOREQM: SECTOREQMID=60, SECTORID=60, ANTCFGMODE=BEAM, RRUNUM=1, RRUCN=0,
RRUSRN=60, RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
//Adding an NR DU cell
ADD NRDUCELL: NrDuCellId=60, NrDuCellName="DUCELL60", DuplexMode=CELL_FDD, CellId=60,
PhysicalCellId=22, FrequencyBand=N1, DINarfcn=424000, ULBandwidth=CELL_BW_20M,
DLBandwidth=CELL_BW_20M, TrackingAreaId=0, TaOffset=0TC, SsbFreqPos=5285, SsbPeriod=MS80,
Sib1Period=MS80, LogicalRootSequenceIndex=100, SmtcPeriod=MS80, SmtcDuration=MS3,
CustomizedBwConfigInd=NOT_CONFIG, DeployPosition=DEPLOY_INTEGRATED;
```

- Step 5** Run the **ADD NRDUCELLTRP** command to add an NR DU cell TRP.

```
//Adding an NR DU cell TRP and specifying the primary and secondary baseband equipment (eCPRI)
ADD NRDUCELLTRP: NrDuCellTrpld=60, TrpType=DEFAULT, NrDuCellId=60, TxRxMode=32T32R,
BasebandEqmId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=380,
CpriCompression=2_COMPRESSION, FrAndDuplexMode=FDD, SecondaryBasebandEqmId=1;
```

- Step 6** Run the **ADD NRDUCELLCOVERAGE** and **ADD NRCELL** commands to add NR DU cell coverage areas and an NR cell.

```
//Adding NR DU cell coverage areas
ADD NRDUCELLCOVERAGE: NrDuCellTrpld=60, NrDuCellCoverageId=0, SectorEqmId=60;
//Adding an NR cell
ADD NRCELL: NrCellId=60, CellName="NRCELL", CellId=60, FrequencyBand=N1, DuplexMode=CELL_FDD;
```

**Step 7** Run the **ACT NRCELL** command to activate the NR cell.

```
//Activating the NR cell  
ACT NRCELL: NrCellId=60;
```

----End

Scenario 3: Inter-BBP Active/Standby Backup (NR TDD) is enabled for distributed massive MIMO cells using the CPRI-based inter-board load sharing topology (single-mode).

**Step 1** Run the **ADD BRD** command to add boards.

```
ADD BRD: CN=0, SRN=0, SN=0, BT=UBBP, BBWS=NR-1;  
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
```

**Step 2** Run the **ADD RRUCHAIN**, **ADD RRU**, and **MOD RRUCHAIN** commands to add an RRU chain/ring, add an AAU, and modify the RRU chain/ring.

```
//Adding RRU chains/rings  
ADD RRUCHAIN: RCN=0, TT=LOADBALANCE, HSRN=0, HSN=0, HPN=0, TSRN=0, TSN=1, TPN=0,  
PROTOCOL=CPRI, CR=AUTO, USERDEFERATENEGOSW=OFF;  
ADD RRUCHAIN: RCN=1, TT=LOADBALANCE, HSRN=0, HSN=0, HPN=1, TSRN=0, TSN=1, TPN=1,  
PROTOCOL=CPRI, CR=AUTO, USERDEFERATENEGOSW=OFF;  
//Configuring breakpoints  
MOD RRUCHAIN: RCN=0, BRKPOS1=0, BRKPOS2=0;  
MOD RRUCHAIN: RCN=1, BRKPOS1=0, BRKPOS2=0;  
//Adding AAUs  
ADD RRU: CN=0, SRN=60, SN=0, TP=TRUNK, RCN=0, PS=0, RT=MRRU, RS=NO, RXNUM=4, TXNUM=4,  
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF;  
ADD RRU: CN=0, SRN=61, SN=0, TP=TRUNK, RCN=1, PS=0, RT=MRRU, RS=NO, RXNUM=4, TXNUM=4,  
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDetectSW=OFF, DORMANCYSW=OFF;  
//Removing the breakpoints  
MOD RRUCHAIN: RCN=0, BRKPOS1=255, BRKPOS2=255;  
MOD RRUCHAIN: RCN=1, BRKPOS1=255, BRKPOS2=255;
```

**Step 3** Run the **ADD BASEBANDEQM** command to add the baseband equipment.

```
ADD BASEBANDEQM: BASEBANDEQMID=0, BASEBANDEQMType=ULDL, UMTSDEMMode=NULL, SN1=0;  
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMType=ULDL, UMTSDEMMode=NULL, SN1=1;
```

**Step 4** Run the **ADD SECTOR** and **ADD NRDUCELL** commands to add sectors and NR DU cells.

```
//Adding sectors (CPRI)  
ADD SECTOR: SECTORID=60, SECNAME="sector-60", ANTNUM=4, ANT1CN=0, ANT1SRN=60, ANT1SN=0,  
ANT1N=R0A, ANT2CN=0, ANT2SRN=60, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=60, ANT3SN=0,  
ANT3N=R0C, ANT4CN=0, ANT4SRN=60, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,  
SECTOREQMID=60;  
ADD SECTOR: SECTORID=61, SECNAME="sector-61", ANTNUM=4, ANT1CN=0, ANT1SRN=61, ANT1SN=0,  
ANT1N=R0A, ANT2CN=0, ANT2SRN=61, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=61, ANT3SN=0,  
ANT3N=R0C, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,  
SECTOREQMID=61;  
//Adding an NR DU cell  
ADD NRDUCELL: NrDuCellId=60, NrDuCellName="DUCELL60", DuplexMode=CELL_TDD, CellId=60,  
PhysicalCellId=22, FrequencyBand=N41, DlnArfcn=513000, UlBandwidth=CELL_BW_100M,  
DlBandwidth=CELL_BW_100M, SlotAssignment=4_1_DDSU, SlotStructure=SS2, TrackingAreaId=0,  
SsbFreqPos=6486, LogicalRootSequenceIndex=0, CustomizedBwConfigInd=NOT_CONFIG,  
DeployPosition=DEPLOY_INTEGRATED;
```

**Step 5** Run the **ADD NRDUCELLTRP** command to add an NR DU cell TRP.

```
//Adding an NR DU cell TRP and specifying the primary and secondary baseband equipment (CPRI)  
ADD NRDUCELLTRP: NrDuCellTrpId=60, TrpType=MASTER_DM_MIMO, NrDuCellId=60, TxRxMode=4T4R,  
BasebandEqmId=0, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300,  
CpriCompression=3DOT2_COMPRESSION, FrAndDuplexMode=FR1_TDD, SecondaryBasebandEqmId=1;  
//Configuring a slave TRP  
ADD NRDUCELLTRP: NrDuCellTrpId=61, TrpType=SLAVE, MasterTrpId=60, TxRxMode=4T4R,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=300;
```

**Step 6** Run the **ADD NRDUCELLCOVERAGE** and **ADD NRCELL** commands to add NR DU cell coverage areas and an NR cell.

```
//Adding NR DU cell coverage areas
ADD NRDUCELLCOVERAGE: NrDuCellTrpld=60, NrDuCellCoverageId=0, SectorEqmId=60;
ADD NRDUCELLCOVERAGE: NrDuCellTrpld=61, NrDuCellCoverageId=1, SectorEqmId=61;
//Adding an NR cell
ADD NRCELL: NrCellId=60, CellName="NRCELL", CellId=60, FrequencyBand=N41, DuplexMode=CELL_TDD;
```

**Step 7** Run the **ACT NRCELL** command to activate the NR cell.

```
//Activating the NR cell
ACT NRCELL: NrCellId=60;
```

----End

## Activation Command Examples (LampSite)

**Step 1** Run the **ADD BRD** command to add a board.

```
ADD BRD: CN=0, SRN=0, SN=1, BT=UBBP, BBWS=NR-1;
ADD BRD: CN=0, SRN=0, SN=3, BT=UBBP, BBWS=NR-1;
```

**Step 2** Run the **ADD RRUCHAIN**, **ADD RHUB**, and **ADD RRU** commands to add main and branch chains, RHUBs, and pRRUs, respectively.

```
//Adding main chains with Topo Type set to TRUNK_CHAIN
ADD RRUCHAIN: RCN=0, TT=TRUNK_CHAIN, AT=LOCALPORT, CONNPORTNUM=2, CONNSN1=1,
CONNNPN1=0, CONNSN2=3, CONNNPN2=0;
ADD RRUCHAIN: RCN=1, TT=TRUNK_CHAIN, AT=LOCALPORT, CONNPORTNUM=2, CONNSN1=1,
CONNNPN1=1, CONNSN2=3, CONNNPN2=1;
//Adding RHUBs
ADD RHUB: CN=0, SRN=60, SN=0, RCN=0, PS=0;
ADD RHUB: CN=0, SRN=70, SN=0, RCN=1, PS=0;
//Adding branch chains
ADD RRUCHAIN: RCN=61, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=60, HSN=0, HPN=0;
ADD RRUCHAIN: RCN=62, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=60, HSN=0, HPN=1;
ADD RRUCHAIN: RCN=71, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=70, HSN=0, HPN=6;
ADD RRUCHAIN: RCN=72, TT=CHAIN, BM=COLD, AT=LOCALPORT, HSRN=70, HSN=0, HPN=7;
//Adding pRRUs
ADD RRU: CN=0, SRN=61, SN=0, TP=BRANCH, RCN=61, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;
ADD RRU: CN=0, SRN=62, SN=0, TP=BRANCH, RCN=62, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;
ADD RRU: CN=0, SRN=71, SN=0, TP=BRANCH, RCN=71, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;
ADD RRU: CN=0, SRN=72, SN=0, TP=BRANCH, RCN=72, PS=0, RT=MPMU, RS=LN, RXNUM=4, TXNUM=4;
```

**Step 3** Run the **ADD BASEBANDEQM** command to add the baseband equipment.

```
ADD BASEBANDEQM: BASEBANDEQMID=1, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=1;
ADD BASEBANDEQM: BASEBANDEQMID=3, BASEBANDEQMTYPE=ULDL, UMTSDEMMode=NULL, SN1=3;
```

**Step 4** Run the **ADD SECTOR** and **ADD NRDUCELL** commands to add sectors and NR DU cells.

```
//Adding sectors
ADD SECTOR: SECTORID=61, SECNAME="sector-61", ANTNUM=4, ANT1CN=0, ANT1SRN=61, ANT1SN=0,
ANT1N=R0A, ANT2CN=0, ANT2SRN=61, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=61, ANT3SN=0,
ANT3N=R0C, ANT4CN=0, ANT4SRN=61, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=61;
ADD SECTOR: SECTORID=62, SECNAME="sector-62", ANTNUM=4, ANT1CN=0, ANT1SRN=62, ANT1SN=0,
ANT1N=R0A, ANT2CN=0, ANT2SRN=62, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=62, ANT3SN=0,
ANT3N=R0C, ANT4CN=0, ANT4SRN=62, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=62;
ADD SECTOR: SECTORID=71, SECNAME="sector-71", ANTNUM=4, ANT1CN=0, ANT1SRN=71, ANT1SN=0,
ANT1N=R0A, ANT2CN=0, ANT2SRN=71, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=71, ANT3SN=0,
ANT3N=R0C, ANT4CN=0, ANT4SRN=71, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=71;
ADD SECTOR: SECTORID=72, SECNAME="sector-72", ANTNUM=4, ANT1CN=0, ANT1SRN=72, ANT1SN=0,
ANT1N=R0A, ANT2CN=0, ANT2SRN=72, ANT2SN=0, ANT2N=R0B, ANT3CN=0, ANT3SRN=72, ANT3SN=0,
ANT3N=R0C, ANT4CN=0, ANT4SRN=72, ANT4SN=0, ANT4N=R0D, CREATESECTOREQM=TRUE,
SECTOREQMID=72;
ADD NRDUCELL: NrDuCellId=0, NrDuCellName="2.6G_100M_8:2", DuplexMode=CELL_TDD, CellId=0,
PhysicalCellId=1, FrequencyBand=N41, ULNarfcn=525000, DLNarfcn=525000, ULBandwidth=CELL_BW_100M,
DLBandwidth=CELL_BW_100M, CellRadius=800, SlotAssignment=8_2_DDDDDDDSUU, SlotStructure=SS54,
```

```
LampSiteCellFlag=YES, TrackingAreaId=0, SsbFreqPos=6561, LogicalRootSequenceIndex=1,  
DeployPosition=DEPLOY_INTEGRATED;
```

**Step 5** Run the **ADD NRDUCELLTRP** command to add an NR DU cell TRP.

```
//Configuring the master TRP for the distributed massive MIMO cell and specifying the primary and  
secondary baseband equipment  
ADD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=MASTER_DM_MIMO, NrDuCellId=0, TxRxMode=4T4R,  
BasebandEqmId=1, PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240,  
CpriCompression=3DOT2_COMPRESSION, SecondaryBasebandEqmId=3;  
//Configuring slave TRPs  
ADD NRDUCELLTRP: NrDuCellTrpId=1, TrpType=SLAVE, MasterTrpId=0, TxRxMode=4T4R,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240;  
ADD NRDUCELLTRP: NrDuCellTrpId=2, TrpType=SLAVE, MasterTrpId=0, TxRxMode=4T4R,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240;  
ADD NRDUCELLTRP: NrDuCellTrpId=3, TrpType=SLAVE, MasterTrpId=0, TxRxMode=4T4R,  
PowerConfigMode=TRANSMIT_POWER, MaxTransmitPower=240;
```

**Step 6** Run the **ADD NRDUCELLCOVERAGE** and **ADD NRCELL** commands to add NR DU cell coverage areas and an NR cell.

```
//Adding NR DU cell coverage areas  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=0, NrDuCellCoverageId=61, SectorEqmId=61;  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=1, NrDuCellCoverageId=62, SectorEqmId=62;  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=2, NrDuCellCoverageId=71, SectorEqmId=71;  
ADD NRDUCELLCOVERAGE: NrDuCellTrpId=3, NrDuCellCoverageId=72, SectorEqmId=72;  
//Adding an NR cell  
ADD NRCELL: NrCellId=0, CellName="2.6G_100M_8:2", CellId=0, FrequencyBand=N41,  
DuplexMode=CELL_TDD;
```

**Step 7** Run the **ACT NRCELL** command to activate the NR cell.

```
//Activating the NR cell  
ACT NRCELL: NrCellId=0;
```

----End

## Command Examples for Triggering a Cell Reestablishment on the Primary BBP

**Step 1** If the cell has been switched over to the secondary BBP, run the **STR NRDUCELLREDEPLOY** command to manually trigger a reestablishment on the primary BBP.

```
STR NRDUCELLREDEPLOY: DeployType=ALL;
```

----End

## Deactivation Command Examples (Macro)

**Step 1** Run the **MOD NRDUCELLTRP** command to remove the binding relationship between the NR DU cell TRP and the secondary BBP.

```
MOD NRDUCELLTRP: NrDuCellTrpId=60, TrpType=DEFAULT, SecondaryBasebandEqmId=255;
```

----End

## Deactivation Command Examples (LampSite)

**Step 1** Run the **MOD NRDUCELLTRP** command to remove the binding relationship between the NR DU cell TRP and the secondary BBP.

```
//Removing the binding relationship between the TRP of the distributed massive MIMO cell and the  
secondary BBP  
MOD NRDUCELLTRP: NrDuCellTrpId=0, TrpType=MASTER_DM_MIMO, SecondaryBasebandEqmId=255;
```

----End

### 9.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 9.4.3 Activation Verification

On the MAE-Access or LMT, perform the following steps to check whether Inter-BBP Active/Standby Backup has taken effect.

**Step 1** Check the status of the cell.

Run the **DSP NRCELL** command to check the cell status. If **NR Cell State** is **Normal**, the cell has been activated.

Run the **DSP NRDUCELLTRP** command to list the configuration of TRPs for the NR DU cell. The value of **Serving BBP Information** indicates the information about the serving BBP of the NR DU cell TRP.

**Step 2** Check whether the cell is functional.

If the UEs can successfully access the cell, the cell is functional.

**Step 3** Change the BBP that serves the cell.

Run the **BLK BRD** command to block the BBP on which the cell is established, triggering cell reestablishment.

**Step 4** Perform steps 1 and 2.

If the parameter values and operation results are the same as those in the preceding steps, and the serving BBP for the NR DU cell TRP changes to the secondary BBP, the function has taken effect and cell reestablishment has succeeded.

----End

## 9.4.4 Network Monitoring

None

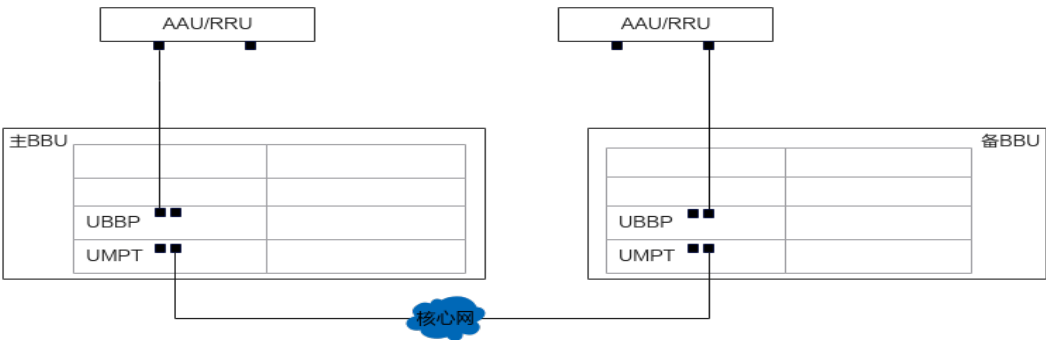


# 10 Dual-Plane Function in Intra-Frequency Dual-Network Redundancy Networking in ToB Scenarios (NR TDD)

## 10.1 Principles

In 5GtoB scenarios, if a single point of failure (SPOF) occurs on a board (baseband processing unit or main control board) in a BBU, an RF module (AAU or RRU), or a transmission link, cells will be unavailable. In this case, services in the coverage area will be abnormal or interrupted. To solve this problem, "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is introduced. [Figure 10-1](#) shows the typical networking when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" applies.

**Figure 10-1** Dual-plane function in intra-frequency dual-network redundancy networking



- The boards in the same slots of the active and standby BBUs must be of the same model, and so must the RF modules connected to the BBUs. The RF modules must share the same coverage area.
- The active and standby BBUs share the same software version and are connected to the same core network.

- The primary and backup gNodeBs must be connected through the Xn interface. The primary gNodeB is configured with the active BBU, and the backup gNodeB is configured with the standby BBU.

"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is controlled by the **DUAL\_PLANE\_TOB\_SW** option of the **gNBRedundancy.gNBReliabilitySwitch** parameter. Backup NR cells can be added on both the primary and backup gNodeBs using the **NRCELLREDUNDANCY** MO. This function takes effect only if the following conditions are met: (1) The preceding option is selected on both the primary and backup gNodeBs. (2) Both primary and standby cells are activated. If the backup gNodeB cannot obtain heartbeat messages from the primary gNodeB for a period longer than the transmission fault detection duration specified by the **gNBRedundancy.FaultDetectionDuration** parameter, a fast switchover to the standby BBU is triggered. If cell unavailability due to a fault on the primary gNodeB persists for a period longer than the time-to-trigger for reporting a fault specified by the **gNBRedundancy.TimeToTriggerForFaultRpt** parameter, a fast switchover to the standby BBU is triggered.

After this function is enabled, if an SPOF occurs on a board in the active BBU, an RF module, or a transmission link on the active BBU side, which causes cell unavailability, while the standby BBU works properly, the standby BBU and the RF module connected to it can be used to quickly activate a standby cell for radio service restoration. Besides device and transmission link faults, any of the following operations performed on the primary gNodeB triggers a fast switchover of the faulty cell to the standby BBU:

- Manually remove a component (such as a board, transmission optical fiber, or power cable).
- Perform a high-risk O&M operation that will trigger cell reset and cause the cell to be unavailable for a long time (such as blocking, powering off, or resetting a board, blocking a transmission link, and performing upgrade and reconstruction).
- Run the **SWP NRCELLREDUNDANCY** command.

The transmission-link-fault-triggered switchover can be sped up by configuring the **SCTPTEMPLATE** MO and setting **GUPTRANSPARA.UPSTATFASW** to **ENABLE**.

To ensure the success rate of the active/standby switchover and no impact on the performance during the switchover, a check can be performed on the consistency of the configurations or licenses between the active and standby BBUs. Such a check can be performed only if **SYSTEMOM.DualBBUConsChkSw** is set to **ON** on both the primary and backup gNodeBs. If this parameter is set to **ON**, the NE will check the consistency of the configurations and licenses between the active and standby BBUs. In the case of an inconsistency, ALM-26405 Dual Network Active/Standby Inconsistency is reported, which can be cleared by referring to the alarm reference.

## 10.2 Network Analysis

### 10.2.1 Benefits

If an SPOF occurs on a board in the active BBU, an RF module, or a transmission link on the active BBU side, which causes cell unavailability, while the standby BBU

works properly, the "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" function can be used to enable a fast switchover to the standby BBU to restore network services.

## 10.2.2 Impacts

The impacts between this function and other functions are described in [10.3.2.3 Function Impacts](#).

- If the backup gNodeB is powered on but does not provide network services, the service-related KPIs measured by the backup gNodeB are empty.
- During an active/standby switchover triggered by "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios", network services are interrupted for a short period of time. After the switchover is complete, UEs need to re-access the activated standby cell for service restoration.

## 10.3 Requirements

### 10.3.1 Licenses

This function is not under license control.

### 10.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

#### 10.3.2.1 Prerequisite Functions

| RAT                    | Function Name                           | Function Switch             | Reference | Description                                                                                                                                     |
|------------------------|-----------------------------------------|-----------------------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Low - frequency NR TDD | Intra-mode Dual-site Ring Topology Mode | <b>EQUIPMENT.IMDSRTMODE</b> | None      | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" requires this parameter to be set to <b>NONE</b> . |

### 10.3.2.2 Mutually Exclusive Functions

| RAT                  | Function Name            | Function Switch                                                                                        | Reference                                                                                | Description                                                                                                                                         |
|----------------------|--------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Timing carrier shutdown  | <b>TIMING_CARRIER_SHUT DOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter | <i>Energy Conservation and Emission Reduction</i> in <i>5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "timing carrier shutdown" cannot be both enabled.  |
| Low-frequency NR TDD | Power saving scheduling  | <b>POWER_SAVING_SCHEDULE_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter    | <i>Energy Conservation and Emission Reduction</i> in <i>5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "power saving scheduling" cannot be both enabled.  |
| Low-frequency NR TDD | Deep symbol power saving | <b>DEEP_SYMBOL_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter     | <i>Energy Conservation and Emission Reduction</i> in <i>5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "deep symbol power saving" cannot be both enabled. |

| RAT                  | Function Name                                               | Function Switch                                                                                                                                  | Reference                                                                         | Description                                                                                                                                                                            |
|----------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Energy saving based on flexible frequency-domain scheduling | <b>FLEX_FREQ_SCH_ENERGY_SAVING_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                        | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "energy saving based on flexible frequency-domain scheduling" cannot be both enabled. |
| Low-frequency NR TDD | RF channel intelligent shutdown                             | <b>RF_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                                        | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "RF channel intelligent shutdown" cannot be both enabled.                             |
| Low-frequency NR TDD | Intelligent carrier shutdown                                | <b>INTRA_GNB_MULTI_CARRIER_SD_SW</b> or <b>INTER_GNB_MULTI_CARRIER_SD_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intelligent carrier shutdown" cannot be both enabled.                                |

| RAT                  | Function Name                           | Function Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Reference                                                                         | Description                                                                                                                                                        |
|----------------------|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Fast carrier shutdown                   | <ul style="list-style-type: none"> <li>For capacity-layer cells: <ul style="list-style-type: none"> <li><b>INTRA_GNB_MULTI_CARR_SD_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter selected (intra-base-station multi-carrier scenarios) or <b>INTER_GNB_MULTI_CARR_SD_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter selected (inter-base-station multi-carrier scenarios)</li> <li><b>FAST_CARRIER_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingExtSwitch</b> parameter selected</li> </ul> </li> <li>For basic-layer cells: <b>FAST_CARR_SHUT_SERV_ASSUR_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingExtSwitch</b> parameter selected</li> </ul> | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "fast carrier shutdown" cannot be both enabled.                   |
| Low-frequency NR TDD | LTE and NR intelligent carrier shutdown | <b>LNR_SMART_CARRIER_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <i>Multi-RAT Coordinated Energy Saving</i>                                        | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR intelligent carrier shutdown" cannot be both enabled. |

| RAT                  | Function Name                              | Function Switch                                                                                                                                                                                                                                                  | Reference                                          | Description                                                                                                                                                           |
|----------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | LTE and NR coordinated symbol power saving | <b>MULTI_RAT_SYMBOL_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                                                                                                                                          | <i>Multi-RAT Coordinated Energy Saving</i>         | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR coordinated symbol power saving" cannot be both enabled. |
| Low-frequency NR TDD | LTE and NR bundled scheduling              | <b>LNR_BUNDLED_SCHEDULING_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                                                                                                                                             | <i>Multi-RAT Coordinated Energy Saving</i>         | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE and NR bundled scheduling" cannot be both enabled.              |
| Low-frequency NR TDD | LNR coordinated channel shutdown           | <b>MULTI_RAT_RF_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter                                                                                                                                                              | <i>Multi-RAT Coordinated Energy Saving</i>         | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LNR coordinated channel shutdown" cannot be both enabled.           |
| Low-frequency NR TDD | LTE FDD and NR Uplink Spectrum Sharing     | <ul style="list-style-type: none"> <li>NR: <b>LTE_NR_UL_SPECTRUM_SHARING_SW</b> option of the <b>NRDUCellAlgoSwitch.SpectrumCloudSwitch</b> parameter</li> <li>LTE: <b>SpectrumCloud.SpectrumCloudSwitch</b> set to <b>LTE_NR_UPLINK_SPECTRUM_SHR</b></li> </ul> | <i>LTE FDD and NR SUL Dynamic Spectrum Sharing</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "LTE FDD and NR Uplink Spectrum Sharing" cannot be both enabled.     |

| RAT                  | Function Name                       | Function Switch                                                                                         | Reference                                                                         | Description                                                                                                                                                    |
|----------------------|-------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Baseband shutdown for energy saving | <b>BB_SHUTDOWN_POWER_SAVING_SW</b> option of the <b>gNodeBParam.BaseStationDeepDormancySw</b> parameter | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "baseband shutdown for energy saving" cannot be both enabled. |
| Low-frequency NR TDD | TTI-level channel shutdown          | <b>TTI_RF_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingSwitch</b> parameter           | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "TTI-level channel shutdown" cannot be both enabled.          |
| Low-frequency NR TDD | RF channel dynamic muting           | <b>RF_CHN_DYN_SHUTDOWN_SW</b> option of the <b>NRDUCellAlgoSwitch.PowerSavingExtSwitch</b> parameter    | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "RF channel dynamic muting" cannot be both enabled.           |



| RAT                  | Function Name                                                             | Function Switch                                                                                             | Reference                                  | Description                                                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Intra-frequency multi-carrier shutdown                                    | <b>INTRA_FREQ_MULTI_CARR_SD_SW</b> option of the <b>NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch</b> parameter | None                                       | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intra-frequency multi-carrier shutdown" cannot be both enabled.                                                 |
| Low-frequency NR TDD | Adaptive BBU deep dormancy                                                | <b>BBP_DEEP_DORMANCY_SW</b> option of the <b>gNBReliabilityAlgo.DormancySwitch</b> parameter                | <i>Multi-RAT Coordinated Energy Saving</i> | If "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is configured for a gNodeB, adaptive BBU deep dormancy cannot be enabled.                                         |
| Low-frequency NR TDD | Backup power saving policy                                                | <b>NRDUCellPowerSaving.BackupPwrSavingPolicy</b>                                                            | None                                       | When "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is enabled, <b>NRDUCellPowerSaving.BackupPwrSavingPolicy</b> cannot be set to <b>CUSTOMIZED_BACKUP_POLICY</b> . |
| Low-frequency NR TDD | Dual-plane function in intra-frequency dual-network redundancy networking | <b>DUAL_PLANE_SW</b> option of the <b>gNBRedundancy.gNBReliabilitySwitch</b> parameter                      | None                                       | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "dual-plane function in intra-frequency dual-network redundancy networking" cannot be both enabled.              |

| RAT                  | Function Name                                                                 | Function Switch                                                                                | Reference | Description                                                                                                                                                                                              |
|----------------------|-------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Primary/Backup function in intra-frequency dual-network redundancy networking | <b>PRIMARY_BACKUP_SW</b><br>option of the <b>gNBRedundancy.gNBReliabilitySwitch</b> parameter  | None      | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "primary/backup function in intra-frequency dual-network redundancy networking" cannot be both enabled. |
| Low-frequency NR TDD | Dual-BBU backup in dual-network redundancy networking                         | <b>DUAL_BBU_BACKUP_SW</b><br>option of the <b>gNBRedundancy.gNBReliabilitySwitch</b> parameter | None      | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "dual-BBU backup in dual-network redundancy networking" cannot be both enabled.                         |

| RAT                  | Function Name                          | Function Switch                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Reference                                  | Description                                                                                                                                                       |
|----------------------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | RF module fast deep dormancy           | <ul style="list-style-type: none"> <li>NR load-based RF module fast deep dormancy:<br/><b>FAST_DEEP_DORM_SW</b> option of the <b>RRUEXTINFO.RRUDOPTFEATSW (5G gNodeB, LTE eNodeB)</b> parameter and <b>NR_FAST_DORMANCY_BY_LOAD_SW</b> option of the <b>NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch</b> parameter</li> <li>LNR load-based RF module fast deep dormancy:<br/><b>FAST_DEEP_DORM_SW</b> option of the <b>RRUEXTINFO.RRUDOPTFEATSW (5G gNodeB, LTE eNodeB)</b> parameter and <b>LNR_FAST_DORMANCY_BY_LOAD_SW</b> option of the <b>NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch</b> parameter</li> </ul> | <i>Multi-RAT Coordinated Energy Saving</i> | "Dual-BBU backup in dual-network redundancy networking" and "NR or LNR load-based RF module fast deep dormancy" cannot be both enabled.                           |
| Low-frequency NR TDD | Intra-frequency multi-carrier shutdown | <b>INTRA_FREQ_MULTI_CARR_SD_SW</b> option of the <b>NRDUCellPwrSaveAlgo.PowerSavingAlgoSwitch</b> parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | None                                       | "Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" and "intra-frequency multi-carrier shutdown" cannot be both enabled. |

10.3.2.3 Function Impacts

| RAT                  | Function Name              | Function Switch                                                | Reference                                                                         | Description                                                                                                                                                                                                                                                                                           |
|----------------------|----------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | Adaptive baseband shutdown | <b>BBP.CELLDEPLOY</b> set to <b>ADAPT_BBP_SHUTDOWN</b>         | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | Adaptive baseband shutdown does not take effect when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is in effect.                                                                                                                                       |
| Low-frequency NR TDD | Power saving first         | <b>BBP.CELLDEPLOY</b> set to <b>POWERSAVING</b>                | <i>Energy Conservation and Emission Reduction in 5G RAN Feature Documentation</i> | Power saving first does not take effect when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is in effect.                                                                                                                                               |
| Low-frequency NR TDD | RF module deep dormancy    | <b>RRU.DORMANCYSW</b> set to <b>ON</b> or <b>ON_DEEP_SUPER</b> | <i>Multi-RAT Coordinated Energy Saving</i>                                        | When RF module deep dormancy is enabled, RF modules enter the deep dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module deep dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled. |

| RAT                  | Function Name            | Function Switch                                                      | Reference                                  | Description                                                                                                                                                                                                                                                                                              |
|----------------------|--------------------------|----------------------------------------------------------------------|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Low-frequency NR TDD | RF module super dormancy | <b>RRU.DORMANCYSW</b> set to <b>ON_SUPER</b> or <b>ON_DEEP_SUPER</b> | <i>Multi-RAT Coordinated Energy Saving</i> | When RF module super dormancy is enabled, RF modules enter the super dormancy state, which causes an active/standby BBU switchover failure. Therefore, RF module super dormancy and "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" cannot be both enabled. |

### 10.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

#### Base Station Models

5900 series base stations (macro base stations)

#### BBUs

Only the BBU5900 and BBU5910 support this function.

#### Boards

- Baseband processing units: Only the NR TDD-capable UBBPg and UBBPh series boards support this function.
- Main control boards: Only the UMPTg, UMPTha, and UMPT hb series boards support this function.

#### RF Modules

All NR TDD-capable RF modules that work in low frequency bands support this function. For details, see the technical specifications of RF modules in *3900 & 5900 Series Base Station Product Documentation*.

## 10.3.4 Networking

- The primary and backup gNodeBs must be configured with the same license file that contains the ESNs of both the active and standby BBUs and the same configuration policy (for example, configuring parameters in the **SET FIXTERMLICSWITCH** command). For details about the license file and configuration policy, see *License Management* in *5G RAN Feature Documentation*.
- Both the active and standby base stations must be configured with clock sources.
- Only SA networking is supported.
- The combination of the **SUBRACK.CN** and **SUBRACK.SRN** parameter settings of the active and standby BBUs cannot be the same. In addition, the **SN** parameters of the main control boards in the active and standby BBUs must be set to the same value, and so must the **SN** parameters of the baseband processing units.
- The main control board in the active BBU and that in the standby BBU cannot be connected through a CI cable.
- The primary and backup gNodeBs with their NG interfaces connected to the same core network both have established an Xn link. For details about how to configure an NG/Xn link, see *NG and Xn Self-Management* in *5G RAN Feature Documentation*.
- To ensure that a check on the consistency between the active and standby BBUs can take effect, a dual-plane control link (DPCL) must be added between the primary and backup gNodeBs by running the **ADD DPCL** command.
- To prevent performance from being affected during an active/standby switchover, the parameters specifying the IDs of the cells served by the primary and backup gNodeBs must be set to the same value, and the switch or parameter settings related to the service functions of the primary and backup gNodeBs must be the same. The following uses the **NRDUCellTrp** and **NRDUCell** MOs as an example:
  - The settings of the parameters (such as **NRDUCellTrp.MaxTransmitPower** and **NRDUCellTrp.FrAndDuplexMode**) must be the same between the **NRDUCellTrp** MOs configured for the primary and backup gNodeBs.
  - The settings of all the parameters except **NRDUCell.PhysicalCellId** (such as **NRDUCell.DuplexMode**, **NRDUCell.FrequencyBand**, **NRDUCell.DlNarfcn**, **NRDUCell.UlNarfcn**, **NRDUCell.UlBandwidth**, **NRDUCell.DlBandwidth**, and **NRDUCell.LogicalRootSequenceIndex**) must be the same between the **NRDUCell** MOs configured for the primary and backup gNodeBs.
- It is recommended that the **NRDUCell.PhysicalCellId** parameter be set to different values for the paired cells on the primary and backup gNodeBs. All cells on the primary and backup gNodeBs must be added to the neighboring cell lists of neighboring cells.

## 10.3.5 Cells

"Dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" takes effect only for cells of the primary operator. To configure an

**NRCellRedundancy** MO, the cell corresponding to the MO must meet either of the following conditions:

- No **NRDUCellOp** MO is configured for the cell.
- The value of **gNBOperator.OperatorType** of the operator specified by **NRDUCellOp.OperatorId** in at least one **NRDUCellOp** MO is **PRIMARY\_OPERATOR**.

10.3.6 Others

None

10.4 Operation and Maintenance

10.4.1 Data Configuration

10.4.1.1 Data Preparation

**Table 10-1** describes the parameters used for function activation. This table does not describe parameters related to cell establishment.

**Table 10-1** Parameters used for activation

| Parameter Name                  | Parameter ID            | Setting Notes                                                                                                                                                                                 |
|---------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Network Element Name            | <b>NE.NENAME</b>        | Set this parameter based on the network plan.                                                                                                                                                 |
| Redundancy BTS Group Identifier | <b>NE.RBTSGROUPID</b>   | Set this parameter based on the network plan. This parameter must be set to the same value for the active and standby BBUs, which is different from the value specified for other BBU groups. |
| Redundancy BTS Group Name       | <b>NE.RBTSGROUPNAME</b> | Set this parameter based on the network plan. This parameter must be set to the same value for the active and standby BBUs, which is different from the value specified for other BBU groups. |

| Parameter Name                          | Parameter ID                                | Setting Notes                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-----------------------------------------|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Link ID                                 | <b>DPCL.LINKID</b>                          | <p>Set this parameter based on the network plan.</p> <ul style="list-style-type: none"> <li>The DPCL link supports only the new transmission model, that is, the transmission model for which <b>GTRANSPARA.TRANS CFGMODE</b> is set to <b>NEW</b>.</li> <li>The OMCH IP address can be reused as the local IP address of the DPCL, which can also be planned based on operator requirements and transport network conditions. A maximum of two IP addresses can be added.</li> </ul> |
| VRF Index                               | <b>DPCL.VRFIDX</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| IP Version                              | <b>DPCL.IPVERSION</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| First Local IPv4 Address                | <b>DPCL.LOCIP</b>                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Second Local IPv4 Address               | <b>DPCL.SECLOCIP</b>                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| First Peer IPv4 Address                 | <b>DPCL.PEERIP</b>                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Second Peer IPv4 Address                | <b>DPCL.SECPEERIP</b>                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Intra-mode Dual-site Ring Topology Mode | <b>EQUIPMENT.IMDSRTMODE</b>                 | Set this parameter to <b>NONE</b> when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is enabled. The modification of this parameter can take effect only after the NE is restarted (by running the <b>RST BTSNODE</b> command).                                                                                                                                                                                                        |
| Peer gNodeB ID                          | <b>gNBRedundancy.PeergNBId</b>              | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Local gNodeB Type                       | <b>gNBRedundancy.LocalgNBType</b>           | Set this parameter based on the network plan.                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| gNodeB Reliability Switch               | <b>gNBRedundancy.gNBReliabilitySwitch</b>   | Select the <b>DUAL_PLANE_TOB_SW</b> option when "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" is enabled.                                                                                                                                                                                                                                                                                                                              |
| Fault Detection Duration                | <b>gNBRedundancy.FaultDetectionDuration</b> | <p>Set this parameter based on the network plan.</p> <p>To prevent frequent switchovers caused by intermittent transmission link disconnections, it is recommended that this parameter be set to a value greater than or equal to <b>2</b>.</p>                                                                                                                                                                                                                                       |



| Parameter Name                               | Parameter ID                                  | Setting Notes                                                                                                    |
|----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Time-to-Trigger for Fault Reporting          | <b>gNBRedundancy.TimeToTriggerForFaultRpt</b> | Retain the default value.                                                                                        |
| Dual BBU Consistency Check Switch            | <b>SYSTEMOM.DualBBUConsChkSw</b>              | Set this parameter to <b>ON</b> when a check on the consistency between the active and standby BBUs is required. |
| RTO Initial Value                            | <b>SCTPTEMPLATE.RTOINIT</b>                   | Set this parameter to <b>220</b> .                                                                               |
| RTO Max Value                                | <b>SCTPTEMPLATE.RTOMAX</b>                    | Set this parameter to <b>221</b> .                                                                               |
| Max Association Retransmission               | <b>SCTPTEMPLATE.MAXASSOCRETR</b>              | Set this parameter to <b>2</b> .                                                                                 |
| Max Path Retransmission                      | <b>SCTPTEMPLATE.MAXPATHRETR</b>               | Set this parameter to <b>1</b> .                                                                                 |
| Heart-beat Interval                          | <b>SCTPTEMPLATE.HBINTER</b>                   | Set this parameter to <b>1000</b> .                                                                              |
| RTO Min Value                                | <b>SCTPTEMPLATE.RTOMIN</b>                    | Set this parameter to <b>220</b> .                                                                               |
| User-plane Path Status Fast Awareness Switch | <b>GUPTRANSPARA.UPSTATFASTSW</b>              | Set this parameter to <b>ENABLE</b> .                                                                            |

### 10.4.1.2 Using MML Commands

Before using MML commands, refer to [10.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

## Activation Command Examples

- Activation command examples on the primary gNodeB side

**Step 1** Set the ID and name of the backup base station group for "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios".

```
SET NE: NENAME="HUAWEI", RBTSGROUPID="001", RBTSGROUPNAME="BBUBackupGroup001";
```

**Step 2** Set flags of the same RAT between subracks. The modification of **Intra-mode Dual-site Ring Topology Mode** can take effect only after the NE is restarted (by running the **RST BTSNODE** command).

```
SET EQUIPMENT: IMDSRTMODE=NONE;
```

**Step 3** Configure a DPCL. The following uses two IP addresses as an example.

```
ADD DPCL: LINKID=0, VRFID=0, IPVERSION=IPv4, LOCIP="192.168.4.8", SECLOCIP="192.168.5.8",  
PEERIP="192.168.4.9", SECPEERIP="192.168.5.9";
```

**Step 4** Enable "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios".

```
ADD GNBREDUNDANCY: PeergNBId=1000001, LocalgNBType=PRIMARY,  
gNBReliabilitySwitch=DUAL_PLANE_TOB_SW-1, FaultDetectionDuration=3, TimeToTriggerForFaultRpt=0;
```

**Step 5** Add a backup NR cell.

```
ADD NRCELLREDUNDANCY: NrCellId=0, PeerNrCellId=0;
```

### NOTE

Ensure that the mapping between the primary and standby cells is correct. That is, the cells specified by the **NrCellId** parameter of the primary gNodeB and the **PeerNrCellId** parameter of the backup gNodeB are primary and standby cells with the same coverage.

**Step 6** Activate a cell.

```
ACT NRCELL: NrCellId=0;
```

**Step 7** (Optional) Modify parameters in the **SCTPTEMPLATE** MO to speed up the switchover upon a transmission link fault.

```
MOD SCTPTEMPLATE: SCTPTEMPLATEID=0, RTOINIT=220, RTOMAX=221, MAXASSOCRETR=2,  
MAXPATHRETR=1, HBINTER=1000, RTOMIN=220;
```

**Step 8** (Optional) Set **User-plane Path Status Fast Awareness Switch** to **ENABLE** to speed up the switchover upon a transmission link fault.

```
SET GUPTRANSPARA: UPSTATFASW=ENABLE;
```

**Step 9** Initiate a consistency check between active and standby BBUs.

```
SET SYSOM: DUALBBUCONSCHKSW=ON;
```

----End

- Activation command examples on the backup gNodeB side

**Step 1** Set the ID and name of the backup base station group for "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios".

```
SET NE: NENAME="HUAWEI", RBTSGROUPID="001", RBTSGROUPNAME="BBUBackupGroup001";
```

**Step 2** Set flags of the same RAT between subracks. The modification of **Intra-mode Dual-site Ring Topology Mode** can take effect only after the NE is restarted (by running the **RST BTSNODE** command).

```
SET EQUIPMENT: IMDSRTMODE=NONE;
```

**Step 3** Configure a DPCL. The following uses two IP addresses as an example.

```
ADD DPCL: LINKID=0, VRFIDX=0, IPVERSION=IPv4, LOCIP="192.168.4.9",  
SECLOCIP="192.168.5.9", PEERIP="192.168.4.8", SECPEERIP="192.168.5.8";
```

**Step 4** Enable "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios".

```
ADD GNBREDUNDANCY: PeergNBId=1, LocalgNBType=BACKUP,  
gNBReliabilitySwitch=DUAL_PLANE_TOB_SW-1, FaultDetectionDuration=3, TimeToTriggerForFaultRpt=0;
```

**Step 5** Add a backup NR cell.

```
ADD NRCELLREDUNDANCY: NrCellId=0, PeerNrCellId=0;
```

#### NOTE

Ensure that the mapping between the primary and standby cells is correct. That is, the cells specified by the **NrCellId** parameter of the primary gNodeB and the **PeerNrCellId** parameter of the backup gNodeB are primary and standby cells with the same coverage.

**Step 6** Activate a cell.

```
ACT NRCELL: NrCellId=0;
```

**Step 7** (Optional) Modify parameters in the **SCTPTEMPLATE** MO to speed up the switchover upon a transmission link fault.

```
MOD SCTPTEMPLATE: SCTPTEMPLATEID=0, RTOINIT=220, RTOMAX=221, MAXASSOCRETR=2,  
MAXPATHRETR=1, HBINTER=1000, RTOMIN=220;
```

**Step 8** (Optional) Set **User-plane Path Status Fast Awareness Switch** to **ENABLE** to speed up the switchover upon a transmission link fault.

```
SET GUPTRANSPARA: UPSTATFASW=ENABLE;
```

**Step 9** Initiate a consistency check between active and standby BBUs.

```
SET SYSOM: DUALBBUCONSCHKSW=ON;
```

----End

## Deactivation Command Examples

- Deactivation command examples on the primary gNodeB side

**Step 1** Terminate the consistency check between active and standby BBUs.

```
SET SYSOM: DUALBBUCONSCHKSW=OFF;
```

**Step 2** (Optional) Set **User-plane Path Status Fast Awareness Switch** to **DISABLE**.

```
SET GUPTRANSPARA: UPSTATFASW=DISABLE;
```

**Step 3** Remove the **GNBREDUNDANCY** MO.

```
RMV GNBREDUNDANCY: PeergNBId=1000001;
```

**Step 4** Remove the **NRCELLREDUNDANCY** MO.

```
RMV NRCELLREDUNDANCY: NrCellId=0;
```

----End

- Deactivation command examples on the backup gNodeB side

**Step 1** Terminate the consistency check between active and standby BBUs.

```
SET SYSOM: DUALBBUCONSCHKSW=OFF;
```

**Step 2** (Optional) Set **User-plane Path Status Fast Awareness Switch** to **DISABLE**.

```
SET GUPTRANSPARA: UPSTATFASW=DISABLE;
```

**Step 3** Remove the **GNBREDUNDANCY** MO.

```
RMV GNBREDUNDANCY: PeergNBId=1;
```

**Step 4** Remove the **NRCELLREDUNDANCY** MO.

```
RMV NRCELLREDUNDANCY: NrCellId=0;
```

----End

### 10.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 **NOTE**

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

## 10.4.2 Activation Verification

To check whether "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios" has taken effect, perform the following steps:

**Step 1** Enable a UE to access a cell served by the primary gNodeB.

**Step 2** Run the **DSP NRCELLREDUNDANCY** command to query the primary/backup status of the cell.

**Step 3** Block the baseband processing unit serving the cell that the UE accesses.

Run the **BLK BRD** command to block the board. EVT-29905 NR Cell Switchover between gNodeBs is reported on both the primary and backup gNodeBs, which carries an NR cell ID, indicating that the cell is successfully switched over to the backup gNodeB and starts to provide services. The UE can access the NR cell.

**Step 4** Run the **DSP NRCELLREDUNDANCY** command again to query the primary/backup status of the cell. The command output indicates that the primary/backup status of the cell changes.

----End

To check whether a consistency check between active and standby BBUs has been performed, perform the following steps:

Run the **LST SYSOM** command on both the primary and backup gNodeBs. If the values of **Dual BBU Consistency Check Switch** are both **ON** in the command output, the consistency check between active and standby BBUs has been performed. One hour after the check is performed, if an inconsistency of the configurations and licenses between the active and standby BBUs is detected, ALM-26405 Dual Network Primary/Backup Inconsistency is reported on the backup gNodeB, which can be cleared by referring to the alarm reference. The backup gNodeB refers to the gNodeB for which **gNBRedundancy.LocalgNBType** is set to **BACKUP**.

## 10.4.3 Network Monitoring

During a switchover triggered by "dual-plane function in intra-frequency dual-network redundancy networking in ToB scenarios", EVT-29905 NR Cell Switchover

between gNodeBs will be reported, indicating that the NR cell switchover is successful.

# 11 Cold Backup of RF Modules (NR TDD)

---

## 11.1 Principles

When an RF module is faulty, the cells served by this faulty module will be affected. When two RF modules are configured in a base station to cover the same area, Cold Backup of RF Modules enables the cells or carriers served by the master RF module to be reestablished and served by the backup RF module when the master RF module is faulty. The base station services can be recovered within 3 minutes, improving base station reliability. If cells are successfully reestablished, EVT-29904 NR DU Cell TRP RF Unit Switchover will be reported. Currently, only 1:1 backup of AAUs using eCPRI is supported when the **SECTOREQM.ANTCFGMODE** parameter is set to **BEAM** in low-frequency NR TDD scenarios.

When both master and backup RF modules are available, the master RF module is preferentially used to serve the cells. When only one of the master and backup RF modules is available, cells are served by the available RF module. If the master RF module changes from unavailable to available, the base station will not proactively trigger reestablishments for cells served by the backup RF module and change the serving RF module to the master RF module. Cells are reestablished and preferentially served by the master RF module only when the base station is upgraded or reset. [Figure 11-1](#) and [Figure 11-2](#) show the recommended topologies for this function.

Figure 11-1 Intra-board load sharing topology

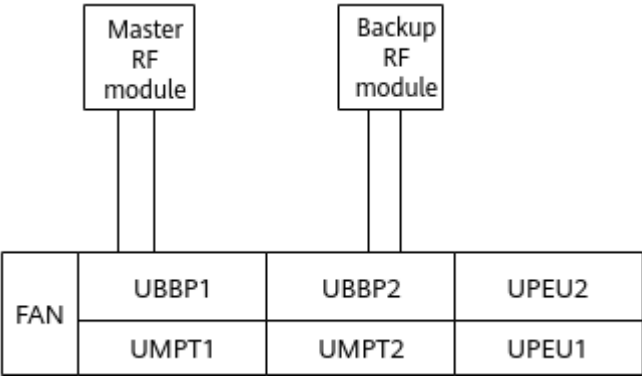
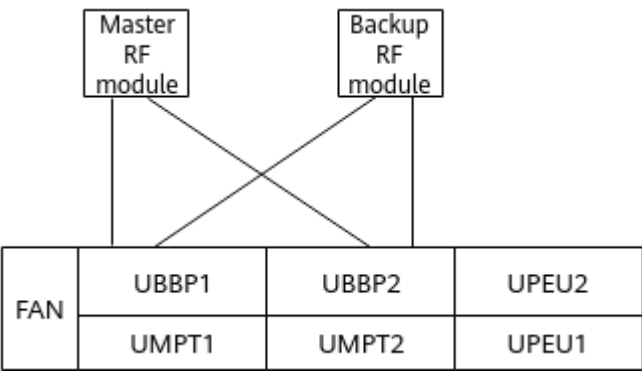


Figure 11-2 Inter-board load sharing topology



## 11.2 Network Analysis

### 11.2.1 Benefits

None

### 11.2.2 Impacts

The impacts between this function and other functions are described in [11.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 11.3 Requirements

### 11.3.1 Licenses

None

## 11.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 11.3.2.1 Prerequisite Functions

None

### 11.3.2.2 Mutually Exclusive Functions

| RAT                  | Function Name           | Function Switch       | Reference                                  |
|----------------------|-------------------------|-----------------------|--------------------------------------------|
| Low-frequency NR TDD | RF module deep dormancy | <b>RRU.DORMANCYSW</b> | <i>Multi-RAT Coordinated Energy Saving</i> |

### 11.3.2.3 Function Impacts

None

## 11.3.3 Hardware

Only NR TDD-only AAUs support this function.

The models of the master and backup AAUs must be the same.

## 11.3.4 Others

None

## 11.4 Operation and Maintenance

### 11.4.1 When to Use

When cells are faulty due to RF module faults, Cold Backup of RF Modules enables cells to be reestablished and served by an available RF module.

Cold Backup of RF Modules ensures that cell services recover automatically, reduces service interruption duration, and improves reliability.

### 11.4.2 Data Configuration



### 11.4.2.1 Data Preparation

**Table 11-1** describes the configuration parameters that must be set in the **RFHAGRP** MO.

**Table 11-1** Configuration parameters that must be set for the master and backup RF modules

| Parameter Name        | Parameter ID          | Setting Notes                                                                                                         |
|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Master RF Cabinet No. | <b>RFHAGRP.MRFCN</b>  | Indicates the cabinet number of the master RF module in an RF HA group. Set this parameter based on the network plan. |
| Master RF Subrack No. | <b>RFHAGRP.MRFSRN</b> | Indicates the subrack number of the master RF module in an RF HA group. Set this parameter based on the network plan. |
| Master RF Slot No.    | <b>RFHAGRP.MRFSN</b>  | Indicates the slot number of the master RF module in an RF HA group. Set this parameter based on the network plan.    |
| Backup RF Cabinet No. | <b>RFHAGRP.BRFCN</b>  | Indicates the cabinet number of the backup RF module in an RF HA group. Set this parameter based on the network plan. |
| Backup RF Subrack No. | <b>RFHAGRP.BRFSRN</b> | Indicates the subrack number of the backup RF module in an RF HA group. Set this parameter based on the network plan. |
| Backup RF Slot No.    | <b>RFHAGRP.BRFSN</b>  | Indicates the slot number of the backup RF module in an RF HA group. Set this parameter based on the network plan.    |

**Table 11-2** describes the configuration parameters that must be set in the **SECTOREQM** MO.

**Table 11-2** Configuration parameters that must be set for the master AAU sector equipment

| Parameter Name      | Parameter ID                 | Setting Notes                                                                                |
|---------------------|------------------------------|----------------------------------------------------------------------------------------------|
| Antenna Config Mode | <b>SECTOREQM.ANTCFG MODE</b> | When this parameter is set to <b>BEAM</b> , 1:1 backup of AAUs using eCPRI can be performed. |

### 11.4.2.2 Using MML Commands

Before using MML commands, refer to [11.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

- Step 1** Run the **ADD RRUCHAIN** command to add the master and backup AAU chains/rings.

```
ADD RRUCHAIN: RCN=10, TT=LOADBALANCE, AT=LOCALPORT, HSRN=0, HSN=5, HPN=0, TSRN=0, TSN=5,
TPN=1, PROTOCOL=eCPRI, CR=AUTO, USERDEFRATENEGOSW=OFF;
ADD RRUCHAIN: RCN=11, TT=LOADBALANCE, AT=LOCALPORT, HSRN=0, HSN=5, HPN=2, TSRN=0, TSN=5,
TPN=3, PROTOCOL=eCPRI, CR=AUTO, USERDEFRATENEGOSW=OFF;
```

- Step 2** Run the **MOD RRUCHAIN** and **ADD RRU** commands to add the master and backup AAUs.

```
MOD RRUCHAIN: RCN=10, BRKPOS1=0, BRKPOS2=0;
MOD RRUCHAIN: RCN=11, BRKPOS1=0, BRKPOS2=0;
ADD RRU: CN=0, SRN=80, SN=0, TP=TRUNK, RCN=10, PS=0, RT=AIRU, RS=NO, RXNUM=64, TXNUM=64,
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDETECTSW=OFF, DORMANCYSW=OFF;
ADD RRU: CN=0, SRN=81, SN=0, TP=TRUNK, RCN=11, PS=0, RT=AIRU, RS=NO, RXNUM=64, TXNUM=64,
MNTMODE=NORMAL, RFDCPWROFFALMDetectSW=OFF, RFTXSIGNDETECTSW=OFF, DORMANCYSW=OFF;
MOD RRUCHAIN: RCN=10, BRKPOS1=255, BRKPOS2=255;
MOD RRUCHAIN: RCN=11, BRKPOS1=255, BRKPOS2=255;
```

#### NOTE

Ensure that the coverage area of the backup AAU is the same as that of the master AAU (by adjusting the mechanical tilts of the master and backup AAUs).

- Step 3** Run the **ADD RFHAGRP** command to add an RF HA group. Specify the master and backup RF modules.

```
ADD RFHAGRP: MRFCN=0, MRFSRN=80, MRFSN=0, BRFCN=0, BRFSRN=81, BRFSN=0;
```

- Step 4** Run the **ADD SECTOR** command to add the master AAU sector.

```
ADD SECTOR: SECTORID=80, SECNAME="80", LOCATIONNAME="80", USERLABEL="80", ANTNUM=0,
CREATESECTOREQM=FALSE;
```

- Step 5** Run the **ADD SECTOREQM** command to add the master AAU sector equipment.

```
ADD SECTOREQM: SECTOREQMID=80, SECTORID=80, ANTFCGMODE=BEAM, RRUCN=0, RRUSRN=80,
RRUSN=0, BEAMSHAPE=SEC_120DEG, BEAMLAYERSPLIT=None, BEAMAZIMUTHOFFSET=None;
```

----End

### Deactivation Command Examples

- Step 1** Run the **RMV RFHAGRP** command to remove the RF HA group.

```
RMV RFHAGRP: MRFCN=0, MRFSRN=80, MRFSN=0;
```

- Step 2** Run the **MOD RRUCHAIN** and **RMV RRU** commands to remove the backup AAU.

```
MOD RRUCHAIN: RCN=11, BRKPOS1=0, BRKPOS2=1;
RMV RRU: CN=0, SRN=81, SN=0;
```

**Step 3** Run the **RMV RRUCHAIN** command to remove the backup AAU chain/ring.

```
RMV RRUCHAIN: RCN=11;
```

----End

### 11.4.2.3 Using the MAE-Deployment

This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

## 11.4.3 Activation Verification

On the MAE-Access or LMT, perform the following steps to check whether Cold Backup of RF Modules has taken effect.

**Step 1** Check the status of the cell.

Run the **DSP NRCELL** command to check the cell status. If **NR Cell State** is **Normal**, the cell has been activated.

**Step 2** Check the antenna information of the cell.

Run the **DSP NRDUCELLANTENNA** command to check the antenna information of the cell. Multiple records are displayed in the command output. For the record in which the value of **RF Work State** is **Normal**, the value of **RF Module Information** indicates the RF module information of the master RF module. For the record in which the value of **RF Work State** is -, the value of **RF Module Information** indicates the RF module information of the backup RF module.

**Step 3** Check whether the cell is functional.

If the UEs can successfully access the cell, the cell is functional.

**Step 4** Block the master RF module.

Run the **BLK BRD** command to block the RF module serving the cell, and trigger the cell to be served by another RF module.

**Step 5** Perform steps 1 to 3.

If the parameter values and operation results are the same as those in the preceding steps, and in the antenna information query results, the record in which the value of **RF Work State** is **Normal** corresponds to the backup RF module, the feature has taken effect and cell reestablishment has succeeded. In this case, EVT-29904 NR DU Cell TRP RF Unit Switchover is reported.

----End

## 11.4.4 Network Monitoring

None

# 12 RRU Channel Cross Connection Under MIMO (LTE/NR)

---

## 12.1 Principles

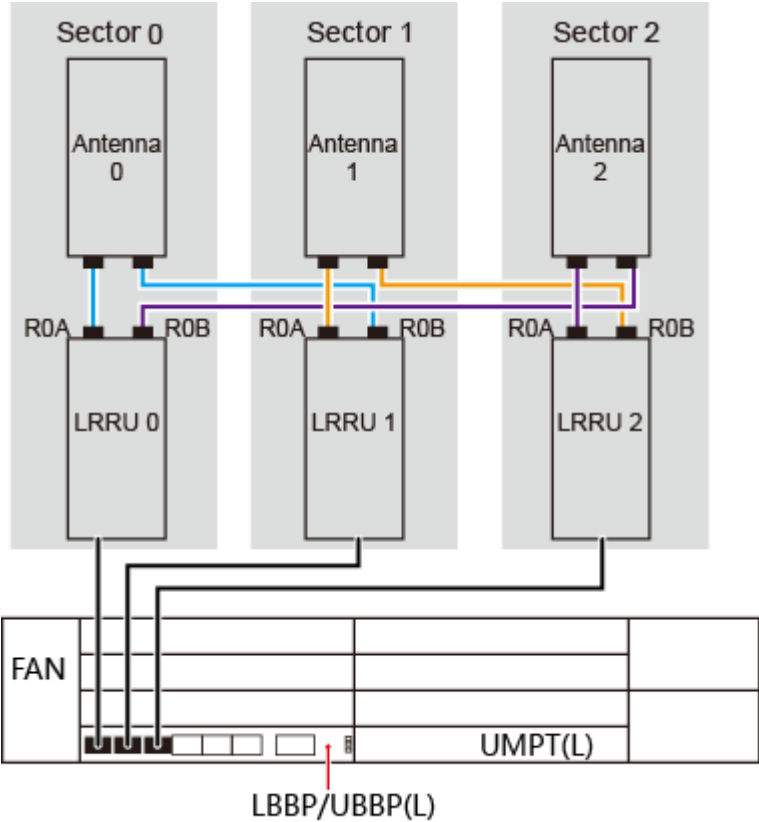
In sparsely populated areas, RRUs or RFUs may be installed on top of a tower and base stations are installed in inaccessible areas. This makes subsequent equipment maintenance difficult. If any RRU or RFU is faulty, an entire sector remains unavailable for a long time. With RRU Channel Cross Connection Under MIMO enabled on the eNodeB or gNodeB, the failure in one RRU or RFU will not lead to service interruption of the entire sector. This feature increases RRU or RFU reliability without increasing hardware costs.

As shown in [Figure 12-1](#) (using three sectors as an example), a baseband processing unit (BBP) is connected to multiple RRUs. In this case, the data on two TX/RX channels of a cell is transmitted over two different optical fibers and processed by two different RRUs. When a fault occurs on an optical fiber or an RRU has a hardware fault, the antenna mode changes from 2T2R to 1T1R to keep the cell working normally. This prevents long-time service interruption and increases system reliability.

### NOTE

The implementation method of this feature on the gNodeB is the same as that on the eNodeB. The following figure uses the eNodeB as an example.

Figure 12-1 RF cable connections for RRU channel cross connection under MIMO



CIB01C3020

## 12.2 Network Analysis

### 12.2.1 Benefits

None

### 12.2.2 Impacts

The impacts between this function and other functions are described in [12.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 12.3 Requirements

### 12.3.1 Licenses

None

## 12.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 12.3.2.1 Prerequisite Functions

None

### 12.3.2.2 Mutually Exclusive Functions

| RAT     | Function Name                     | Function Switch                  | Reference                         |
|---------|-----------------------------------|----------------------------------|-----------------------------------|
| LTE FDD | GSM and LTE Dynamic Power Sharing | <b>CellAlgoSwitch.GLPwrShare</b> | GSM and LTE Dynamic Power Sharing |

### 12.3.2.3 Function Impacts

None

## 12.3.3 Hardware

- This feature applies only to macro base stations and LampSite base stations. It is implemented in the same way on macro base stations and LampSite base stations.
- The RRUs or RFUs must be of the same model and support the same set of frequency bands.
- The number of RRUs or RFUs must be greater than or equal to two.
- The difference in length of optical fibers for RRUs must be less than or equal to 100 m.
- The LRRUs or LRFUs must form a star topology and connect to the same BBP.

#### NOTE

In multimode base stations where the dual-star topology is used, RRUs must be connected to the same BBP.

## 12.3.4 Cells

- Cells enabled with RRU channel cross-connection under MIMO must operate on the same frequency and have the same bandwidth.
- Cells enabled with RRU channel cross-connection under MIMO must be 2T2R cells.

## 12.3.5 Others

None

## 12.4 Operation and Maintenance

This section involves only LTE.

### 12.4.1 When to Use

It is recommended that RRU Channel Cross Connection Under MIMO be enabled when RRUs/RFUs are installed on top of a tower. Cross-connections between a BBP and RRUs/RFUs enable the data on two TX/RX channels of a cell to be transmitted using two optical fibers and to be processed by two RRUs/RFUs. When an optical fiber or an RRU/RFU has a hardware fault, the antenna mode changes from 2T2R to 1T1R to keep the cell working normally. This prevents long-time service interruption and increases system reliability.

### 12.4.2 Precautions

The precautions for deploying RRU Channel Cross Connection Under MIMO are as follows:

- The RRUs or RFUs must be of the same model and support the same set of frequency bands.
- Two or more RRUs or RFUs are required.
- The antenna mode must be 2T2R for the sectors on which RRU Channel Cross Connection Under MIMO applies. Two transmit and two receive channels of each sector must be configured on different RRUs or RFUs, and the RRUs or RFUs must be correctly connected to antennas.
- The RRUs or RFUs must form a star topology and connect to the same BBP.
- For LBBPc boards, optical fibers that connect the LBBPc boards and RRUs or RFUs should have the same length if possible. However, if this condition cannot be satisfied, any differences in lengths must be less than or equal to 100 m (328.08 ft). This restriction does not apply to LBBPd boards.
- If faults on the optical fiber or RRU are rectified when the cell has rolled back to 1T1R and is in active mode, the system triggers the cell reestablishment to change the cell configuration from 1T1R to 2T2R only when no user in RRC connected mode exists in the cell.

In multimode scenarios, RRU Channel Cross Connection Under MIMO is supported by LTE and NR. For other RATs, support for this feature depends on the RAT capability.

### 12.4.3 Data Configuration

#### 12.4.3.1 Data Preparation

**Table 12-1** describes the parameters for RRU Channel Cross Connection Under MIMO.

**Table 12-1** Parameters for RRU Channel Cross Connection Under MIMO

| Parameter Name | Parameter ID           | Setting Notes |
|----------------|------------------------|---------------|
| Sector ID      | <b>SECTOR.SECTORID</b> | None          |

### 12.4.3.2 Using MML Commands

Before using MML commands, refer to [12.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

**Step 1** Run the **ADD SECTOR** command to add a sector.

 **NOTE**

Two antennas are generally configured, and R0A and R0B corresponding to antenna channels are configured on different antenna ports on the RRU. Physical cable connections must be consistent with the configurations.

```
ADD SECTOR: SECTORID=0, ANTNUM=2, ANT1CN=0, ANT1SRN=60, ANT1SN=0, ANT1N=R0A, ANT2CN=0,
ANT2SRN=61, ANT2SN=0, ANT2N=R0B, CREATESECTOREQM=TRUE, SECTOREQMID=0;
ADD SECTOR: SECTORID=1, ANTNUM=2, ANT1CN=0, ANT1SRN=61, ANT1SN=0, ANT1N=R0A, ANT2CN=0,
ANT2SRN=62, ANT2SN=0, ANT2N=R0B, CREATESECTOREQM=TRUE, SECTOREQMID=1;
ADD SECTOR: SECTORID=2, ANTNUM=2, ANT1CN=0, ANT1SRN=62, ANT1SN=0, ANT1N=R0A, ANT2CN=0,
ANT2SRN=60, ANT2SN=0, ANT2N=R0B, CREATESECTOREQM=TRUE, SECTOREQMID=2;
```

**Step 2** Run the **ADD CNOOPERATOR** command to add an operator.

```
ADD CNOOPERATOR: CnOperatorId=0, CnOperatorName="mobile",
CnOperatorType=CNOOPERATOR_PRIMARY, Mcc="460", Mnc="01";
```

**Step 3** Run the **ADD CNOOPERATOR TA** command to add a tracking area.

```
ADD CNOOPERATOR TA: TrackingAreaId=0, CnOperatorId=0, Tac=33, NblotTaFlag=BOOLEAN_TRUE;
```

**Step 4** Run the **ADD CELL** command to add a cell.

```
ADD CELL: LocalCellId=0, CellName="MIMO", FreqBand=3, UIEarfcnCfgInd=NOT_CFG, DIEarfcn=1600,
UIBandWidth=CELL_BW_N50, DIBandWidth=CELL_BW_N50, CellId=0, PhyCellId=0, FddTddInd=CELL_FDD,
RootSequenceIdx=0, CustomizedBandWidthCfgInd=NOT_CFG, UePowerMaxCfgInd=NOT_CFG,
MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=2T2R;
ADD CELL: LocalCellId=1, CellName="MIMO", FreqBand=3, UIEarfcnCfgInd=NOT_CFG, DIEarfcn=1600,
UIBandWidth=CELL_BW_N50, DIBandWidth=CELL_BW_N50, CellId=1, PhyCellId=1, FddTddInd=CELL_FDD,
RootSequenceIdx=0, CustomizedBandWidthCfgInd=NOT_CFG, UePowerMaxCfgInd=NOT_CFG,
MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=2T2R;
ADD CELL: LocalCellId=2, CellName="MIMO", FreqBand=3, UIEarfcnCfgInd=NOT_CFG, DIEarfcn=1600,
UIBandWidth=CELL_BW_N50, DIBandWidth=CELL_BW_N50, CellId=2, PhyCellId=2, FddTddInd=CELL_FDD,
RootSequenceIdx=0, CustomizedBandWidthCfgInd=NOT_CFG, UePowerMaxCfgInd=NOT_CFG,
MultiRruCellFlag=BOOLEAN_FALSE, TxRxMode=2T2R;
```

**Step 5** Run the **ADD EUCELLSECTOREQM** command to add cell sector equipment.



```
ADD EUCELLSECTOREQM: LocalCellId=0, SectorEqmId=0;  
ADD EUCELLSECTOREQM: LocalCellId=1, SectorEqmId=1;  
ADD EUCELLSECTOREQM: LocalCellId=2, SectorEqmId=2;
```

**Step 6** Run the **ADD CELLOP** command to add a cell operator.

```
ADD CELLOP: LocalCellId=0, TrackingAreaId=0, MMECfgNum=CELL_MME_CFG_NUM_0;  
ADD CELLOP: LocalCellId=1, TrackingAreaId=0, MMECfgNum=CELL_MME_CFG_NUM_0;  
ADD CELLOP: LocalCellId=2, TrackingAreaId=0, MMECfgNum=CELL_MME_CFG_NUM_0;
```

**Step 7** Run the **ACT CELL** command to activate the cell.

```
ACT CELL: LocalCellId=0;  
ACT CELL: LocalCellId=1;  
ACT CELL: LocalCellId=2;
```

----End

## Deactivation Command Examples

**Step 1** Run the **RMV CELL** command to remove the cell.

```
RMV CELL: LocalCellId=0;
```

**Step 2** Run the **RMV SECTOR** command to remove the sector.

```
RMV SECTOR: SECTORID=0;
```

----End

### 12.4.3.3 Using the MAE-Deployment

This function can be activated for a single base station or a batch of base stations on the MAE-Deployment. For detailed operations, see Feature Configuration Using the MAE-Deployment.

## 12.4.4 Activation Verification

For details, see the "Activation Verification" section in *Cell Management*.

## 12.4.5 Network Monitoring

None

# 13 Power Supply Reliability

---

## 13.1 Principles

### 13.1.1 Power Supply Reliability for a Base Station

#### Protection Against Overvoltage and Overcurrent

The base station supports a wide input voltage range and provides protection against overcurrent.

- The base station supports a wide input voltage range. For details about the supported voltage range, see section "Engineering Specifications of Cabinets" in chapter "Product Specifications" of *3900 & 5900 Series Base Station Technical Description*.
- In AC input scenarios, the power supply unit (PSU) provides protection against overcurrent and overvoltage after the protection function is enabled. Once overcurrent or overvoltage occurs, the PSU stops supplying power.
- In DC input scenarios, the direct current distribution unit (DCDU) provides a circuit breaker or fuse for each DC output. Once short-circuit or overload occurs on a DC output, the corresponding circuit breaker or fuse is disconnected automatically. This does not affect upper-level equipment.

#### Enhanced Power Supply for Huawei AC-Powered Base Stations Equipped with the PMU

In addition to basic power supply functions, the features in the following table are provided for Huawei AC-powered base stations equipped with the PMU to improve power supply reliability.

**Table 13-1** Enhanced power supply for Huawei AC-powered base stations equipped with the PMU

| Function                                 | Feature ID and Name                                                                                                                                                                                                                                                                                                                         | Description                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intelligent battery management           | <p>GSM: GBFD-510710 Intelligent Battery Management</p> <p>UMTS: WRFD-140220 Intelligent Battery Management</p> <p>LTE FDD: LOFD-001071 Intelligent Battery Management</p> <p>LTE TDD: TDLOFD-001071 Intelligent Battery Management</p> <p>LTE NB-IoT and NR: This is a basic function for base stations and does not have a feature ID.</p> | <p>Intelligent battery management provides the following functions: automatic switching between different charge-and-discharge modes, self-protection under high temperature, and battery runtime display.</p> <p>For details about the principles and engineering guidelines of intelligent battery management, see <i>Power Supply Management</i>.</p>                           |
| Automatic battery and load disconnection | <p>GSM: GBFD-111601 BTS Power Management</p> <p>This is a basic function for UMTS, LTE, and NR base stations and does not have a feature ID.</p>                                                                                                                                                                                            | <p>Automatic battery and load disconnection provides the following functions: automatic battery disconnection under low voltage, automatic battery disconnection under high temperature, and automatic load disconnection.</p> <p>For details about the principles and engineering guidelines of automatic battery and load disconnection, see <i>Power Supply Management</i>.</p> |
| Intelligent diesel generator management  | <p>This is a basic function for GSM, UMTS, LTE, and NR base stations and does not have a feature ID.</p>                                                                                                                                                                                                                                    | <p>Base stations supplied with solar power support intelligent diesel generator management. Using either RS485 or dry contact ports, the PMU monitors the status, fuel level, and faults of the diesel generator.</p> <p>For details about the principles and engineering guidelines of intelligent diesel generator management, see <i>Power Supply Management</i>.</p>           |

| Function                                            | Feature ID and Name                                                                                                                                                    | Description                                                                                                                                                                                                |
|-----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Intelligent shutdown of carriers due to PSU failure | GSM: GBFD-117804<br>Intelligent Shutdown of TRX Due to PSU Failure<br><br>This is a basic function for UMTS, LTE, and NR base stations and does not have a feature ID. | When some PSUs become faulty and the remaining PSUs cannot meet the base station's power requirements, the base station enters energy saving mode to reduce power consumption if this function is enabled. |

#### Intelligent Shutdown of TRX Due to PSU Failure:

In scenarios where a base station uses the AC power input, the PSU converts the AC power to DC power and then supplies the DC power to boards in the base station. Generally, multiple PSUs are required to supply sufficient power to a base station and these PSUs work in parallel. If one or several PSUs are faulty, the load of the PSUs that work properly increases. As a result, all PSUs may stop working due to overcurrent protection and all the services carried on the base station may be interrupted. To prevent this problem, Intelligent Shutdown of TRX Due to PSU Failure is introduced to the base station. The procedure for Intelligent Shutdown of TRX Due to PSU Failure is as follows:

- If some PSUs are faulty, the PSUs that work properly cannot provide the base station with sufficient power. In this situation, the base station shuts down the power amplifiers of all TRXs and derates the TRXs to reduce power consumption. In this way, the TRXs that are not shut down can work properly.
- After an alarm related to the PSU fault is reported, the base station shuts down the power amplifiers of all TRXs and derates the TRXs based on the power supply capability of the PSUs that work properly. This ensures that base station power consumption does not exceed the power supply capability supported by the remaining PSUs after the power amplifiers of all TRXs are shut down.
- When all PSUs recover, the power amplifiers of all TRXs are powered on.

For a GBTS, the Intelligent Shutdown of TRX Due to PSU Failure feature is enabled when the **BTSPSUFP.CFGFLAG** parameter is set to **YES**. In this case, if one PSU is faulty, the power of all TRXs decreases by 3 dB (50% of the original power). If two or more PSUs are faulty, the power amplifiers of all TRXs are shut down.

For an eGBTS, the Intelligent Shutdown of TRX Due to PSU Failure function is enabled when the **EQUIPMENT.PSUFP (LTE eNodeB, 5G gNodeB)** parameter is set to **ON**. In addition, the Intelligent Shutdown of the TRX Due to PSU Failure function takes effect only when the enhanced backup power saving function and the deep backup power saving function are enabled.

- The enhanced backup power saving function is controlled by the **GBTSENERGYMGTPARA.ENBAKPWRSAVALLOW** parameter. When this parameter is set to **YES**, the enhanced backup power saving function is enabled. After the enhanced backup power saving function is enabled, if one PSU is faulty, the base station enters the enhanced energy saving state upon receiving a backup power saving message. In this case, the power of all TRXs is reduced by 3 dB to reduce power consumption.

- The deep backup power saving function is controlled by the **GBTSENERGYMGTPARA.DEEPBACKPWRSAVALLOW** parameter. When this parameter is set to **YES**, the deep backup power saving function is enabled. After the deep backup power saving function is enabled, if two or more PSUs are faulty, the base station enters the deep energy saving state upon receiving a deep backup power saving message and shuts down the power amplifiers of all TRXs. As a result, all TRXs are out of service.

For a NodeB/eNodeB/gNodeB, the Intelligent Shutdown of the TRX Due to PSU Failure function is enabled when the **EQUIPMENT.PSUFP** parameter is set to **ON**. In this case, if one PSU is faulty, the power of all TRXs decreases by 3 dB (50% of the original power). When two or more PSUs are faulty, the power amplifiers of all TRXs are shut down.

### 13.1.2 Power Supply Reliability for a BBU

Power supply reliability for a BBU includes good environment adaptability, improved fault handling mechanism, and sound power consumption management for BBU boards.

- Good environment adaptability
  - Wide range of input voltage: The BBU supports -48 V DC power input and an actual input voltage range of -57 V DC to -38.4 V DC.
  - Wide range of operating temperatures: The BBU supports an operating temperature range of -20°C to 60°C.
  - Satisfying indoor protection: The BBU does not require an additional surge protection unit.
- Improved fault handling mechanism
  - Protection against reverse connection: When the input positive and negative poles are reversely connected, the power board is not powered on, preventing the power board from being damaged.
  - Protection against undervoltage: When the input voltage is lower than the lower threshold of the operating voltage range, the power board stops working, preventing the power board from being damaged. When the input voltage becomes normal, the power board restarts.
  - Protection against output overload: When the power supply requirements of the BBU exceed the power supply capability of power boards, the power board enters hiccup protection mode, preventing the power board, power-consuming devices, and system from being damaged. In this case, the BBU will be reset.
  - Protection against output short-circuits: If an output short-circuit occurs, the power board enters hiccup protection mode, preventing the power board, power-consuming devices, and system from being damaged. In this case, the BBU will be reset.
  - Protection against output overvoltage: If the output overvoltage occurs, the power board enters hiccup protection mode, preventing the power board, power-consuming devices, and system from being damaged. In this case, the BBU will be reset.
  - Protection against overtemperature: The power board stops working when its temperature is too high and restarts when its temperature

returns to the normal operating temperature. In this case, the BBU will be reset.

 **NOTE**

Hiccup protection mode: When a power board experiences a fault that may damage itself, the power board stops supplying power and at the same time continues detecting whether the fault is rectified. Once the fault is rectified, the power board resumes the power supply.

- Sound power consumption management for BBU boards
  - When the power supply capability of power boards in the BBU is insufficient because of board expansion or power board failure, the BBPs with a low power-on priority are powered off, preventing power overload in the BBU.
  - After a BBU is restarted due to insufficient power supply, the BBU attempts to power on the BBPs after it is powered on again. If the BBU is restarted for a second time due to insufficient power supply after powering on BBPs, some BBPs will not be powered on after the BBU is powered on for the third time. This ensures the power supply to other boards in the BBU.

Power supply reliability for a BBU is a basic function and does not require any software configuration.

## 13.2 Network Analysis

### 13.2.1 Benefits

None

### 13.2.2 Impacts

The impacts between this function and other functions are described in [13.3.2.3 Function Impacts](#).

This function has no impact on the network.

## 13.3 Requirements

### 13.3.1 Licenses

None

### 13.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

### 13.3.2.1 Prerequisite Functions

None

### 13.3.2.2 Mutually Exclusive Functions

| RA<br>T | Function<br>Name                         | Function Switch                                   | Referenc<br>e                                                   | Description                                                                                                                                                 |
|---------|------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GS<br>M | GSM Low<br>Power<br>Consumpt<br>ion Mode | <b>GBTSENERGYMGTPARA.LOW<br/>PWRCONSUMPMODESW</b> | <i>Intellige<br/>nt Power<br/>Consum<br/>ption<br/>Decrease</i> | When GSM Low<br>Power<br>Consumption<br>Mode is enabled,<br>the Intelligent<br>Shutdown of TRX<br>Due to PSU<br>Failure feature<br>does not take<br>effect. |

### 13.3.2.3 Function Impacts

| RA<br>T                          | Function<br>Name                                         | Function Switch                                                                                                                                                                                                   | Referenc<br>e         | Description                                                                                                                                                     |
|----------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GS<br>M                          | SOC level-<br>based low<br>power<br>consumpti<br>on mode | <ul style="list-style-type: none"> <li><b>IPWRM.SOCLVLADJSW</b> set to ON</li> <li><b>GBTSENERGYMGTPARA.S<br/>OCLVLLLOWPWARMODESW</b></li> </ul>                                                                  | <i>Green<br/>Site</i> | Intelligent<br>Shutdown of TRX<br>Due to PSU<br>Failure does not<br>take effect as<br>long as the<br><b>IPWRM.SOCLVL<br/>ADJSW</b><br>parameter is set<br>to ON |
| UM<br>TS                         | SOC level-<br>based low<br>power<br>consumpti<br>on mode | <ul style="list-style-type: none"> <li><b>IPWRM.SOCLVLADJSW</b> set to ON</li> <li><b>SOCLVL_BASED_LOW_PW<br/>R_SW</b> option of the<br/><b>NODEBALGPARA.LOWPW<br/>RCONSUMEMODEEXTSW</b><br/>parameter</li> </ul> | <i>Green<br/>Site</i> |                                                                                                                                                                 |
| LTE<br>FD<br>D<br>LTE<br>TD<br>D | SOC level-<br>based low<br>power<br>consumpti<br>on mode | <ul style="list-style-type: none"> <li><b>IPWRM.SOCLVLADJSW</b> set to ON</li> <li><b>SOCLVL_BASED_LOW_PW<br/>R_SW</b> option of the<br/><b>CellLowPower.LowPwrCo<br/>nsumeModeExtSw</b><br/>parameter</li> </ul> | <i>Green<br/>Site</i> |                                                                                                                                                                 |

### 13.3.3 Hardware

- The PMU and PSU must be provided by Huawei.
- For details about how to install the PMU and PSU, see the section related to monitoring signal cable installation in the base station-specific installation guide.

### 13.3.4 Others

None

## 13.4 Operation and Maintenance

### 13.4.1 When to Use

For macro base stations and LampSite base stations, basic power modules such as PMU and PSU must be configured to ensure the power supply and power distribution functions. It is recommended that the Intelligent Shutdown of Carriers Due to PSU Failure feature be activated for Huawei AC-powered base stations configured with the PMU and PSU.

### 13.4.2 Data Configuration

#### 13.4.2.1 Data Preparation

[Table 13-2](#) describes the GBTS data preparation.

**Table 13-2** GBTS data preparation

| Parameter Name                        | Parameter ID                                    | Description                                                                                                                                                                                        |
|---------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tran. Cabinet Configured              | <b>BTSPSUFP.<i>ISTRANCABCON</i></b>             | When this parameter is set to <b>CONFIG</b> , which is recommended, the power consumption of the transmission cabinet must be considered. You are advised to set this parameter to <b>ENABLE</b> . |
| Service Priority Policy               | <b>BTSPSUFP.<i>SRVPRI</i><br/><i>POLICY</i></b> | This parameter specifies which type of service should be preferentially processed when the PSU is faulty. You are advised to set this parameter to <b>GSM_PRIOR</b> .                              |
| Board Parameter Configuration Enabled | <b>BTSPSUFP.<i>CFGFL</i><br/><i>AG</i></b>      | If the Intelligent Shutdown of TRX Due to PSU Failure feature needs to be enabled, you are advised to set this parameter to <b>YES</b> .                                                           |

[Table 13-3](#) describes the eGBTS data preparation.



**Table 13-3** eGBTS data preparation

| Parameter Name                       | Parameter ID                                    | Description                                                                                                                                                          |
|--------------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PSU Fault Shutdown TRX Enabled       | <b>EQUIPMENT.PSUF P (LTE eNodeB, 5G gNodeB)</b> | It is recommended that this parameter be set to <b>OFF</b> . To enable the Intelligent Shutdown of TRX Due to PSU Failure feature, set this parameter to <b>ON</b> . |
| Enhanced Backup Power Saving Allowed | <b>GBTSENERGYMGT PARA.ENBAKPWR SAVALLOW</b>     | It is recommended that this parameter be set to <b>NO</b> . To enable the enhanced backup power saving function, set this parameter to <b>YES</b> .                  |
| Deep Backup Power Saving Allowed     | <b>GBTSENERGYMGT PARA.DEEPBAKP WRSALLOW</b>     | It is recommended that this parameter be set to <b>NO</b> . To enable the deep backup power saving function, set this parameter to <b>YES</b> .                      |

**Table 13-4** describes the NodeB/eNodeB/gNodeB data preparation.

**Table 13-4** NodeB/eNodeB/gNodeB data preparation

| Parameter Name                 | Parameter ID                                    | Description                                                                                                                                                          |
|--------------------------------|-------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PSU Fault Shutdown TRX Enabled | <b>EQUIPMENT.PSUF P (LTE eNodeB, 5G gNodeB)</b> | It is recommended that this parameter be set to <b>OFF</b> . To enable the Intelligent Shutdown of TRX Due to PSU Failure feature, set this parameter to <b>ON</b> . |

### 13.4.2.2 Using MML Commands

Before using MML commands, refer to [13.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

### Activation Command Examples

#### GBTS

Run the **SET BTSPSUFP** command to enable the Intelligent Shutdown of TRX Due to PSU Failure feature and set related parameters.

```
SET BTSPSUFP: IDTYPE=BYID, BTSID=0, CFGFLAG=YES,  
ISTRANCABCON=CONFIG,SRVPRIPOLICY=GSM_PRIOR;
```

### eGBTS

Run the **SET EQUIPMENT** and **SET GBTSENERGYMGTPARA** commands to enable the Intelligent Shutdown of TRX Due to PSU Failure feature.

```
SET EQUIPMENT: PSUFP=ON;  
SET GBTSENERGYMGTPARA: ENBAKPWRSALLOW=YES, DEEPBAKPWRSALLOW=YES;
```

### NodeB/eNodeB/gNodeB

Run the **SET EQUIPMENT** command to enable the Intelligent Shutdown of TRX Due to PSU Failure feature. In this step, set **PSU Fault Shutdown TRX Enabled** to **ON(ON)**.

```
SET EQUIPMENT: PSUFP=ON;
```

## Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

### GBTS

On the BSC side, run the **SET BTSPSUFP** command with **Board Parameter Configuration Enabled** set to **NO(NO)**.

```
SET BTSPSUFP: IDTYPE=BYID, BTSID=0, CFGFLAG=NO;
```

### eGBTS

Run the **SET EQUIPMENT** and **SET GBTSENERGYMGTPARA** commands to disable the Intelligent Shutdown of TRX Due to PSU Failure feature.

```
SET EQUIPMENT: PSUFP=OFF;  
SET GBTSENERGYMGTPARA: ENBAKPWRSALLOW=NO, DEEPBAKPWRSALLOW=NO;
```

### NodeB/eNodeB/gNodeB

Run the **SET EQUIPMENT** command to disable the Intelligent Shutdown of TRX Due to PSU Failure feature. In this step, set **PSU Fault Shutdown TRX Enabled** to **OFF(OFF)**.

```
SET EQUIPMENT: PSUFP=OFF;
```

## 13.4.2.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

## 13.4.3 Activation Verification

### GBTS

**Step 1** Run the **LST BTSPSUFP** command to query the parameters about TRX shutdown due to PSU failures.

Expected result:

- The value of **Board Parameter Configuration Enabled** is **YES**.
- The value of **Tran. Cabinet Configured** is consistent with that configured in the activation procedure.
- The value of **Service Priority Policy** is **GSM\_PRIOR(GSM Traffic Prior)**.

**Step 2** Remove some PSUs from the site. Ensure that the power supplied by remaining PSUs cannot meet requirements of all TRXs at the site but can meet requirements of other devices.

**Step 3** Run the **DSP CHNSTAT** command to query the channel status of all TRXs at the site. In this step, set **Object Type** to **SITE(Site)**.

Expected result: Some channels on TRXs are available and some are not. The site works properly and is not powered off due to insufficient power supply.

----End

## eGBTS/NodeB/eNodeB/gNodeB

**Step 1** Run the **LST EQUIPMENT** command to verify that the value of **PSU Fault Shutdown TRX Enabled** is **ON**.

**Step 2** Use a power meter to test the power of the BCCH TRX and record the result.

**Step 3** Remove one PSU from the base stations so that the power of all TRXs is halved.

**Step 4** Use a power meter to test the power of the BCCH TRX, which is expected to be half of that tested in step 2.

----End

## 13.4.4 Network Monitoring

None

# 14 Other Functions

This function is implemented in the same way on macro base stations and LampSite base stations.

## 14.1 RF Channel Cooperation (GSM/UMTS)

With the development of mobile communications, wireless network coverage increasingly extends to remote areas along with a rapid increase in demand for network services.

However, the terrain, climate, or traffic conditions in remote areas may be extreme. As a result, network maintenance is difficult and services may be interrupted if a remote radio unit (RRU) is faulty. To facilitate site maintenance, RF Channel Cooperation is introduced. With RF Channel Cooperation, when one RF channel becomes faulty, the system automatically switches the services carried on the faulty RF channel to a functional RF channel. This shortens the period of service interruption caused by a fault in the RF channel and improves system reliability.

**Table 14-1** describes the features involved in RF Channel Cooperation. For details about these features, see the corresponding feature parameter description.

**Table 14-1** Features involved in RF Channel Cooperation

| RAT  | Feature                     | Feature Parameter Description |
|------|-----------------------------|-------------------------------|
| GSM  | GBFD-113801 TRX Cooperation | TRX Cooperation               |
| UMTS | WRFD-040203 RRU Redundancy  | RRU Redundancy                |

## 14.2 Heat Dissipation Reliability for Fans

Fans are used for inner and outer air circulation, allowing heat to dissipate from the equipment through a ventilation channel. When a fan on a ventilation channel

is faulty, heat dissipation will be affected. Fans do not support redundancy design in hardware due to inconvenient installation. To ensure adequate heat dissipation, the following functions are provided:

- When the FMU works in smart temperature control mode, the FMU adjusts the rotation speed of fans based on the temperature control parameters delivered by the BBU. If a fan becomes faulty, ALM-25673 Fan Stalled is reported and the policy for adjusting the rotation speed of other fans remains unchanged.
- When the FMU works in temperature control mode and cannot obtain the temperature information of the equipment, the FMU adjusts the rotation speed of fans based on the ambient temperature. If a fan becomes faulty, ALM-25673 Fan Stalled is reported and other fans in the same fan group rotate at full speed to ensure heat dissipation.
- When the TCU cannot obtain the temperature at the air exhaust vent of the cabinet, fans in the TCU rotate at full speed. If a fan becomes faulty, ALM-25673 Fan Stalled is reported and other fans in the same fan group rotate at full speed to ensure heat dissipation.
- When a fan in the FAN unit of the BBU becomes faulty, ALM-26110 BBU Fan Stalled and ALM-26111 BBU Fan Not at Full Speed are reported and other fans rotate at full speed to ensure heat dissipation.

 **NOTE**

If only one fan in the BBU5900A is stalled and there is no ambient overtemperature risk, fans that are not stalled do not rotate at full speed. Instead, the rotation speed of the fans is adjusted based on the temperature control mode (specified by the **TCU.TCMODE (5G gNodeB, LTE eNodeB)** parameter).

- When the control signals for a fan in the FMU or TCU are unavailable, the fan in the FMU or TCU rotates at full speed.
- When ALM-26101 Inter-Board CANBUS Communication Failure is reported, fans in the BBU rotate at full speed.

## 14.3 Power Supply Redundancy

Power supply redundancy consists of power supply redundancy for a base station and power supply redundancy for a BBU.

### 14.3.1 Power Supply Redundancy for a BBU

The BBU supports 1+1 backup mode for power boards.

Currently, only the UPEUe, UPEUc, or UPEUd board can work in 1+1 backup mode. When the configured power consumption exceeds the power supply capability of a UPEU board, 1+1 backup mode is not supported.

In the normal working state, the two power boards share the power load. When a power board becomes faulty, the power load on the faulty board automatically switches to the other board, avoiding service interruption.

To work in 1+1 backup mode, power boards in the BBU must meet the following requirements:

- Each power board can undertake the power load of the whole BBU.
- The two power boards are of the same type and have the same specifications.

Power Supply Redundancy for a BBU is a basic function and does not require any software configurations.

### 14.3.2 Power Supply Redundancy

PSUs in a base station should be configured in N+1 backup mode. After power supply redundancy for a base station is enabled for a Huawei AC-powered base station equipped with a PMU, ALM-25636 Loss of Power Supply Redundancy is reported if PSUs in the base station are not configured in N+1 mode. The reported alarm alerts the customer to the insufficiency of PSUs.

The GSM, UMTS, LTE, and NR base stations support this feature by default. This feature has no independent feature ID. For the principles and engineering guidelines, see sections "Reporting of ALM-25636 Loss of Power Supply Redundancy" and "Deployment of Reporting of ALM-25636 Loss of Power Supply Redundancy" in *Power Supply Management*.

## 14.4 Anti-Misinsertion Design of Boards

When a board of one type is inserted into a slot for a board of another type, the board cannot connect to the backplane. This prevents the board from being damaged.

## 14.5 Overtemperature Protection for BBU Boards

When the temperature of a BBU board exceeds its maximum operating temperature, the lifespan of the board may be shortened or its reliability may be affected. In the worst-case scenario, the board may be burnt out, imposing safety risks. To prevent this from happening, Huawei provides the Power-Off on Overtemperature function.

### 14.5.1 Overtemperature Power-off for Non-Main-Control Boards

#### Power-off Requirements

Overtemperature power-off for non-main-control boards can be classified into passive power-off and active power-off.

- Passive power-off: The main control board powers off a non-main-control board and reports ALM-26214 Board Powered Off when any of the following conditions is met:
  - A minor overtemperature alarm is generated on a non-main-control board and lasts for more than 24 hours.
  - A severe overtemperature alarm is generated on a non-main-control board and lasts for more than 1 hour.
  - The temperature of a non-main-control board is higher than the overtemperature power-off threshold.

- Active power-off: When detecting that the temperature exceeds the overtemperature power-off threshold, a non-main-control board automatically powers off itself for self-protection and reports ALM-26214 Board Powered Off.

## Power-on Requirements

The overtemperature alarm reported on a non-main-control board can be manually or automatically cleared only when the main control board is not powered off due to overtemperature.

- Automatic mode: When the main control board detects that the temperature of a non-main-control board meets the alarm clearing threshold, the overtemperature alarm is automatically cleared. If the non-main-control board has been powered-off in this case, the main control board powers on the non-main-control board automatically. The following conditions must be met for automatically clearing the overtemperature alarm or powering on a non-main-control board:
  - The fans work properly and no alarm indicating that the fans stall is reported.
  - The temperature of the non-main-control board is 5°C lower than the threshold for a minor overtemperature alarm.
  - No severe overtemperature alarm exists on the main control board.
  - More than 10 minutes have elapsed since the non-main-control board has been powered off.
- Manual mode: Users can forcibly power on a non-main-control board by running an MML command to quickly restore it. In this case, the reported alarms will not be cleared unless the conditions for automatically clearing the alarm are met. If the temperature of the non-main-control board is higher than the overtemperature power-off threshold after it is forcibly powered on, the main control board will power off the non-main-control board again. Otherwise, the non-main-control board will stay in powered-on status.

## Impact of Overtemperature Power-off of Non-Main-Control Boards on Multimode Base Stations

In a multimode base station, a main control board detects the temperature of boards working in the same RAT as itself (such as the BBP and transmission board, excluding the power board and fan board) and does not manage boards working in other RATs or boards that are not configured.

When a non-main-control board is powered off due to overtemperature, services of the peer RAT may be affected or even interrupted in scenarios such as co-transmission or CPRI MUX. The impact of overtemperature power-off on services of the peer RAT is the same as that caused by other faults on the board.

## 14.5.2 Overtemperature Power-off for Main Control Boards

### Power-off Requirements

If the temperature of a main control board is higher than the common overtemperature alarm threshold, a common overtemperature alarm is reported. If

the temperature continues to rise and becomes higher than the severe overtemperature alarm threshold, a severe overtemperature alarm is reported. In this case, all baseband processing units in the BBU subrack are powered off. The main control board powers off itself in the following situations:

- If the temperature of the main control board is higher than the severe overtemperature alarm threshold for more than one hour, the main control board reports ALM-26214 Board Powered Off, powers off all boards except itself and the baseband processing units in the BBU subrack, and then powers off itself.
- If the temperature of the main control board is higher than the overtemperature power-off threshold, the main control board reports ALM-26214 Board Powered Off and powers off itself.

## Power-on Requirements

If a main control board is powered off due to overtemperature, users must troubleshoot the fault onsite and then power on the main control board.

## Impact of Overtemperature Power-off of Main Control Boards on Multimode Base Stations

- In a multimode base station, the impact of overtemperature power-off of a main control board on services is the same as that caused by a reset or fault of the main control board.
- In a separate-MPT multimode base station, each main control board only manages itself and other boards working in the same RAT. In a co-MPT multimode base station, the active main control board manages all the boards in the BBU subrack. In this case, the active main control board reports alarms for all the boards and powers off them at the same time when necessary, without considering the RAT priority.

# 14.6 Surge Protection Design

The surge protection design of Huawei products complies with related standards. Different surge protection solutions are provided for different ports.

## 14.6.1 Standards

| No. | File Code  | File Name                                                                                  |
|-----|------------|--------------------------------------------------------------------------------------------|
| 1   | IEC62305-1 | Protection against lightning – Part 1: General principles                                  |
| 2   | IEC62305-2 | Protection against lightning – Part 2: Risk management                                     |
| 3   | IEC62305-3 | Protection against lightning – Part 3: Physical damage to structures and life hazard       |
| 4   | IEC62305-4 | Protection against lightning – Part 4: Electrical and electronic systems within structures |



| No. | File Code       | File Name                                                                                                                                                                                 |
|-----|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5   | ITU-T K.56      | Protection of radio base stations against lightning discharges                                                                                                                            |
| 6   | ITU-T K.35      | Bonding configurations and earthing at remote electronic sites                                                                                                                            |
| 7   | ITU-T Handbook  | ITU-T Earthing and Bonding Handbook                                                                                                                                                       |
| 8   | IEC 60364-5-54  | Electrical installations of buildings – Part 5-54<br>Selection and erection of electrical equipment –<br>Earthing arrangement, protective conductors and<br>protective bonding conductors |
| 9   | YD 5098         | Specifications on Engineering Design of Lightning<br>Protection and Earthing for Telecommunication<br>Bureaus (Stations)                                                                  |
| 10  | GB50689-2011    | Code for design of lightning protection and<br>earthing engineering for telecommunication<br>bureaus (stations)                                                                           |
| 11  | ITU-T K.97      | Lightning protection of distributed base stations                                                                                                                                         |
| 12  | ETSI EN 300 253 | Earthing and bonding of ICT equipment powered<br>by -48 VDC in telecom and data centres                                                                                                   |

## 14.6.2 Surge Protection Capability of Ports

The following table lists the surge protection capability of different ports.

| No. | Port Type    | Surge Protection Capability                                                                                                                                                                                              |
|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | AC           | A BTS3900A or BTS5900A meets the requirement for 30 kA (8/20 $\mu$ s) surge protection and no external surge protector is required. A BTS3900 or BTS5900 meets the requirement for 5 kA (8/20 $\mu$ s) surge protection. |
| 2   | DC           | The requirement for 4 kV (1.2/50 $\mu$ s) surge protection is met in indoor scenarios and the requirement for 20 kA (8/20 $\mu$ s) surge protection is met in outdoor scenarios.                                         |
| 3   | Antenna port | The built-in 40 kA surge protection design meets the surge protection requirements in all scenarios and no additional surge protector is required.                                                                       |

| No. | Port Type             | Surge Protection Capability                                                                                           |
|-----|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| 4   | E1/T1                 | Different surge protection solutions are provided in indoor and outdoor scenarios and no surge protector is required. |
| 5   | GE/FE                 | Different surge protection solutions are provided in indoor and outdoor scenarios and no surge protector is required. |
| 6   | RGPS                  | Built-in surge protection is used on the equipment and no surge protector is required.                                |
| 7   | GNSS                  | A surge protector is required on the equipment side.                                                                  |
| 8   | Dry contact/<br>RS485 | Different surge protection solutions are provided in indoor and outdoor scenarios and no surge protector is required. |
| 9   | AISG                  | Built-in surge protection is used and no surge protector is required.                                                 |

# 15 Parameters

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The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

### **FAQ 1: How do I find the parameters related to a certain feature from parameter reference?**

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

### **FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?**

- Step 1** Open the EXCEL file of the used reserved parameter list.

- Step 2** On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.
- End

# 16 Counters

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The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

## NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

## **FAQ: How do I find the counters related to a certain feature from performance counter reference?**

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

# 17 Glossary

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For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

# 18 Reference Documents

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1. *TRX Cooperation*
2. *RRU Redundancy*
3. *3900 & 5900 Series Base Station Initial Configuration Guide* in *3900 & 5900 Series Base Station Product Documentation*
4. *Power Supply Management*
5. *Cell Management*
6. *RF Unit and Topology Management*
7. *3900 & 5900 Series Base Station Commissioning Guide*
8. *Transmission Security Overview*
9. *3900 & 5900 Series Base Station Technical Description*
10. *GSM and LTE Dynamic Power Sharing*
11. *Multi-RAT Coordinated Energy Saving*
12. *Distributed Antenna Solutions (Low-Frequency TDD)*
13. *Energy Conservation and Emission Reduction* in *eRAN Feature Documentation*
14. *IPv4 Transmission* or *IPv6 Transmission*
15. *IP eRAN Engineering Guide* or *IP NR Engineering Guide*
16. *S1 and X2 Self-Management* or *X2 and S1 Self-Management* in *NSA Networking*
17. *IPsec*
18. *NG and Xn Self-Management*
19. *eCPRI*
20. *Energy Conservation and Emission Reduction* in *5G RAN Feature Documentation*
21. *License Management*
22. *Intelligent Power Consumption Decrease*
23. *Green Site*
24. *X2 Self-Management* in *SA Networking*